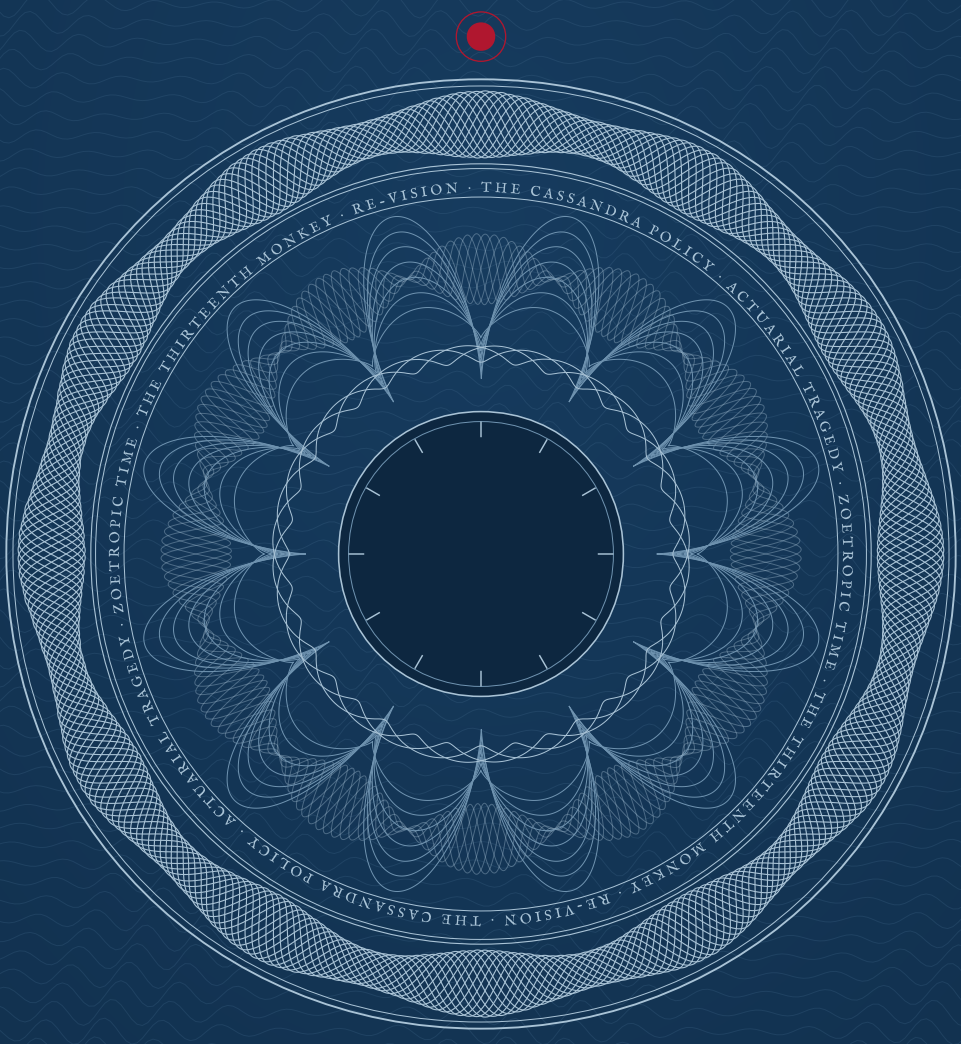
The Cassandra Policy




ACTUARIAL TRAGEDY · ZOETROPIC TIME · THE THIRTEENTH MONKEY · RE-VISION · THE CASSANDRA POLICY · ACTUARIAL TRAGEDY · ZOETROPIC TIME · THE THIRTEENTH MONKEY · RE-VISION · THE CASSANDRA POLICY ·



The Cassandra Policy

Actuarial Tragedy, Zoetropic Time, and the Thirteenth Monkey in Terry Gilliam's Twelve Monkeys

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Abstract

This dissertation proposes a new theoretical paradigm for Terry Gilliam's *Twelve Monkeys* (1995), written by David Webb Peoples and Janet Peoples and credited as "inspired by" Chris Marker's photo-roman *La Jetée* (1962). Against the three paradigms that have governed its reception—the ludic-formal reading of the film as a time-travel puzzle, the psychiatric-hermeneutic reading of it as a study in undecidable madness, and the intermedial reading of it as a Hollywood animation of Marker's modernist stills—it argues that *Twelve Monkeys* is the paradigmatic text of a form here named *actuarial tragedy*: a dramatic structure in which the necessity that Attic tragedy assigned to the gods is reassigned to an insurance rationality, so that catastrophe ceases to be what a hero suffers and a city learns from and becomes instead what a future institution requires, underwrites, and harvests. The argument turns on the film's last line of consequence, in which a scientist from 2035, seated beside the man who carries the virus, introduces herself as being "in insurance." Read back through the whole work, the line discloses what the dissertation calls *the Cassandra Policy*: the institutional design by which a future whose existence is premised on the past's catastrophe dispatches a prophet whose disbelief it does not merely tolerate but structurally requires. The curse that Aeschylus staged as Apollo's erotic vengeance becomes an actuarial clause; and the god who in the first book of the *Iliad* is at once oracle, plague-archer, and healer returns as a council of scientists who send the prophet, need the plague, and await the cure.

Five convergent inquiries sustain the claim. The first reconstructs the ontology of the image that the film inherits from *La Jetée*—the photogram as Barthes's "that-has-been" and anterior future, Bazin's embalmed time, Benjamin's dialectical image—and rereads Marker's single moving shot, a woman's blink, as the *Augenblick* of Kierkegaard, Nietzsche, and the Pauline "twinkling of an eye." The second analyses the metaphysics of the closed timeline in dialogue with Sophocles, Stoic co-fatedness, Boethius, Augustine, David Lewis, the Novikov self-consistency principle, Kierkegaard's repetition "recollected forward," Freud's *Nachträglichkeit*, and Deleuze's ethics of the event. The third reads the asylum as the clinic in which disbelief is manufactured, through Foucault, Goffman, Szasz, Laing, and Ian Hacking's looping kinds, and through two historical Cassandras: Ignaz Semmelweis, whom the

film names, and the Hiroshima reconnaissance pilot Claude Eatherly, whom Günther Anders defended as the one sane man in an insane order. The fourth argues that the film's emblem, a ring of twelve monkeys, is overdetermined as zodiac, clock face, zoetrope, and Aeschylean chorus of twelve, and that the human appears within it as *the thirteenth monkey*: Nietzsche's promising animal become the insuring animal, the primate that steps out of cyclical animal time into calculable history and is returned to the ring by plague. The fifth reconstructs the genealogy of actuarial reason from the plague-born London Bills of Mortality through John Graunt, Edmond Halley—astronomer, and author of the first scientific life table—and Adolphe Quetelet to Ulrich Beck's risk society and François Ewald's philosophy of insurance, and sets the film's future against the enlightened doomsaying of Hans Jonas, Günther Anders, and Jean-Pierre Dupuy.

The dissertation concludes that *Twelve Monkeys* locates the last freedom available within a fixed universe not in revision, the alteration of what has happened, but in *re-vision*, the transformed look by which an unalterable image is seen otherwise. Cinema —whose reel, as the protagonist observes, "never changes," yet appears different to a changed spectator—is at once the film's model of this freedom and its proof. Sustained comparison across Attic tragedy, scripture, the plague narrative from Thucydides to Camus, speculative fiction from Mary Shelley and H. G. Wells to Kurt Vonnegut, Ted Chiang, and the series *Dark*, and across painting, comics, and the time-loop video game, situates *Twelve Monkeys* as an indispensable text for thinking catastrophe in the age of pandemics, risk, and the Anthropocene.

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Chapter One

Prolegomena

The Cassandra Policy and the Idea of Actuarial Tragedy

'Tis the times' plague, when madmen lead the blind.

— Shakespeare, *King Lear*, IV.i

Apollo, Apollo, god of the ways, my destroyer.

— Cassandra, in Aeschylus, *Agamemnon* (trans. mine)

I.I
The Scene at the End of the Jetty

Twelve Monkeys ends where *La Jetée* ends and where, in both films, everything began: at the threshold of an airport, before the eyes of a child. A man runs, is shot, and falls. A woman in a blonde wig kneels over him in the posture that Western devotional art has reserved, since the German *Vesperbild* of the early fourteenth century, for the Virgin bent over the body of her son. And a boy, held at the edge of the action by his parents, watches a scene whose meaning he will not possess for decades and whose content is his own death. In Marker's film this image is the whole of the protagonist's destiny, and its sense arrives only with the narrator's final sentences: the moment the child had been permitted to witness, and which had never ceased to obsess him, was the moment of his death. Gilliam and the Peoples preserve this structure with an almost liturgical fidelity, but they append to it a coda that Marker's film neither has nor needs. The camera leaves the concourse for the cabin of an airliner. Dr. Peters—the laboratory assistant whose case of vials has just passed through airport security, and who is about to seed a virus along an itinerary of world cities—settles into his seat beside a composed, middle-aged woman whom the attentive spectator recognises, with a small shock, as one of the scientists of 2035. She offers her hand, her name, Jones, and her profession. She is, she says, "in insurance." The film returns once more to the boy, whose upturned face follows the departing aircraft, and the credits roll beneath Louis Armstrong's "What a Wonderful World."

The coda has been read in two incompatible ways, and their incompatibility is the hinge of this dissertation. On the first reading, which the film's affect solicits and much popular commentary prefers, the scientist's presence on the aircraft signals that the mission has succeeded after all: the future has identified the true source of the plague and placed an agent at its side, and the catastrophe may yet be averted. On the second, which the film's metaphysics enforces, nothing has been averted, because nothing can be. Cole has insisted throughout, against the scepticism of his psychiatrist and with the flat authority of a man reporting what he has already lived through, that the past is unalterable; that the five billion dead are already dead; that he has been sent not to save anyone but to gather information, so that the survivors beneath the earth might obtain an unmutated sample of the virus and fashion a cure that would permit them to reclaim the surface. On this reading the scientist has come for the sample, not for Peters. The plague will happen because it has happened, and the future has arranged only to be present at the scene in order to collect.

This dissertation does not choose between these readings by adjudicating plot logic. It argues that the word the film places in the scientist's mouth makes the choice, and that the choice is a third possibility which the two readings jointly occlude. Insurance is neither prevention nor passivity. The insurer does not extinguish the fire, nor does he merely watch the house burn; he converts the fire in advance into a calculable loss, and he converts the loss after the event into a claim. Insurance is a technology of time: it binds an uncertain future to a present contract, prices the not-yet in the currency of the now, and makes the occurrence of the disaster the condition under which the contract performs. The insurer's interest in a catastrophe that never arrives is exhausted by the premium; his interest in a catastrophe that does arrive is absolute, because only then does the policy become what it promised to be. When the woman from 2035 calls herself an insurer, she names with deadpan exactitude the relation between her world and ours. The future does not save the past; it underwrites it. The society beneath the ruined city is constituted by the catastrophe that destroyed its predecessor, and its agent has come not to cancel the loss on which her existence depends but to ensure that the loss yields its indemnity: a pure sample, a cure, a return to the surface, a humanity refounded on the ruins of the old.

From this terminal disclosure the whole film can be reread, and the rereading yields what this dissertation calls *the Cassandra Policy*. Cole is, as the film all but announces, a Cassandra: a man who knows the future, speaks it, and is not believed—confined, sedated, diagnosed, his prophecy entered into the clinical record as the speech of a paranoid schizophrenic. The film's opening intertitles—five billion dead of a virus in 1997, the surface abandoned, the animals once more in possession of the world—are attributed to interviews with a "clinically diagnosed paranoid schizophrenic" conducted in a Baltimore hospital in 1990, and the sequences that follow disclose that patient to be Cole himself. The prophecy enters history already classified as madness. In the *Agamemnon*, Cassandra's affliction is the private vengeance of a god she refused: Apollo gave her prophecy as a lover's gift and, spurned, ensured that no one would heed it. In *Twelve Monkeys* the curse has been institutionalised. The future that sends Cole does not need him to be believed; it needs him not to be. A believed prophecy might avert the catastrophe, and an averted catastrophe would void the future that issued it—its survivors, its science, its authority, its prison, its very need for a machine that travels in time. The disbelief of the past is therefore not an obstacle to the mission but its actuarial premise. What tragedy presented as the caprice of a god appears here as a clause in a contract: a structural guarantee, written into the logic of a closed timeline, that the warning will be delivered and will fail.

The divinity who presides over this transformation is the one Cassandra apostrophises, in Aeschylus, with a pun upon his name: *Apollōn*, *apollōn*—Apollo, my

destroyer. He is the god of prophecy; but the *Iliad* opens with the same god as Smintheus, lord of mice, loosing pestilential arrows upon the Achaean camp for nine days because his priest has been dishonoured; and he is also Paian, the healer, father of Asclepius. Oracle, plague-archer, physician: these three offices, which the archaic imagination united in a single figure, are the offices the scientists of 2035 resume. They dispatch the prophet; they require the plague, or at least its uncorrupted sample; and they await the cure. *Twelve Monkeys* is, among other things, the return of an Apollonian complex in the idiom of late modernity, and its Smintheus, the plague-god whose emblem is a rodent, has been rationalised into an underwriter.

Yet the scene at the end of the jetty contains something the policy does not cover. Kneeling over Cole, Kathryn Railly lifts her eyes and finds the boy, and the boy looks back. The exchange alters nothing in the chain of events: the child will grow up, be imprisoned, be sent back, be shot. But it alters the image. A woman who had spent the film diagnosing prophecy as pathology now possesses the knowledge the prophet possessed, and she regards the child who will become him with a look in which recognition and grief are no longer distinguishable. The film has prepared this moment in a scene set, appropriately, in a cinema. Hiding from the police in a Hitchcock retrospective, Cole watches *Vertigo* and remarks that the film "never changes," that it cannot change, and that it nonetheless seems different each time because the viewer is different and sees different things. That remark is the film's theory of itself and the hinge of the ethics this dissertation attributes to it. In a universe whose past is fixed, freedom cannot take the form of *revision*, the alteration of what has happened. It can take only the form of *re-vision*—the term is Adrienne Rich's, and it is developed below—the transformed look by which an unalterable image is seen otherwise. *Twelve Monkeys* stages the contest between two relations to a fixed catastrophe: the actuarial relation, which calculates it, and the visionary relation, which sees it. Between the insurer in the aircraft and the woman on the concourse, the film distributes the possibilities open to a civilisation that has learned it cannot change its past and must decide what to do with its knowledge of it.

Stated in its most compressed form, the thesis of this dissertation is that *Twelve Monkeys* is the paradigmatic instance of actuarial tragedy, a form defined by five features. First, tragic necessity is administered by an institution rather than decreed by a god. Second, the catastrophe is the founding condition of the very authority that investigates and narrates it. Third, the prophet's disbelief is structurally required by the future that sends him. Fourth, the human appears as the animal that insures itself against its own end and, in the same gesture, secures that end, since the laboratory that promises the cure is the laboratory that breeds the plague. Fifth, the only freedom that survives is the freedom of re-vision. The remainder of this chapter situates

the film genealogically and historically, reviews the state of the scholarship, defines the method and the conceptual lexicon of the argument, and sets out the architecture of the chapters that follow.

1.2
From Photo-Roman to Studio Picture

The Genealogy of an Inspiration

La Jetée runs some twenty-eight minutes and is composed almost entirely of still photographs. Marker called it a *photo-roman*, borrowing the name of a popular genre of captioned photographic romance and converting it into an instrument of metaphysical inquiry. Over a sequence of black-and-white stills, joined by cuts and dissolves, a narrator tells the story. Paris has been destroyed in the Third World War; the survivors inhabit the galleries beneath the Palais de Chaillot, where the victors conduct experiments upon prisoners. Space being closed to humanity, the only hope lies in time: the experimenters, whose whispered German drifts across the soundtrack, seek to send subjects into the past and the future in quest of food, medicine, and energy. Most subjects go mad or die. The protagonist is chosen because he is fixated upon an image from his childhood—a Sunday on the observation pier at Orly, a woman's face at the end of the jetty, a sudden commotion, a man falling. Injected and blindfolded, he travels into a peacetime past and finds the woman. They walk in gardens, contemplate the cross-section of a sequoia scored with historical dates, wander through a museum of stuffed animals. He is then projected into the future, whose inhabitants grant him a power source sufficient to set human industry in motion again. Marked for liquidation by his captors once his usefulness is exhausted, he is offered refuge by the people of the future, but asks instead to be returned to the world of his childhood and to the woman. He reaches the jetty at Orly, runs toward her, and is shot by an agent of the camp who has followed him. The narrator's last sentences deliver the anagnorisis: there was no escaping time, and the moment granted to the child was the moment of the man's death.

Into this sequence Marker inserts a single shot of continuous motion. The woman, asleep, opens her eyes and blinks, as birdsong swells on the soundtrack. The shot lasts a few seconds; it has become one of the most discussed moments in the history of cinema, and the next chapter is devoted to it. Here it suffices to note that *La Jetée* stakes its entire metaphysics on the difference between the still and the moving image, and that it localises this difference in a blink. The gesture should be read

against Marker's career. Born Christian-François Bouche-Villeneuve in 1921 and active under a lifelong pseudonym, he had co-directed with Alain Resnais *Les statues meurent aussi* (1953), a film on the colonial expropriation of African art that French censorship suppressed for its anticolonial argument; he had made essay-films from Beijing, Siberia, and Cuba; and in the year of *La Jetée* he was at work on *Le Joli Mai* (1963), a collective portrait of Paris in its first spring of peace after the Algerian War. His proximity to the so-called Left Bank group—Resnais, Agnès Varda, the novelist Jean Cayrol—places *La Jetée* within a cinema obsessed with the persistence of the past, from *Hiroshima mon amour* (1959) and *L'Année dernière à Marienbad* (1961) to Resnais's own experiment in time travel, *Je t'aime, je t'aime* (1968).

The decisive intertext is Hitchcock. The sequoia of *La Jetée* restages the scene in *Vertigo* (1958) in which Madeleine, before a felled redwood in Muir Woods, traces with a gloved finger the rings that mark her birth and her death; Marker's protagonist, before a similar section, points to a place beyond the tree, outside its rings altogether, and hears himself say that this is where he comes from. Two decades later the narrator of *Sans Soleil* (1983) recounts a pilgrimage to the San Francisco locations of *Vertigo* and proposes that it is the only film capable of rendering a memory that is impossible, even insane; and in a 1994 essay for *Positif*, "A Free Replay," Marker suggested that the second half of Hitchcock's film might be read as the fantasy of a second chance, the extra game a pinball machine awards to the player who has lost. Memory, madness, and the second chance: the three themes that *Twelve Monkeys* will develop are already bound together in Marker's lifelong conversation with *Vertigo*. A further irony, which Marker can scarcely have overlooked, concerns the casting of *La Jetée* itself. The chief experimenter, the man who sends the subject into time, is played by Jacques Ledoux, curator of the Belgian Royal Film Archive. Within the fiction, the machine that sends a man into the past is operated by the custodian of an archive of moving images.

The road from this work to a Universal picture starring Bruce Willis runs through the least romantic channel of the industry, the acquisition of remake rights. The producer Robert Kosberg initiated the project and secured Marker's agreement; Universal took it up with Charles Roven producing; and David Webb Peoples and Janet Peoples were engaged to write it. The credit that resulted—the film is "inspired by" Marker's—is a juridical formula, a way of naming a debt without conceding an adaptation. Read literally, however, it is also a theory of the remake. To inspire is to breathe into, *inspirare*: the remake proposes to breathe motion into a work composed of stills, to do on the scale of a feature what Marker did in the space of a blink. The etymological chain deserves to be spelled out, because the film spells it out in its own materials. *Anima*, breath, gives *animal*, the breathing being, and *animation*, the imparting of motion to the inert image. The survivors of 2035 cannot breathe the air of the sur-

face and must cross it sealed in translucent suits; the virus that has driven them underground is airborne; and when Cole, released into 1996, rides in Railly's car, he thrusts his head into the rushing air and drinks it with a dog's abandon. Breath is the film's sacrament and its medium of death, and the remake "inspired by" *La Jetée* undertakes to breathe into Marker's photographs the very element whose contamination ends the world.

The choice of writers was not incidental. David Peoples had co-written *Blade Runner* (1982) with Hampton Fancher, adapting Philip K. Dick's *Do Androids Dream of Electric Sheep?* (1968), a novel set in a world from which animals have all but vanished and in which artificial animals serve as tokens of status and piety. *Blade Runner* is already a meditation on the authenticity of memory and on the photograph as its fragile guarantor: the replicants cling to snapshots of a past they never lived, and Deckard submits a photograph found in a replicant's lodgings to an image-analysis machine that enlarges, rotates, and penetrates it until it yields a face. *Twelve Monkeys* restages this scene when Railly takes a magnifying glass to a photograph of a wounded soldier from 1917 and finds Cole. The Peoples had also collaborated with the director Jon Else on *The Day After Trinity* (1980), the Academy Award-nominated documentary on J. Robert Oppenheimer and the Manhattan Project—a portrait of the physicist who, in the words he took from the *Bhagavad Gita*, had become death, the destroyer of worlds. David Peoples had written *Unforgiven* (1992), a western about the impossibility of escaping one's past that dismantles the genre's myth of redemptive violence, and had co-written *Leviathan* (1989), in which a deep-sea crew is contaminated by a mutagen recovered from a sunken Soviet vessel. Scientific apocalypse, contagion, the inescapable past, the photograph as evidence of a memory that may be false: the thematic repertoire of *Twelve Monkeys* is legible in its writers' prior work before Gilliam touched it.

Gilliam, for his part, had by 1995 built one of the most consistent bodies of work in postwar cinema around a single antagonism: the imagination of the solitary dreamer against the rationalised system that would correct, classify, or destroy him. The cut-out animations he made for *Monty Python's Flying Circus*, assembled from Victorian engravings and Renaissance paintings, had already practised the art of animating the still and the dead. *Time Bandits* (1981) sent a boy through holes in the fabric of creation, ended with the annihilation of his parents, and left him alone with his knowledge. *Brazil* (1985), his masterpiece of bureaucratic dystopia, ended with its hero strapped into a chair at the centre of a cavernous torture chamber, escaping into a fantasy from which the system can no longer retrieve him because he has gone mad. *The Adventures of Baron Munchausen* (1988) pitted a liar against the Age of Reason in the person of a functionary who has a soldier executed for heroism, and *The Fisher*

King (1991) made a homeless visionary's delusion the form taken by unbearable grief. His visual lineage—Bosch and Bruegel, the *Carceri* of Piranesi, Doré's engravings, the expressionist sets of *The Cabinet of Dr. Caligari* (1920), whose asylum may be run by the murderer and narrated by a patient, the labyrinthine bureaucracy of Welles's *The Trial* (1962)—is a lineage of crowded, oblique, and carceral space. *Twelve Monkeys* was, after *The Fisher King*, only the second feature Gilliam directed from a screenplay in which he had no writing hand, and Keith Fulton and Louis Pepe's documentary of its production, *The Hamster Factor and Other Tales of Twelve Monkeys*, records the cost of that discipline. Its title alludes to the director's insistence on a hamster running in its wheel in the background of a shot, a detail no audience would consciously register. This dissertation proposes, half in earnest, that the hamster is the film's unwitting emblem: a rodent—Smintheus's creature—running in a circle that goes nowhere, a living zoetrope, the prisoner of a loop that the eye mistakes for progress.

The film that emerged is the product of a collaboration whose structure Nietzsche anticipated in *The Birth of Tragedy* (1872). The Peoples' screenplay is Apollonian in Nietzsche's precise sense: it belongs to the order of the dream-image and the *principium individuationis*, of measure, boundary, and clear causal line; its closed loop has the lucidity of a geometric demonstration, and its governing image is literally a dream, the recurring vision of the airport that Cole cannot stop seeing. Gilliam's direction is Dionysian. It dissolves the individual into the crowd, the ward, the herd, the chorus; it distorts the frame with the short focal lengths and canted angles that Roger Pratt's camera makes habitual; it clutters every space with the debris of obsolete technologies; and in Brad Pitt's Jeffrey Goines it releases into the asylum a manic, intoxicated, animal-loving god whose followers will open the gates of the zoo. Nietzsche held that Attic tragedy was born when the Apollonian image and the Dionysian intoxication were forced into union, and that its matrix was the satyr chorus, the band of goat-men in whom the human and the animal had not yet been divided. *Twelve Monkeys* reproduces this genesis with uncanny literalness. Its Dionysian band is named for twelve animals; Chapter Five will show that twelve was the number of the Aeschylean chorus; and the score that Paul Buckmaster built on Astor Piazzolla's *Suite Punta del Este*, a bandoneon figure that turns and returns without resolving, supplies the music from which, Nietzsche insisted, tragedy is born.

I.3

1962/1995

Two Conjunctures of the End

The two films belong to two conjunctures of the end, and the difference between them is philosophically decisive. *La Jetée* was completed in 1962, the year of the Évian Accords that ended the Algerian War in March and of the Cuban missile crisis in October. Its underground camp, its experiments upon prisoners, and its whispering German condense three catastrophes into a single image: the memory of the Nazi camps, which Resnais had confronted in *Nuit et brouillard* (1955); the present of a colonial war whose practice of torture and internment French cinema could barely name; and the imminence of a nuclear war that, for some days in October, seemed less imminent than scheduled. The same year saw Rachel Carson open *Silent Spring* with "A Fable for Tomorrow," a parable of an American town struck silent by a blight that no enemy and no witchcraft had caused, because its people had done it to themselves; J. G. Ballard publish *The Drowned World*, in which a warming planet returns its inhabitants to a Triassic memory sedimented in their own nervous systems; Thomas Kuhn publish *The Structure of Scientific Revolutions*, a theory of how normal science suppresses the anomaly until the anomaly overwhelms it; and Ken Kesey publish *One Flew Over the Cuckoo's Nest*. It saw, too, a second French short film about the time of a man's death, Robert Enrico's *La Rivière du hibou*, adapted from Ambrose Bierce's "An Occurrence at Owl Creek Bridge," in which a hanged man's escape unfolds within the instant of his execution. And it stood a year after the annus mirabilis of the critique of the asylum: 1961 brought Foucault's *Folie et déraison*, Erving Goffman's *Asylums*, and Thomas Szasz's *The Myth of Mental Illness*.

The point of this inventory is not ornamental. It establishes that the time-travel fable and the critique of psychiatric confinement were born together, out of a single historical anxiety about rational institutions that had learned to administer death. Marker's film does not thematise madness directly; its hero is chosen precisely because his obsession immunises him against the madness that destroys the other subjects. *Twelve Monkeys* reunites the two strands that 1962 produced side by side. It returns the time traveller to the asylum, renders the experimental subject indistinguishable from the psychiatric patient, and joins Marker's photograph to Kesey's ward.

The Peoples' screenplay was filmed in Philadelphia and Baltimore in 1995 and released at the turn of 1995 and 1996, and its conjuncture is the migration of the apocalyptic imagination from the bomb to the microbe. The Cold War had ended; Fran-

cis Fukuyama had proclaimed the end of history (1992); and the dread that the superpower standoff had concentrated was redistributed across a new object, the emergent virus. Richard Preston's *The Hot Zone* (1994) had made a bestseller of the 1989 discovery of an Ebola-family virus among macaques imported into a quarantine facility in Reston, Virginia, within commuting distance of Washington; Laurie Garrett's *The Coming Plague* (1994) mapped a planet primed for pandemic; Wolfgang Petersen's *Outbreak*, released in March 1995, made a monkey smuggled out of Zaire the vector of a haemorrhagic fever in a California town; and in the spring of 1995 an Ebola epidemic broke out in Kikwit, in Zaire itself. The American AIDS epidemic was then at its deadliest. On 20 March 1995 the apocalyptic sect Aum Shinrikyo released sarin in the Tokyo subway, having also attempted to cultivate biological agents in its own laboratories; in April a truck bomb destroyed the federal building in Oklahoma City; in September the manifesto of the Unabomber, a former mathematician waging war on industrial society, appeared in the *Washington Post*. By the year of the film's production the monkey had become the emblem of zoonotic threat, the laboratory a site of apocalyptic intention, and the scientist turned misanthrope a figure from the evening news.

What changed between 1962 and 1995 was not merely the agent of the apocalypse but its temporal form, and this is the historical thesis on which the dissertation rests. The nuclear apocalypse is an *instant*. It arrives as a flash, is decided by a sovereign hand upon a button, and is total, visible, and simultaneous; its proper medium is the photograph, and the bomb was itself a kind of camera, printing the shadows of the vaporised onto the stone of Hiroshima. Its temporality is that of the *Augenblick*, the blink of an eye in which everything changes. The viral apocalypse is a *curve*. It is invisible, distributed, and cumulative; it is decided by no one and administered by everyone; it is known only through incidence rates, mortality tables, and projections; and its proper medium is statistics. Foucault's lecture of 17 March 1976 at the Collège de France drew the political consequence: sovereign power, the right to take life or let live, culminates in the atomic bomb, whereas biopower, the power to make live and let die, takes as its object the population, with its birth rates, its morbidity, and its epidemics. The passage from *La Jetée* to *Twelve Monkeys* is the passage from the photographic instant to the actuarial curve, from the sovereign apocalypse to the biopolitical one. Marker's film, made of stills punctured by a single blink, is the formal correlate of the bomb. Gilliam's film, made of a moving image bent into a ring, is the formal correlate of the epidemic.

One further difference completes the contrast and anticipates the argument of Chapter Six. Ulrich Beck, whose *Risikogesellschaft* appeared in 1986, the year of Chernobyl, proposed what he presented as an operational criterion for the arrival of

the risk society: it begins where private insurance withdraws, where hazards become too vast, too uncertain, or too unbounded to be priced. The nuclear catastrophe is the paradigm of the uninsurable. It destroys the insured, the insurer, and the archive in which any claim might be registered, which is why Derrida, writing in 1984, could call nuclear war a "fabulously textual" event, one that can exist only in discourse because it would leave no one to remember it. The viral catastrophe of *Twelve Monkeys* is of another kind. It kills people and spares things. In 2035 the streets and buildings of Philadelphia stand intact beneath the snow, colonised by beasts; only their inhabitants are gone. It is, in the precise sense, a catastrophe that leaves the archive standing, and therefore a catastrophe against which one can be insured. The future of *La Jetée* sends its subject into the past in search of food and energy, the bare means of survival. The future of *Twelve Monkeys* sends its subject in search of a sample, an item of evidence, a claim. Between the two films the apocalypse has become insurable, and the time traveller has become a loss adjuster.

1.4
Philadelphia, or the Archive of Dispositifs

The choice of Philadelphia as the city of 1996 and of 2035, with Baltimore as the city of 1990, was in part a matter of production economics. But cities are not neutral backdrops, and Philadelphia condenses, with an almost uncanny completeness, the institutions through which *Twelve Monkeys* thinks. Foucault called a *dispositif* the heterogeneous ensemble of discourses, institutions, architectural forms, regulations, laws, administrative measures, and scientific statements that responds to an urgency and organises a field of power. Philadelphia is an archive of the dispositifs the film assembles, and they are worth enumerating, because the enumeration is itself an argument.

It is, first, the city of liberty, where the Declaration of Independence was signed and where the bell that gave American liberty its icon is still displayed—an irony the film need not state when it shows the prisoners of 2035 caged and hosed down beneath the same streets. It is, second, the city of the American hospital and of American psychiatry. The Pennsylvania Hospital, founded in 1751 by Benjamin Franklin and Thomas Bond, was the first hospital in the British colonies, and from its opening it received the insane, lodged in cells in its basement. Benjamin Rush, a signer of the Declaration and the figure to whom American psychiatry traditionally traces its paternity, practised there and designed the "tranquilizing chair," an apparatus that immobilised the patient's head, trunk, and limbs and enclosed his field of vision. The

chair in which Cole is strapped and raised before the panel of scientists in 2035, and the restraints applied to him in the hospital of 1990, have their ancestor here.

It is, third, the city of the penitentiary. Eastern State Penitentiary, opened in 1829 to John Haviland's radial plan, was the paradigmatic institution of the "separate system," in which prisoners lived, worked, and prayed in solitude so that isolation might produce repentance. Dickens, who visited in 1842, wrote in *American Notes* that he held "this slow and daily tampering with the mysteries of the brain, to be immeasurably worse than any torture of the body." Foucault, in *Discipline and Punish* (1975), treats the Philadelphia model as one of the founding experiments of carceral modernity. The production of *Twelve Monkeys* used the penitentiary, long since closed and preserved as a ruin, for scenes of its asylum, so that the psychiatric ward of 1990 is quite literally lodged within the architecture of the separate system. Philadelphia was, fourth, the city of the experimental prisoner. From 1951 into the 1970s, at Holmesburg Prison, the dermatologist Albert Kligman of the University of Pennsylvania conducted hundreds of experiments upon inmates—skin trials for cosmetic and pharmaceutical firms, exposure studies for chemical manufacturers and for the army—with prisoners paid as "volunteers"; Allen Hornblum's *Acres of Skin* (1998) later documented the programme. When the authorities of 2035 call their conscripted prisoners "volunteers" and send them, with the promise of a reduced sentence, into a contaminated world and an unknown time, the film does not invent its scandal. It remembers one. In the very year the film was shot, the President's Advisory Committee on Human Radiation Experiments delivered its final report on Cold War experiments performed on prisoners, patients, and soldiers without meaningful consent.

It is, fifth, a city of epidemics. In 1793, when Philadelphia was the capital of the republic, yellow fever killed roughly a tenth of its population; the federal government fled, Rush bled and purged his patients with notorious zeal, and Charles Brockden Brown made the epidemic the infernal setting of *Arthur Mervyn* (1799–1800). In the autumn of 1918 a Liberty Loan parade drew some two hundred thousand people into the streets in the midst of the influenza pandemic, and within weeks the city was counting its dead in the thousands, among the highest tolls of any American city. The detour that *Twelve Monkeys* takes when a technical error deposits Cole in the trenches of the First World War is therefore a detour into the very war that incubated the deadliest pandemic of the modern era. The film's ostensibly gratuitous excursion to 1917 is a visit to the prehistory of its own plague.

It is, sixth, the city of the first American zoo. The Philadelphia Zoological Garden opened in 1874, and it is from a zoo that the Army of the Twelve Monkeys releases its animals into the streets. It is, seventh, a birthplace of the cinema's prehistory: between 1884 and 1886, under the auspices of the University of Pennsylvania, Eadweard Muy-

bridge produced the photographic sequences published as *Animal Locomotion* (1887), decomposing the movements of men, women, horses, and the beasts of the Philadelphia zoo into series of still images—the protocinematic act of which *La Jetée*, a film made of stills, is the late and melancholy inversion. And it is, finally, a cradle of American insurance. Franklin founded the Philadelphia Contributionship for the Insurance of Houses from Loss by Fire in 1752, the oldest continuously operating property insurer in the United States, and the Presbyterian Ministers' Fund, established in Philadelphia in 1759, is generally counted the first life insurance company in America. Franklin, who founded both the city's hospital and its first insurer, also invented the lightning rod, the device by which a modern subject first insured his roof against the wrath of Zeus.

To read the city in this way is not to claim that Gilliam or the Peoples designed these correspondences. It is to claim that the film, in choosing Philadelphia, inherited an archive, and that the archive thinks alongside it. Liberty and confinement, hospital and penitentiary, experimental prisoner and epidemic, zoo and chronophotograph, fire policy and lightning rod: these are not the backdrop of *Twelve Monkeys* but its conceptual furniture. Philadelphia, moreover, means brotherly love. That the city of *philadelphia* should be the point of departure for the misanthropic angel of extinction, and that Dr. Peters's itinerary of death should begin in the city named for the love of human beings for one another, is an irony the film leaves silent and this dissertation will not.

1.5
The State of the Question

Scholarship on *Twelve Monkeys* has been organised, explicitly or tacitly, by three paradigms, each of which captures something real and each of which leaves the film's central structure unexamined. The first is ludic-formal. It treats the film as a puzzle whose rules the spectator must infer, and it places *Twelve Monkeys* in the genealogy of what Thomas Elsaesser called the "mind-game film," the cycle of the late 1990s and early 2000s—*The Matrix*, *Fight Club*, *Memento*, *Donnie Darko*—in which the audience is played with as much as the protagonist. Philosophers of time travel, for their part, cite the film as cinema's most rigorous representation of a self-consistent timeline: one in which, as David Lewis argued in "The Paradoxes of Time Travel" (1976), a traveller may in one sense be able to kill his grandfather and yet will certainly fail to, because the past contains only what in fact occurred; and which Igor Novikov and his collaborators, Kip Thorne among them, formalised around 1990 as

the principle that the only events admissible in a universe containing closed timelike curves are those that are globally self-consistent. The ludic-formal paradigm asks whether the loop holds. It does not ask whom the loop serves.

The second paradigm is psychiatric-hermeneutic. It reads the film as an extended experiment in undecidability: for most of its length the spectator, like Railly, cannot know whether Cole is a traveller from the future or a schizophrenic whose delusions happen to cohere. Auteurist criticism has assimilated this ambiguity to the pattern of Gilliam's work as a whole—the holy fool against the system, in the formula that supplied the title of Bob McCabe's study *Dark Knights and Holy Fools* (1999)—while anti-psychiatric readings, drawing on Foucault and R. D. Laing, have found in it a critique of the clinic's power to declare unreason. The limitation of this paradigm is that the film resolves its ambiguity. The bullet Railly extracts from Cole's leg is of First World War manufacture; the soldier in the 1917 photograph wears his face; the boy reported trapped in a well turns out, as Cole foretold, to have been hiding in a barn all along. The question the film poses is therefore not epistemic but political. It is not whether Cole is mad, but what a society does with a true prophecy, and why the institutions that hear it are constitutively unable to believe it.

The third paradigm is intermedial and adaptive. It compares *Twelve Monkeys* with *La Jetée*, usually to the latter's advantage, and it draws on the rich theoretical literature on the still image in film: Raymond Bellour's account of the "pensive spectator" whom the arrested image releases from the forward drive of narrative; Laura Mulvey's *Death 24x a Second* (2006), on stillness, mortality, and the delayed cinema of the digital age; Garrett Stewart's *Between Film and Screen* (1999), on the photogram as the repressed substrate of motion; Catherine Lupton's *Chris Marker: Memories of the Future* (2005) and Janet Harbord's monograph on *La Jetée* (2009). Remake theory, as systematised by Constantine Verevis, supplies a vocabulary for the film's relation to its source. This paradigm has the merit of taking the medium seriously, but it has tended to treat the medium as a question of value—Marker's austere modernism against Gilliam's baroque studio picture—rather than as the site where the film's metaphysics is actually worked out.

Above these paradigms stands one essay that has done more than any other to shape psychoanalytic thought about time travel in cinema: Constance Penley's "Time Travel, Primal Scene, and the Critical Dystopia" (1986), which reads *La Jetée* and *The Terminator* as fantasies of the primal scene—the scenario in which the subject imagines himself present at his own origin. Penley's insight remains indispensable, and this dissertation extends it in a direction her essay did not pursue. In *La Jetée*, and still more insistently in *Twelve Monkeys*, the scene the child witnesses is not the scene of his conception but the scene of his death. It is a thanatic primal scene, and the woman

who kneels over the fallen man in the posture of the Pietà is at once lover and mother, so that the Oedipal fantasy and the devotional image of mourning collapse into a single figure.

What the three paradigms share, despite their differences, is a set of blind spots. They do not ask about the institution that sends the traveller—who it is, what it wants, what it stands to gain or lose. They neglect the animal, although the film's opening intertitles end with the return of the animals to dominion and its title names a band of beasts. They treat the insurance line as a punchline. And they separate medium from metaphysics, missing the fact that the film's most explicit statement about time is a remark about cinema made in a cinema. The 1990s, moreover, produced a cycle of asylum Cassandras whose comparison sharpens these questions. In *Terminator 2: Judgment Day* (1991), Sarah Connor is confined to a state hospital for prophesying a nuclear Judgment Day that the film dates, as *Twelve Monkeys* dates its plague, to 1997; but James Cameron's film declares that there is "no fate but what we make," and its timeline is mutable, whereas the Peoples' is not. The television series that Syfy derived from the film between 2015 and 2018 likewise converted its fixed timeline into a branching and revisable one, and the conversion clarifies, by contrast, how radical the original's commitment was. This dissertation reads the film's institutions—clinic, laboratory, prison, zoo, airport, cinema, insurer—as a single apparatus for the management of time, and it brings tragic theory, animal studies, the history of probability, and the theory of risk into the same frame as the philosophy of film.

1.6
Method

Constellation, Afterlife, Apparatus

The method of this dissertation is constellational in the sense Walter Benjamin gave the term in the epistemo-critical prologue to his study of the German mourning play (1928): ideas stand to phenomena as constellations stand to stars, neither containing them nor reducible to them, but drawing them into a figure in which each acquires a significance it could not have alone. The method suits a film whose final figure is an astrophysicist and whose emblem is a ring. It requires, however, a discipline of distinction. The dissertation distinguishes throughout between attested sources and constellational interlocutors. Attested sources are those the film names, quotes, or demonstrably derives from: Marker's film; Hitchcock's *Vertigo* and *The Birds*, screened in the retrospective where Cole and Railly hide; the Cassandra myth,

on which Railly lectures; Semmelweis, whom Jeffrey invokes; the Peoples' earlier work. Constellational interlocutors are figures—Halley, Agamben, Dupuy, Kafka's ape—whose ideas illuminate the film without any claim that its makers consulted them. To conflate the two would be to manufacture influence; to refuse the second would be to commit what W. K. Wimsatt and Monroe Beardsley called the intentional fallacy (1946), confining a work's meaning to the designs of its makers. A film of this density thinks beyond its authors, because the problems it engages—time, catastrophe, madness, the animal—have histories that it inherits whether or not it cites them.

The second methodological resource is Aby Warburg's theory of the *Nachleben*, the afterlife or survival of images, and of the *Pathosformel*, the formula of gesture in which an intensity of feeling is preserved and transmitted across centuries and media. *Twelve Monkeys* is saturated with pathos formulas whose survival this dissertation traces: the Pietà, the falling man, the raving prophetess, the caged beast, the child who witnesses. Warburg's unfinished *Mnemosyne Atlas* (1924–1929), panels of photographs arranged so that their juxtapositions would think, is the art-historical cousin of Marker's essay-films and of the method attempted here. Warburg belongs to the constellation for a further reason. Confined between 1921 and 1924 in Ludwig Binswanger's sanatorium at Kreuzlingen, he delivered there, in April 1923, a lecture on the serpent ritual of the Pueblo peoples of the American Southwest, in part to demonstrate to his physicians that he had recovered his reason. The lecture ends with a photograph of a man dressed as Uncle Sam strolling beneath an electric wire, and with a lament that the modern capture of lightning destroys the *Denkraum*, the space for reflection, that myth had opened between human beings and their terrors. The historian of the afterlife of images, obliged to prove his sanity by lecturing on snakes and lightning to his doctors, belongs to the world of *Twelve Monkeys*; and his lament for the abolished distance is the lament the film addresses to a future that can travel to any moment of the past and has lost the capacity to contemplate any of them.

Third, the dissertation treats film as a mode of philosophical thought rather than as a source of illustrations for philosophical theses. Stanley Cavell's *The World Viewed* (1971) grounds the ontology of film in the photographic automatism that presents the world to us in our absence, a world past; Gilles Deleuze's two volumes on cinema (1983, 1985) argue that films think in images and that modern cinema gives a direct image of time; Stephen Mulhall's *On Film* (2002) defends the claim that certain films philosophise in their own medium. *Twelve Monkeys* is a case in point. Its claims about time are made through its handling of images: the still photograph magnified until it surrenders a face, the dream repeated until its content shifts, the cinema in which a film about a man who remakes a woman into the image of a dead one is watched by a

woman who, in the same sequence, remakes herself into the image of a man's dream. The film does not illustrate a theory of time. It constructs one, with the means proper to cinema.

Fourth, the dissertation draws on the tradition of historical epistemology that runs from Georges Canguilhem's *The Normal and the Pathological* (1943; enlarged 1966) through Foucault to Ian Hacking. Hacking's essay "The Looping Effects of Human Kinds," published in 1995, the year *Twelve Monkeys* was shot, argued that the classifications of the human sciences act upon the people they classify, who change in response, so that the classification must change in turn: the kinds of the human sciences are moving targets, caught in a loop. The coincidence of terms is the conceptual hinge of Chapter Four. *Twelve Monkeys* joins the time loop of physics to the looping effect of psychiatry. Railly's diagnostic category, the "Cassandra complex," loops through the patient it describes, who comes to wish he fitted it, since only his madness would spare the world; and it loops back through the psychiatrist whose professional authority rested upon it, who ends by falling under it herself.

Finally, the comparisons this dissertation undertakes range across media, because the problems the film addresses have been worked out in every medium, and each medium isolates a different variable. Attic tragedy and the Hebrew and Christian scriptures supply the archetypes of prophecy and plague; the plague narrative, from Thucydides and Boccaccio through Defoe, Mary Shelley, Poe, and Camus, supplies the social phenomenology of contagion; the short story, from Bierce and Kafka to Borges, Heinlein, and Ted Chiang, supplies the purest models of fixed time; the novel, from Wells to Vonnegut and Atwood, its longest elaborations. Painting supplies the iconography: Bruegel's *Triumph of Death* (c. 1562), Dürer's Apocalypse woodcuts (1498), Goya's etching of reason asleep among owls and bats (1799), Klee's *Angelus Novus* (1920). Architecture supplies the spaces: Piranesi's imaginary prisons, Bentham's panopticon, Haviland's penitentiary. Comics supply, in *Watchmen* and *The Sandman*, figures for whom all of time is simultaneously visible; television, in *Doctor Who*'s "fixed points" and in *Dark*, the serial elaboration of the closed loop; and the video game, in *The Legend of Zelda: Majora's Mask* (2000), *Braid* (2008), and *Outer Wilds* (2019), the one medium in which the spectator is condemned, like Cole, to repeat the catastrophe until she understands it. These comparisons are not decorative. Each is a controlled variation that tests what in *Twelve Monkeys* is necessary and what contingent: whether the past is mutable or fixed, whether the future redeems or underwrites, whether the witness of the end is human or animal.

1.7
A Lexicon for Actuarial Tragedy

The argument deploys five concepts, defined here and developed across the chapters that follow.

Actuarial tragedy. Aristotle located the soul of tragedy in its plot and the summit of plot in the coincidence of reversal and recognition, *peripeteia* and *anagnorisis*, of which the *Oedipus Tyrannus* is the paradigm. Hegel relocated it in the collision of two ethical powers, each justified and each one-sided; Benjamin, in the mourning play of the German baroque, found a form in which history appears as natural history, a landscape of ruins presided over by a melancholy sovereign. George Steiner's *The Death of Tragedy* (1961) argued that tragedy cannot survive in a culture that believes every disaster remediable, observing with characteristic asperity that more lenient divorce laws would not have altered Agamemnon's fate, nor social psychiatry answered Oedipus. *Twelve Monkeys* answers Steiner by showing that tragedy returns precisely within a world of total remedy, as the irreparable loss that the machinery of compensation itself requires. Actuarial tragedy is defined by five displacements. Necessity is administered rather than decreed: the role of *Moira* is played by an institution with a laboratory, a prison, and a budget. The catastrophe is constitutive rather than incidental: the authority that investigates it exists because of it. Recognition is distributed asymmetrically: the hero's anagnorisis, which arrives at the moment of his death, coincides with knowledge the institution possessed from the beginning. The chorus is displaced by the archive: the collective voice that comments on the action and fails to prevent it is replaced by recordings, photographs, and data—and in *Twelve Monkeys* the degraded telephone message that sets the scientists on the trail of the Army of the Twelve Monkeys turns out to have been left by Railly in 1996, as a consequence of the investigation itself, so that the future's founding evidence is generated by its own intervention and is, moreover, mistaken. Finally, *katharsis* is replaced by indemnity: the end of the action is not the purgation of pity and fear in the spectator but the compensation of a loss for the institution.

The Cassandra Policy. The word *policy* conceals two etymologies, and the concept requires both. *Policy* in the sense of governance descends, through Old French *policie*, from Latin *politia* and Greek *politeia*, the constitution of a city. *Policy* in the sense of an insurance contract descends by a different route: from Middle French *police*, Italian *polizza*, and Medieval Latin *apodixa*, a receipt, from Greek *apodeixis*, a demonstration or proof. *Apodeixis* is the word with which Western historiography begins: Herodotus opens his work by announcing it as the *apodexis*, the setting-forth,

PLATE 1

Two roads to "policy"



The word *policy* descends by two routes, as Section 1.7 sets out: from the Greek *politeia*, the constitution of a city, to governance; and from *apodeixis*, a demonstration or proof, by way of a receipt, to the insurance contract. Cassandra's curse severs the second root from *peithō*, persuasion.

of his inquiry. An insurance policy is thus, etymologically, a demonstration—a document that proves in advance what will be owed when what is feared comes to pass. The Cassandra Policy names a governance of time founded on a proof that cannot persuade. Cassandra's curse, in its Aeschylean form, is precisely the separation of *apodeixis* from *peithō*, demonstration from persuasion: she is right, she shows that she is right, and no one is moved. The *Oresteia* ends, in the *Eumenides*, when Athena wins over the Furies by persuasion and founds a court; *Twelve Monkeys* ends within an institution that has abolished the need for persuasion altogether, since it requires the past's disbelief. Cole's proofs—the boy in the well, the antique bullet, the photograph—are demonstrations without persuasive force for anyone but Railly, and they persuade her too late to matter except as re-vision.

Zoetropic time. The zoetrope, devised in 1834 by William George Horner under the name *daedaleum* and popularised in the 1860s under the name that stuck, is a slotted drum lined with a strip of sequential images; spun and viewed through the slots, the images animate. Its name joins *zōē*, life, to *tropos*, a turning: it is the wheel of life, and, since *tropos* is also the rhetorician's word for a figure, the trope of life, the figure by which life is produced. Zoetropic time is the time of a closed ring of still images that produces, by repetition and rotation, the appearance of linear living movement. Within the ring each figure seems to advance; from outside, nothing advances, and the ring simply returns. This is the temporal form of *Twelve Monkeys*: from within, Cole investigates, learns, loves, and runs; from without—the vantage of the scientists, and of the spectator who sees the film a second time—the ring returns to the airport. The emblem of the Army, a stencilled ring of red monkeys, is a zoetrope strip arranged as a clock. Greek supplies a further convergence that the dissertation will exploit: *zōon*, living creature, also means a figure or image, so that the painter is a *zōgraphos*, a writer of living things, and the zodiac is the *zōidiakos kyklos*, the circle of little figures—of animals that are also images. The two great themes of the film, the image inherited from Marker and the animal announced in its epigraph, share a single Greek word.

The thirteenth monkey. To count a ring of twelve one must stand outside it; the counter is the thirteenth term the ring presupposes and excludes. The thirteenth monkey is the human being, considered as the animal that counts, promises, and insures. Nietzsche opened the second essay of *On the Genealogy of Morality* (1887) by asking whether breeding an animal that may promise is not the paradoxical task nature has set itself in the case of the human being; to promise, the animal had first to be made calculable, regular, and necessary, and its memory had to be burned into it with pain, so that guilt, *Schuld*, was born from debt, *Schulden*. The insuring animal is the promising animal socialised: insurance, as François Ewald has shown, is a collect-

ive promise that disposes of the future in advance. The thirteenth monkey steps out of the zoetropic ring of animal time into the linear, calculable time of history, contracts debts to its own future, and cultivates in its laboratories the very agents of its return to the ring. The thirteenth figure recurs throughout the film's constellation: in the clock that strikes an hour outside the dial, as on the April day on which Orwell's *Nineteen Eighty-Four* opens; in the ebony clock of Poe's "Masque of the Red Death," whose midnight admits the uninvited guest; in the thirteenth at table; and in Ophiuchus, the serpent-bearer, the thirteenth constellation crossed by the ecliptic, whom myth identifies with Asclepius, son of Apollo, the physician struck down by Zeus for raising the dead. Socrates' last words, in the *Phaedo*, were a reminder that a cock was owed to Asclepius, and Nietzsche read them as a confession that life is a disease. The thirteenth monkey is the sick animal, the debtor animal, the healer animal: the creature that owes its cure a sacrifice.

Re-vision. Adrienne Rich, in "When We Dead Awaken: Writing as Re-Vision" (1972), defined re-vision as the act of looking back, of seeing with fresh eyes, of entering an old text from a new critical direction, and called it, for women writing within a tradition that had not imagined them, an act of survival. Her title borrows that of Ibsen's last play (1899), and it could serve as a caption for the one moving shot of *La Jetée*, in which a dead image wakes. This dissertation extends Rich's term into metaphysics. In a universe whose past is fixed, revision is impossible and re-vision is the only remaining form of freedom: the transformation not of the event but of the look that receives it. Re-vision has a degenerate form, which Vonnegut gave to the Tralfamadorians of *Slaughterhouse-Five* (1969): since all moments exist forever, one may choose to dwell on the pleasant ones and ignore the rest—a quietism of the remote control. It has affirmative forms: Nietzsche's *amor fati*, the wish that nothing be otherwise, not forward, not backward, not in all eternity; and Deleuze's ethics of the event, which asks us to become worthy of what happens to us, and which takes as its emblem the poet Joë Bousquet, paralysed by a bullet in the First World War, who wrote that his wound existed before him and that he was born to embody it. And it has a political form in Benjamin's claim that the past carries a secret index by which it is referred to redemption—a redemption that cannot change what happened but can change what it means. *Twelve Monkeys* is a film built to be re-seen. Its opening dream means one thing on a first viewing and another on a second, and the spectator who returns to it performs, in her own person, the re-vision the film recommends.

1.8
The Architecture of the Dissertation

Chapter Two, "The Photogram and the Blink," reconstructs the ontology of the image that *Twelve Monkeys* inherits from *La Jetée*. From Bazin's account of photography as embalmed time and Barthes's reading, in *Camera Lucida* (1980), of Alexander Gardner's portrait of the condemned Lewis Payne—a photograph that says at once that its subject is dead and that he is going to die—it derives the anterior future as the tense of both films. It reads Marker's blink through Plato's instant, Paul's "twinkling of an eye," Kierkegaard, Nietzsche's gateway, and Benjamin's now-time; traces the spiral that joins *Vertigo* to *Sans Soleil* and the Orphic gaze that kills; reconstructs a media archaeology in which the camera descends from the gun and the astronomical clock, and in which its first subjects were animals; and shows how *Twelve Monkeys* converts the still into dream, enlargement, monitor, and screen.

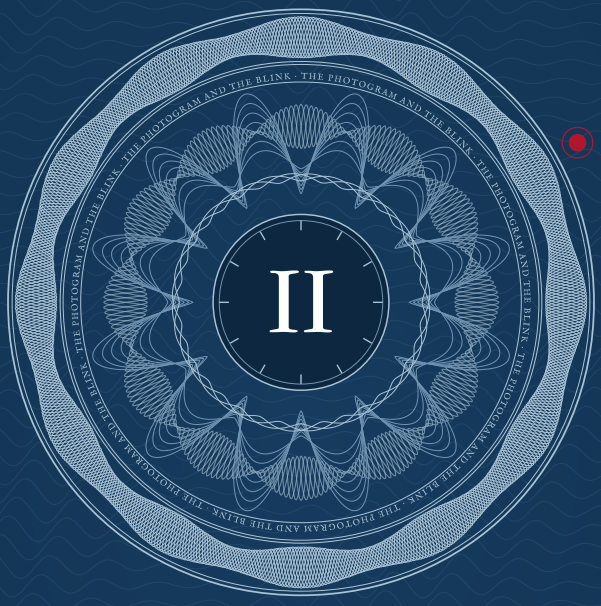
Chapter Three, "The Loop as Fate," analyses the metaphysics of the closed timeline. It reads the film against the *Oedipus Tyrannus*, which opens with a plague and ends with the discovery that the investigator is its cause; the tale of the appointment in Samarra, from the Babylonian Talmud to Somerset Maugham; the Stoic "lazy argument" and Chrysippus's co-fated events; Augustine's refutation of cyclical time; and Boethius, a prisoner awaiting execution who reconciled foreknowledge with freedom. It brings the film into dialogue with McTaggart, Minkowski, Gödel, Lewis, and Novikov; with the bootstrap narratives of Heinlein, the *Planet of the Apes* cycle, the *Terminator* films, *Slaughterhouse-Five*, Chiang's "Story of Your Life," *Donnie Darko*, *Looper*, and *Dark*; with Kierkegaard's *Repetition* and Nietzsche's eternal return; and with Freud's *Beyond the Pleasure Principle*, written out of the war neuroses of the conflict into which Cole is mistakenly thrown, with *Nachträglichkeit*, Lacan's *tuché*, and Deleuze's third synthesis of time.

Chapter Four, "The Asylum and the Oracle," follows Cassandra from Aeschylus, Euripides, Lycophron, and Virgil to Christa Wolf's *Kassandra* (1983) and Melanie Klein's reading of the *Oresteia*, and reads the asylum through Foucault, Goffman, Szasz, Laing, and Deleuze and Guattari. It assembles a counter-archive of true prophets confined as mad—Semmelweis, whose story Jeffrey tells; Stoker's Renfield; the doctor who becomes a patient in his own ward in Chekhov's "Ward No. 6"; Claude Eatherly; Sarah Connor—and reads Jeffrey's tirades on consumption and opinion through Marcuse, Debord, and Baudrillard, and Railly's conversion through Hacking's looping kinds.

Chapter Five, "The Thirteenth Monkey," interprets the ring of twelve monkeys as zodiac, clock, zoetrope, and Aeschylean chorus, with Apollo Smintheus as its presiding deity. It reads Cole through Kafka's Rotpeter, the ape shot in the cheek and below the hip, who limps and imitates in search not of freedom but of a way out; through Agamben's anthropological machine, Kojève's post-historical animal, John Berger's "Why Look at Animals?" and Derrida's cat; and through the zoonotic imaginary of the 1990s, from *The Hot Zone* to *28 Days Later*, whose animal liberationists release infected chimpanzees and so realise the apocalypse that the Army of the Twelve Monkeys only rehearsed as a prank.

Chapter Six, "'I'm in Insurance,'" reconstructs the genealogy of actuarial reason from the plague: the London Bills of Mortality; John Graunt's *Natural and Political Observations* (1662); Halley, the astronomer who predicted his comet's return and in 1693 published the first scientific life table from the burial registers of Breslau; Quetelet's average man; Ewald and Beck. It sets Benjamin's Angel of History against an actuarial angel, and the enlightened doomsaying of Jonas, Anders, and Dupuy against a future that takes the catastrophe as fixed in order to collect on it; it reads Dr. Peters as Malthus's angel, kin to Atwood's Crake and Alan Moore's Adrian Veidt, and the closing song as a Panglossian coda.

Chapter Seven, "Re-vision after the End," reads the boy at the airport through Heraclitus's fragment on lifetime as a child at play, whose kingdom is a child's; considers the remake as zoetropic resurrection and the television series as revision; examines the time-loop games in which nothing persists but the player's knowledge; and asks what *Twelve Monkeys* teaches a world that has since lived through a pandemic, argued over its origins, and watched pandemic catastrophe bonds pay out on a loss they could not prevent.



Chapter Two

The Photogram and the Blink

La Jetée, Vertigo, *and the Ontology of the Unredeemable Image*

In a moment, in the twinkling of an eye, at the last trump: for the trumpet shall sound, and the dead shall be raised incorruptible, and we shall be changed.

— 1 Corinthians 15:52 (Authorised Version)

Behold this gateway, dwarf! It has two faces. Two paths meet here; no one has yet walked either to its end. … The name of the gateway is written above it: "Moment."

— Nietzsche, *Thus Spoke Zarathustra*, III, "On the Vision and the Riddle" (trans. mine)

2.1

The Tense of the Photograph

Embalmed Time and the Anterior Future

Every theory of *La Jetée* begins, whether it acknowledges the debt or not, with André Bazin's "Ontology of the Photographic Image" (1945). Bazin traced the plastic arts to what he called the mummy complex: the primordial wish to preserve the body against the flux of time, to which the Egyptian embalmer answered with natron and linen, and of which painting, sculpture, and finally photography were successive sublimations. Photography satisfied the wish more completely than any prior art because its genesis was mechanical. The image is not made by a hand that interprets but caused by the light the thing reflects, so that the photograph, by the very process of its becoming, shares in the being of its model; it embalms time and rescues it from its own corruption. Cinema, in Bazin's most famous formula, is change mummified: it preserves not the frozen instant but duration itself. *La Jetée* is a film that refuses the second gift in order to meditate on the first. It is cinema that has chosen to be photography, change that has consented to be mummified again, and its subject—a man selected for the strength of an image embalmed in his memory—is the mummy complex become a plot.

Bazin returned to the question in 1951, in a short essay occasioned by a documentary on bullfighting, "Death Every Afternoon." Death, he argued, is the one moment that cannot be repeated; it is the absolute instant, the event that by definition happens once. Cinema can nonetheless repeat it every afternoon, at every screening, and this capacity to make us witness again and again the unrepeatable is what Bazin called a metaphysical obscenity, a profanation of the one thing time does not give twice. The structure shared by *La Jetée* and *Twelve Monkeys* is the narrative form of that obscenity. A man's death, which ought to happen once, happens twice within his own experience—once witnessed by the child, once suffered by the adult—and, in the second film, innumerable times more, in the recurring dream that returns Cole to the concourse night after night. The spectator who watches either film again adds a further repetition, and the films know it. They are built upon the scandal Bazin named: an instant of death that the apparatus of images condemns to recurrence.

Roland Barthes's *Camera Lucida* (1980) supplied the vocabulary in which this scandal can be stated as a tense. The *noeme* of photography, its essence, is the *ça-a-été*, the "that-has-been": whatever else a photograph shows, it certifies that its referent was there, before the lens, at a moment now past. In the second part of the book Barthes discovers a second kind of *punctum*, the detail that wounds, which is no longer a

detail at all but Time itself. Before Alexander Gardner's portrait of Lewis Payne, condemned in 1865 for his part in the conspiracy against Lincoln and photographed in his cell shortly before his hanging, Barthes reads simultaneously that the young man is dead and that he is going to die. The photograph speaks in the anterior future, the tense of *this will have been*; the spectator, Barthes confesses, shudders over a catastrophe that has already taken place, and he generalises the experience: every photograph, whether or not its subject is already dead, is this catastrophe.

Twelve Monkeys is a Lewis Payne portrait extended to the length of a feature. Cole is a convict, serving a sentence in 2035 for crimes the film never specifies, and every image of him carries the double inscription Barthes found in Gardner's plate. He is dead, because the child has already seen his death at the airport; and he is going to die, because the adult is moving, scene by scene, toward the concourse. The spectator who knows the ending, or who has guessed it from the dream, watches a man who is simultaneously a corpse and a fugitive. The film's opening intertitles carry the same tense to the scale of the species. The five billion are announced in the future tense—they will die in 1997—by a speaker for whom they are already dead, and in a document, a clinical transcript of 1990, that preserves the future as a past record. The plague is photographed before it happens; the catastrophe has already occurred.

Barthes himself denied cinema the photograph's melancholy. Film, he argued, is protensive: each image is carried forward by the next and never stops long enough to wound; the photograph, by contrast, is without future, and that is its pathos. A decade earlier, however, in "The Third Meaning" (1970), his notes on stills from Eisenstein, he had proposed the opposite paradox: that the *filmic*, the essence of film, is to be found not in movement but in the photogram, the arrested frame in which an obtuse meaning, irreducible to story, becomes visible. *La Jetée* is the place where these two Barthesian claims meet and contradict one another productively. It is a film composed of photograms, a cinema that has given up protension and acquired the photograph's melancholy, and it nonetheless narrates, drives forward, and ends. It is protensive melancholy: a story told entirely in the tense of the that-has-been about a future that will have been. *Twelve Monkeys* restores protension—it moves, chases, and accelerates toward its climax—but it borrows from *La Jetée* the photograph's futurelessness and installs it at the level of plot. Its protagonist has no future that is not already past. The film moves like cinema and grieves like a photograph.

One further consequence of this photographic ontology concerns the protagonist himself. In *La Jetée* the man is chosen because he carries an image with exceptional intensity; to the experimenters he is a sensitive plate on which a single exposure has been fixed so deeply that it can serve as an anchor in time. The subject is valued as a photographic medium. In *Twelve Monkeys* Cole is chosen, the scientists tell him, for

the strength of his mind and his powers of observation. The criterion has shifted from the retention of an image to the recording of data, and the apparatus has shifted with it. In 2035 Cole is interrogated through a cluster of video monitors mounted on an articulated arm that is thrust toward his face, the scientists' features flickering across its screens; he reports what he saw as a camera reports, and he is evaluated as one. Between the two films the human being has passed from photographic plate to recording device, from the embalmed image to the feed. This passage corresponds exactly to the historical shift, traced in Chapter One, from the sovereign instant of the bomb to the actuarial curve of the epidemic, and the remainder of this chapter follows its consequences.

2.2
The Cinematographic Illusion Undone
Bergson, Deleuze, and the Pensive Spectator

The philosophical charge of *La Jetée*'s stillness becomes fully legible only against Henri Bergson's critique of the cinematograph. In the fourth chapter of *Creative Evolution* (1907), Bergson argued that the intellect apprehends becoming by the same method as the newly invented cinematograph: it takes instantaneous snapshots of a reality that is in fact continuous, and then reconstitutes movement by setting these immobile sections in motion along an abstract, homogeneous, and indifferent time—the time of the projector's crank. Real duration, *durée*, is thereby lost. What the intellect delivers is a succession of stills animated by an external mechanism, never the indivisible flow in which each moment is pregnant with the whole of the past and open onto the new. Bergson named this the cinematographical mechanism of thought, and he held it responsible both for the paradoxes of Zeno and for the whole metaphysical tradition that treated time as a line of points.

La Jetée can be read as Bergson's critique turned into a film, but turned in a direction Bergson did not foresee. Marker removes the mechanism and leaves the snapshots. Deprived of the projector's abstract motion, the stills do not pretend to be duration; they present themselves as what Bergson said they were, immobile sections of a becoming that escapes them. And yet the film does not abandon duration. It relocates it—into the dissolves that bleed one image into the next, into the rhythm of the cuts, the narrator's voice, the music, and above all into the spectator's mind, which supplies between the stills the continuity the stills refuse. Raymond Bellour called this space the *entre-images*, the between-images, and his essay "The Pensive Spec-

tator" (1984) described its effect: when a film stops, or shows a photograph, the spectator is released from the forward drive of narrative and becomes pensive, free to think. *La Jetée* produces a pensive spectator from its first frame to its last. Its duration is not shown; it is induced.

Bergson's earlier masterpiece, *Matter and Memory* (1896), supplies the theory of time travel that *La Jetée* practises. For Bergson the past does not cease to exist when it ceases to act; it survives in itself, virtually and as a whole, and memory is not the storage of traces in the brain but a leap into this preserved past, which we enter, he wrote, by a gradual work of adjustment like the focusing of a camera. His figure for the relation is the inverted cone, whose base is the totality of the past and whose apex touches the plane of present action. The experimenters of *La Jetée* do not transport their subject through a tunnel. They inject him, blindfold him, and send him into the past through his memory: he travels, quite literally, along the cone, from the apex of present misery toward the image fixed in its base. Time travel in *La Jetée* is Bergsonian; it is the intensification of recollection until recollection becomes presence. In *Twelve Monkeys* time travel has been mechanised. Cole is stripped, hosed, strapped, and delivered by an apparatus that hums and sparks, and his arrival is a physical event: he materialises naked in a city street. The shift from memory to machine repeats, at the level of the fiction, the shift from the photograph to the monitor.

Gilles Deleuze's two volumes on cinema (1983, 1985) answer Bergson by arguing that cinema does not give immobile sections plus abstract movement but movement-images, mobile sections of a duration that is itself changing; and that the cinema of the postwar period, by breaking the sensory-motor link between perception and action, began to give direct images of time. Among these time-images Deleuze privileged the crystal-image, in which an actual image and its virtual double—the present that passes and the past that is preserved—become indiscernible, exchanging places like an object and its reflection. Deleuze's cinema of time is largely a cinema of Resnais and Welles, of memory and the powers of the false, and *La Jetée* belongs to that milieu. Within it, however, the film is a limit case, since it subtracts the movement-image altogether in order to let time appear as such: a time-image by subtraction, the crystal reduced to its facets.

Twelve Monkeys is a different and in some respects more curious object. On its surface it is a movement-image film of the most classical kind: an investigation, a pursuit, a sequence of situations that demand action, a protagonist whose perceptions issue in deeds. But this action-image turns around a crystal. The recurring dream of the airport is a virtual image—a memory—that is also, as the film reveals, an actual image, the future event toward which the action drives; and as the film proceeds the two faces of the crystal circulate, the memory taking on the features of the present as the

dream-woman acquires Railly's face, and the present taking on the features of the memory as Railly puts on the blonde wig. The film's classical action is bent into an orbit around a time-image it cannot assimilate. It is this structure, more than any formal experiment, that makes *Twelve Monkeys* a modern film in Deleuze's sense: its actions cannot resolve its situation, because its situation is time itself.

Two further theorists complete the frame. Siegfried Kracauer closed his *Theory of Film* (1960)—subtitled, significantly, *The Redemption of Physical Reality*—with the myth of Perseus, who could behead the Medusa only by looking at her reflection in Athena's polished shield. The cinema screen, Kracauer proposed, is such a shield: the surface on which we can behold horrors, the camps among them, that would turn us to stone if confronted directly. *La Jetée*, with its camp beneath Chaillot and its stills of a ruined Paris, is a film made on Athena's shield, and its stillness is the stillness of what has very nearly been petrified. *Twelve Monkeys* multiplies the shields: the scientists of 2035 see the past only on screens, and the spectator of 1995 sees the end of humanity only as the reflection of a future. Laura Mulvey's *Death 24x a Second* (2006), finally, observed that digital playback—the capacity to pause, return, and repeat at will—has turned the spectator into a possessive and pensive viewer for whom every film is, potentially, a film of stills. The observation casts a retrospective light on Cole's remark in the cinema. A film that never changes, yet changes for the one who watches it again, is the condition of every film in the age of infinite replay. *Twelve Monkeys*, made on the threshold of that age, already addresses the spectator who will rewind it.

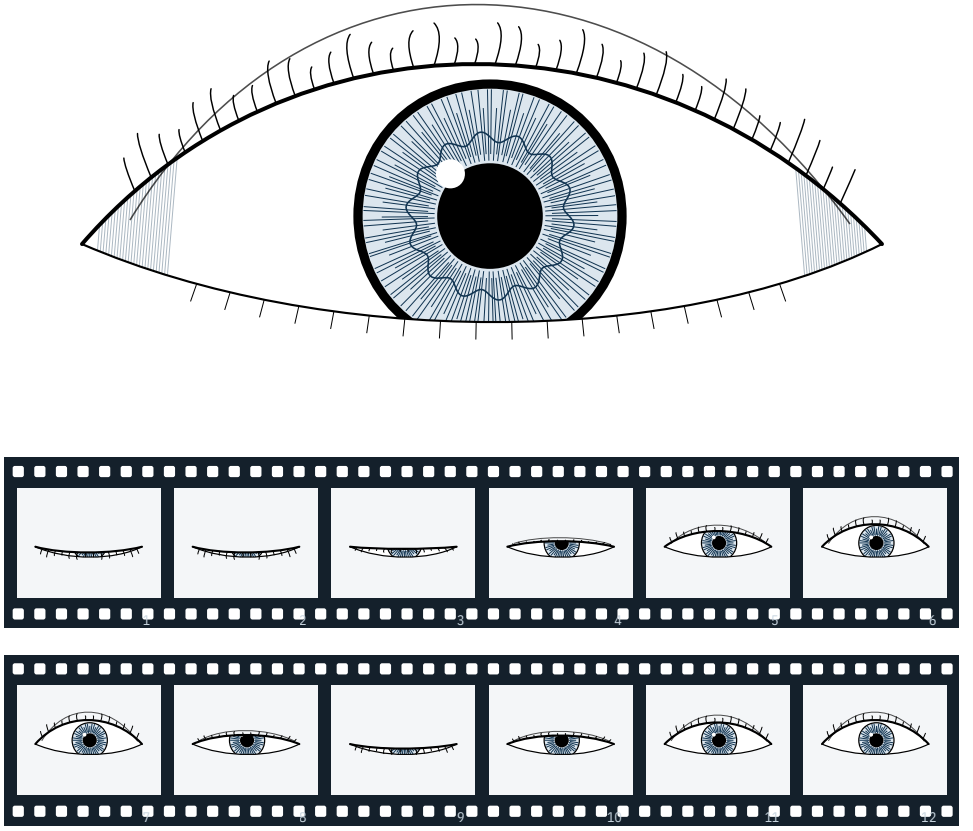
2.3
The Blink

The Instant, the Twinkling of an Eye, and the Gateway Called Moment

Late in *La Jetée*, after the lovers have met and walked and parted and met again, Marker shows the woman asleep. The stills succeed one another more quickly, dissolving into each other as the light changes across her face; birdsong rises; and then, for a few seconds, the image moves. She opens her eyes and blinks, looking toward the camera, toward the man, toward us. Then the stillness resumes. The shot is the only passage of continuous motion in the film, and it is placed with exactitude at the point of the lovers' greatest intimacy and of the man's deepest penetration into the past. What happens in it is the minimal unit of animate life: an eyelid closes and opens. The dead image wakes. To grasp what the blink means philosophically, one must reconstruct

Plate 2

The blink, decomposed



The one passage of continuous motion in *La Jetée*, reduced to the minimal unit of animate life: an eyelid that opens, closes and opens. Above, the open eye; middle, the movement in twelve frames; below, the aperture of the lid traced frame by frame in the manner of Marey's graphic method. A schematic drawing, not a frame from the film.

the long history of the instant in Western thought, a history whose key term, in three languages, is the blink of an eye.

The history begins with a puzzle in Plato's *Parmenides*. If a thing passes from rest to motion, when does the change occur? Not while it is at rest, for then it is not yet moving; not while it is in motion, for then it has already changed. Plato's answer is that the change takes place in *to exaiphnēs*, "the sudden," the instant: a strange and placeless nature that lies between motion and rest and is in no time at all. Marker's blink is the cinematic realisation of the *exaiphnēs*. It is the passage from the stillness of the photograph to the motion of film—from rest to movement—and it occupies, in the economy of *La Jetée*, exactly the placeless position Plato assigned to the instant. It belongs neither to the film of stills nor to an ordinary film of movement; it is the hinge between them, and it has no duration the film can absorb.

The second moment of the history is eschatological. Writing to the Corinthians of the resurrection, Paul declared that the dead would be raised and the living changed *en atomō, en rhipē ophthalmou*: in an atom of time, in the twinkling of an eye. The Greek *atomos* means uncut, indivisible; the Pauline instant of resurrection is the uncuttable moment. Here the correspondence with *La Jetée* becomes almost unnervingly exact. Marker's film is composed entirely of cuts: every image is a section, a photogram severed from the flow. The blink is the one passage of uncut time in the film, its one *atomos*, and in it a sleeping woman—an image of the dead, in a film about a world already destroyed—is raised and changed. Marker stages, within a fable of the last war, the instant of the last trumpet.

Kierkegaard, in *The Concept of Anxiety* (1844), made the Pauline instant the centre of a philosophy of time. The Danish word for the moment, *Øieblikket*, is literally the glance or blink of the eye, and Kierkegaard insisted upon the figure: the moment is not an atom of time, which would be an empty abstraction, but an atom of eternity, the point at which time and eternity touch. Only through the moment does time acquire the distinctions of past, present, and future; without it, time would be an endless and indifferent succession. It is in the moment, moreover, that a human being can relate to the eternal and so become a self. The blink in *La Jetée* has precisely this structure. It does not add a duration to the film's series of stills; it intersects the series vertically, so that for a few seconds the time of the photographs is touched by something not of their order. If the stills are the indifferent succession of what has been, the blink is the eternal present in which the beloved is, for once, simply alive.

Nietzsche's version of the instant is the most radical and, for *Twelve Monkeys*, the most consequential. In "On the Vision and the Riddle," Zarathustra leads the dwarf, the spirit of gravity, to a gateway where two endless paths meet, one stretching back into eternity and one forward; above the gateway is written its name, *Augenblick*—the

moment, literally the blink of an eye. Must not everything that can happen, Zarathustra asks, have already happened along this lane, and must not this very moment return? He adds a detail easily overlooked and important here: the slow spider creeping in the moonlight, and the moonlight itself, and the two of them whispering in the gateway—must they not all have been here before? The spider recurs from *The Gay Science* (1882), where the demon who first announces the eternal return names, among the things that will come back in the same sequence, this spider and this moonlight between the trees. The spider is Nietzsche's emblem of the recurring moment, and its web a figure of the ring. The vision culminates in a scene of disgust and transfiguration: a young shepherd, a heavy black snake hanging from his mouth, bites off its head at Zarathustra's urging, spits it away, and rises laughing, no longer a shepherd, no longer a man.

In the hospital of 1990, Cole catches a spider and eats it. The gesture is usually read as a sign of what life in 2035 has made of him, a forager who consumes what he finds, or as a citation of Renfield, the zoophagous patient of Stoker's *Dracula*, to whom Chapter Four will return. In the constellation proposed here it can also be read as an involuntary mime of Nietzsche's riddle. The man who will relive his death in a loop takes into his mouth the creature Nietzsche made the emblem of eternal recurrence. What he cannot yet do is what the shepherd does: bite through the recurrence and laugh. Whether anyone in the film achieves that transfiguration is the question of Chapter Three, and the provisional answer is that only the look exchanged at the airport approaches it.

Two twentieth-century inheritors of the instant complete its history. Heidegger, in *Being and Time* (1927), named the authentic present *Augenblick*, the moment of vision in which a resolute existence, running ahead toward its own death, takes over its situation; and he named *Wiederholung*, repetition or retrieval, the authentic relation to the past, in which inherited possibilities are explicitly taken up again. Cole's fate can be described in these terms with disquieting accuracy—he is a being whose death has, literally, run on ahead of him—though the film withholds the resoluteness that would make his repetition authentic until, perhaps, the very end, when he chooses to act at the airport knowing what the airport means. Walter Benjamin's *Jetztzeit*, "now-time," is the instant's political form: the present shot through with splinters of messianic time, in which a past moment is blasted out of the continuum of history and seized. Benjamin's dialectical image—what has been coming together in a flash with the now to form a constellation, dialectics at a standstill—is the best name for what *La Jetée* achieves in its blink. The still images, what has been, and the moving image, the now, meet in a flash, and the film stands still at the very moment it moves.

Twelve Monkeys contains no single shot that does what Marker's blink does, since a film made of movement cannot interrupt itself with movement. It displaces the blink onto the look. The film returns again and again to eyes: the child's eyes in the dream, wide and fixed upon the falling man; Cole's eyes opening in the dark of a cell after the dream; the eyes of the scientists on their monitors; Railly's eyes lifting from the dying man to find the boy. Its last image is a child's face raised toward a departing aircraft. Where Marker located the instant in the movement of an eyelid, Gilliam locates it in the meeting of two gazes across a concourse. The blink, the uncut instant in which the dead image wakes, becomes in *Twelve Monkeys* the moment of recognition between a woman and the child who will become the man she has just watched die. It is the film's only *exaiphnēs*, and it is the seed of the ethics of re-vision.

2.4
The Spiral and the Sequoia

Vertigo, *Sans Soleil*, and the Orphic Gaze

If the blink is *La Jetée*'s singular instant, its master image of time is borrowed. *Vertigo* (1958), adapted by Hitchcock from Pierre Boileau and Thomas Narcejac's novel *D'entre les morts* (1954)—"from among the dead," the formula of resurrection—tells of a retired detective hired to follow a woman who believes herself possessed by a dead ancestor. He falls in love with her, fails through his fear of heights to prevent her apparent suicide, collapses into a melancholic stupor that confines him to a sanatorium, and on his release encounters a shopgirl whom he remakes, item by item, into the image of the dead woman—only to learn that she was the same woman all along, the instrument of a murder plot, and to lose her a second time from the same bell tower. Hitchcock's film is thus already a film about time travel by other means: about the attempt to return to a past moment by reconstructing its image, about the woman as an image that can be remade, and about the asylum as the station between the first loss and the second.

Its emblem is the spiral. Saul Bass's title sequence begins on a woman's face, moves to her lips and then to an eye, from whose depths a succession of spiral figures unfurls—curves generated with the help of John Whitney's mechanical analogue computer—while Bernard Herrmann's score turns upon itself in rising and falling figures. The spiral recurs in the coil of hair worn by the woman in the portrait and by her supposed descendant, in the stairwell of the tower, in the famous combination of zoom and track that makes the stairwell stretch and plunge at once. It is the figure of a time

that returns without returning to the same place: a circle displaced along an axis, repetition with a difference. Yeats, in *A Vision* (1925), made the interpenetrating gyres the shape of history, and "The Second Coming" (1920) opens on an animal escaping the human voice as the centre fails: "Turning and turning in the widening gyre / The falcon cannot hear the falconer." The spiral that runs from Yeats through Hitchcock to Marker is the geometry of *Twelve Monkeys*. Its loop is a circle; but its spectator, changed by each passage, travels along the axis.

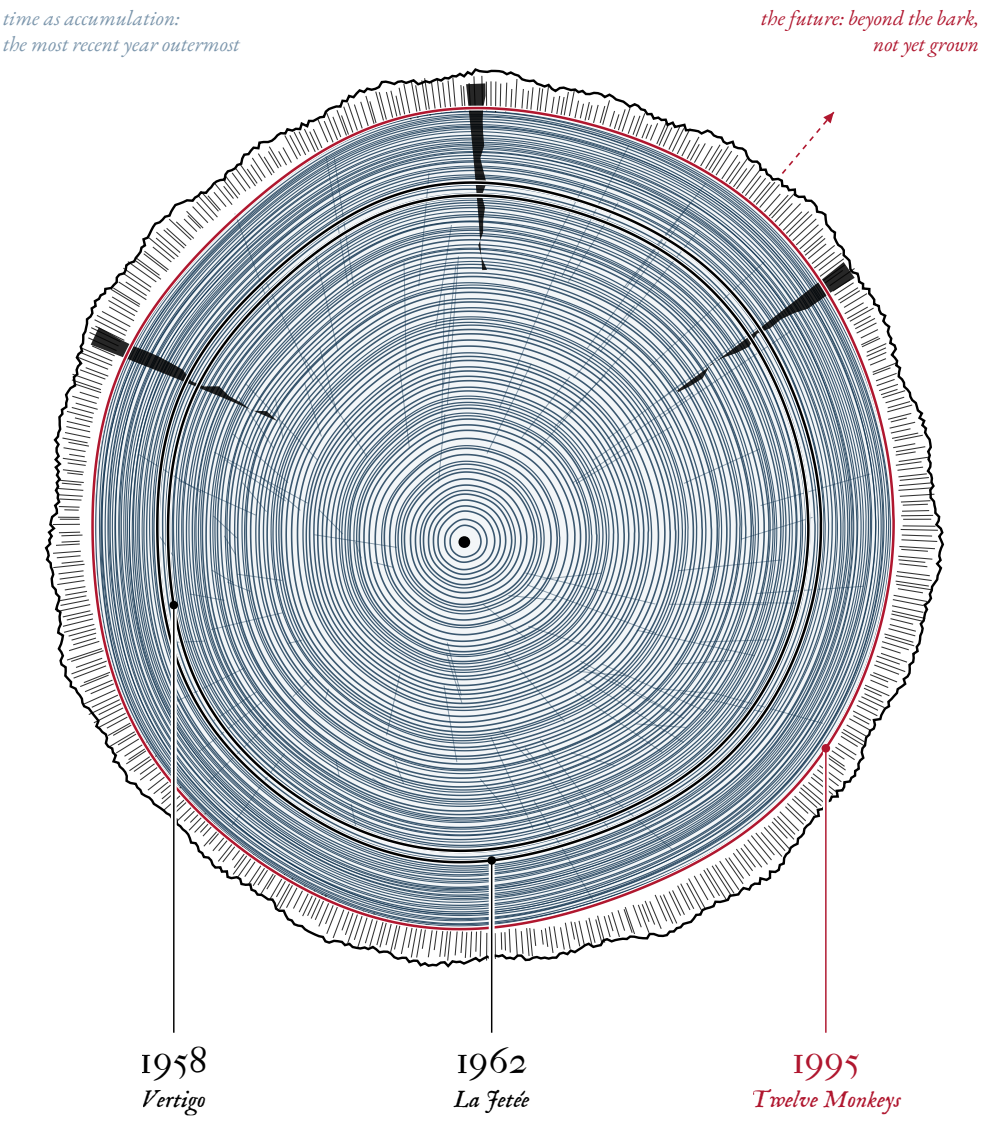
The sequoia scene is the hinge that binds the three films together. In *Vertigo*, Madeleine stands before the cross-section of a felled redwood in Muir Woods, its rings marked with the dates of great events, and traces with a gloved finger the narrow band in which she was born and died; the scene condenses her supposed possession into the image of a life as a thin stripe in the vast accumulation of tree-time. In *La Jetée* the man and the woman stand before a similar section, and the man points not to a ring but beyond the tree altogether, to a place outside its circumference, and says that this is where he comes from. The tree's rings are time as accumulation, each year laid around the last with the most recent outermost; the future is the space beyond the bark, not yet grown and unrecorded. The man from the future is the man from beyond the last ring. In *Twelve Monkeys* the sequoia appears a third time, now as an image on a screen: Cole and Railly, hiding in a cinema, watch Madeleine before the redwood, and the film thereby quotes Marker's quotation of Hitchcock, folding the three works into one another like rings. The scene is the film's *mise en abyme*. A man who comes from beyond the last ring watches a woman trace the ring of her own death in a film that has not changed since 1958, and remarks that it never changes.

What follows completes the spiral. Railly, who must disguise herself, emerges from the shadows of the cinema wearing a blonde wig, and Cole, turning, sees her as the woman of his dream—as, in *Vertigo*, Scottie sees Judy step out of the bathroom of the Empire Hotel, bathed in the green light of a neon sign, transformed at last into Madeleine. But the direction of the transformation has been reversed. In Hitchcock the man remakes the woman into the image of his memory, coercively and item by item; in Gilliam the woman remakes herself, for her own reasons, and in doing so becomes, without intending it, the image the man has carried since childhood. The memory does not produce the woman; the woman produces the memory. This inversion is the first sign that *Twelve Monkeys* reads its sources against the grain of the masculine fantasy that organises *Vertigo* and, more tenderly, *La Jetée*. The woman is not merely the image at the end of the jetty. She is the one who will be left standing when the image has been fulfilled, and who will look.

Marker's own return to *Vertigo*, in *Sans Soleil* (1983), adds a dimension the earlier film only implies. The narrator, reading the letters of a fictive cameraman, recounts a

Plate 3

Time as accumulation



one ring, one year; three films folded like rings

A felled trunk in cross-section, one ring for each year and the most recent outermost. The three films that Section 2.4 folds into one another are marked as rings: *Vertigo* (1958), *La Jetée* (1962) and *Twelve Monkeys* (1995). Beyond the bark lies the future, not yet grown and unrecorded.

pilgrimage to the film's San Francisco locations—the florist's shop, the mission, the museum, the redwoods—and describes *Vertigo* as the only film capable of rendering memory that is impossible, even insane. The phrase joins memory to madness, and its implication is that the only adequate representation of memory is a mad one: a memory that does not merely recall the past but tries to reinhabit it, to make it happen again. The same film proposes that remembering is not the opposite of forgetting but its lining. Together the two propositions define Cole. His memory is insane in exactly Marker's sense, the memory of an event that has not yet occurred, which he is condemned to reinhabit; and his forgetting is the lining of that memory, since he cannot recognise, until the end, what it is he remembers. The madness the clinicians of 1990 diagnose in him is, in Marker's terms, simply the madness of memory itself.

The deepest structure that unites the three films, however, is older than cinema. It is the structure of Orpheus. In the fourth book of Virgil's *Georgics* and the tenth of Ovid's *Metamorphoses*, the singer descends into the underworld to retrieve his dead wife, wins her release on condition that he not look back until they reach the light, looks back, and loses her for ever. *La Jetée*'s protagonist ascends from an underworld —the galleries beneath Chaillot—toward a woman in the past, which is to say behind him; his journey is itself a looking-back, since its vehicle is his obsessive memory; and the fulfilment of the look kills him. Maurice Blanchot's "The Gaze of Orpheus" (1955) gives the myth its modern interpretation: Orpheus's true desire is not to lead Eurydice back into the daylight of ordinary life but to see her in her night, as the dead, as image, and the gaze that ruins the work is the very moment of inspiration the work exists to serve. The man of *La Jetée* chooses exactly this. Offered a future among the people who have survived, he asks to be returned to the image—the woman at the end of the jetty—and the return is his death. Cole repeats the choice in 1996 when he resolves to stay with Railly rather than return to 2035, tearing out the tooth he believes conceals the scientists' tracking device. The Orphic gaze is the refusal of the future in favour of the image.

Rilke's "Orpheus. Eurydice. Hermes." (1904) adds a final turn that illuminates the boy. When Hermes tells Eurydice that Orpheus has turned around, she does not understand and asks only: *Who?* She is already root, Rilke writes, filled with her vast death, and no longer knows the man who loved her. The boy at the airport stands in Eurydice's position. He sees a man fall and feels, at most, the shock of spectacle; he does not know who it is; he will spend his life carrying an image whose subject he cannot name. The Orphic gaze in *Twelve Monkeys* is thus divided among three figures: Cole, who looks back and dies; the boy, who sees and does not know; and Railly, who sees and knows. Only the last of these is a position from which re-vision is possible, and it belongs to the one figure whom the myth itself leaves without a voice.

2.5
To Shoot

The Camera as Gun and the Animal as Its First Subject

English, like several other languages, uses a single verb for photographing and for killing, and both films are built upon the pun. The man of *La Jetée* is shot at the moment the film's images—its photographs, its shots—resolve the meaning of his life. Cole is shot by airport police at the moment the child he was takes the mental exposure he will carry for nearly four decades; and in 1917 he is both shot, in the leg, and shot, in the photograph that Railly will later magnify. The convergence is not an accident of vocabulary. It records a genealogy, reconstructed by the media archaeology of the past half-century, in which the motion-picture camera descends from firearms and from the astronomical observatory, and in which the first subjects of moving pictures were animals.

The genealogy begins not with a gun but with a planet. In 1716 Edmond Halley proposed that astronomers observe the transits of Venus across the face of the sun, timed from widely separated stations, in order to calculate the distance between the earth and the sun; the international expeditions of 1761 and 1769 followed his plan. For the transit of 1874 the French astronomer Jules Janssen built what he called a photographic revolver, a device modelled on the rotating cylinder of a pistol that took a sequence of exposures on a turning plate, and carried it to Japan to record the planet's passage. The first chronophotographic apparatus was built to time a heavenly body. Halley, the astronomer who will return in Chapter Six as the founder of the life table, thus stands at the origin of the cinematic apparatus as well as of actuarial science: the man who calculated the return of a comet and the mortality of Breslau also set in motion the observations for which the first photographic revolver was made. The astrophysicist who takes her seat beside Dr. Peters at the end of *Twelve Monkeys* has a longer genealogy than she knows.

In 1882 the physiologist Étienne-Jules Marey adapted Janssen's principle into a *fusil photographique*, a photographic rifle, with which he could aim at birds in flight and take a rapid series of exposures—twelve in a second—arranged around the circumference of a single rotating plate. The image that resulted is a primal scene of the cinema's prehistory: a ring of twelve pictures of an animal in motion, produced by pulling a trigger. It would be difficult to design a more exact precursor of the emblem of the Army of the Twelve Monkeys, a ring of animal figures, than the plate of Marey's rifle. Marey's subsequent work at the Station Physiologique in Paris decomposed the movements of horses, pelicans, falling cats, and men into superimposed

Plate 4

A ring of twelve exposures



After the plate of Marey's *fusil photographique* (1882), which took twelve exposures in a second around the circumference of a single rotating plate: a ring of animal figures produced by pulling a trigger. The birds are drawn for this diagram; the phases of the wingbeat run clockwise from the top.

sequences, and his aim, like that of the time travellers of both films, was to recover what escapes the eye: the instant that lies between positions, invisible in the flow of perception.

Eadweard Muybridge's work ran in parallel. His sequences of Leland Stanford's horses, made in California in 1878 with a battery of cameras triggered by threads stretched across the track, settled whether a galloping horse ever has all four hooves off the ground and inaugurated the public career of decomposed movement. His zoopraxiscope (1879), the device with which he projected his sequences as moving images, bears in its name the subject of the new art: *zōon*, animal; *praxis*, action; *skopein*, to look—the viewer of animal action. In Philadelphia, as Chapter One recalled, he produced *Animal Locomotion* (1887), including the beasts of the city's zoo. And Muybridge was himself a man who had shot a man: in 1874 he killed his wife's lover, Harry Larkyns, and was acquitted by a jury that judged the killing justifiable. The inventor of decomposed motion carried into his art the knowledge of a single unrepeatable instant he had made with a pistol.

Friedrich Kittler, in *Gramophone, Film, Typewriter* (1986), drew the general conclusion: the intermittent mechanism of the film camera repeats the mechanism of the revolver and the machine gun, so that the history of the movie camera coincides with the history of automatic weapons, and the transport of images reproduces the transport of cartridges. Paul Virilio's *War and Cinema* (1984) traced the corresponding logistics of perception, in which the aerial reconnaissance of the First World War made the camera an instrument of targeting and the battlefield a field of images. Cole's accidental passage through the trenches of 1917 places him at the historical moment when the camera and the gun became, operationally, a single system. He is wounded there by the one and recorded there by the other, and the photograph that results becomes, nearly eighty years later, the proof that he is not mad.

The final term of the genealogy is taxidermy. Carl Akeley, the taxidermist who designed the African Hall of the American Museum of Natural History, also invented a motion-picture camera—the Akeley camera of 1915, prized for the smoothness of its pan—and in Donna Haraway's essay "Teddy Bear Patriarchy" (1984) he appears as the exemplary figure of a culture that hunted with camera and gun alike in order to arrest life in a single moment of truth. Haraway's central exhibit is Akeley's mountain gorilla group, assembled from animals he shot in the 1920s and mounted in a diorama that freezes a family of apes in a posture of eternal vigilance. The diorama is a photograph in three dimensions, and the stuffed ape is its subject. *La Jetée*'s lovers wander through exactly such a world: a museum of animals preserved past death, beasts that time no longer touches. Bazin's mummy complex is here doubled, the photograph embalming the embalmed.

Set against this genealogy, the passage from *La Jetée* to *Twelve Monkeys* acquires a zoological shape. Marker's film, made of stills, shows its lovers among stuffed animals: the arrested image and the arrested beast belong to a single order. Gilliam's film, made of moving images, shows animals that move—escaped from the zoo and wandering a city overpass, and, in 2035, a bear in a snowbound street and a lion roaring from a building's ledge. The photograph is to the moving image as the mounted specimen is to the caged animal, and the caged animal to the animal at large. Each film's medium corresponds to its relation with the animal. In *La Jetée* image and beast are both dead and preserved; in *Twelve Monkeys* both have been set in motion, and both are on the point of escaping human custody. The remake that breathes motion into Marker's stills also opens the museum's cases and the zoo's gates. When the film's opening intertitles declare that the animals will once again rule the world, they announce the end of the long project that runs from Marey's rifle to Akeley's diorama: the project of shooting the animal—killing it, filming it, mounting it—in order to possess its time.

2.6
Dream, Enlargement, Monitor, Screen

What *Twelve Monkeys* Does with the Still

It remains to show how *Twelve Monkeys*, a film that moves, carries the ontology of the still it inherits. It does so not by imitating Marker's form, which a studio feature could not do, but by distributing the functions of the photograph across four devices within its fiction: the dream, the enlargement, the monitor, and the screen. Each preserves one aspect of the photograph's tense, and together they reconstitute, inside a narrative of movement, the metaphysics of the photo-roman.

The first device is the dream. Cole's recurring vision of the airport—the running man, the shot, the blonde woman kneeling, the child watching—is the film's equivalent of the image that marks Marker's protagonist, but it differs in one crucial respect: it changes. As the film proceeds and Cole comes to know Railly, the dream-woman's features resolve into hers, and the scene acquires details it lacked. The dream is a memory revised by its own future. Freud described this process in "Screen Memories" (1899), concluding that what we call memories of childhood may never be memories *from* childhood at all, only memories *about* it, formed when they were later aroused and stamped with the concerns of that later time. *Twelve Monkeys* literalises Freud's suspicion and gives it a temporal twist he could not have imagined. Cole's

memory is reshaped by later experience, as every screen memory is; but the later experience is the very event the memory records, approached from its other side. The revision of the memory is not a distortion of the past. It is the past coming into focus as its future arrives.

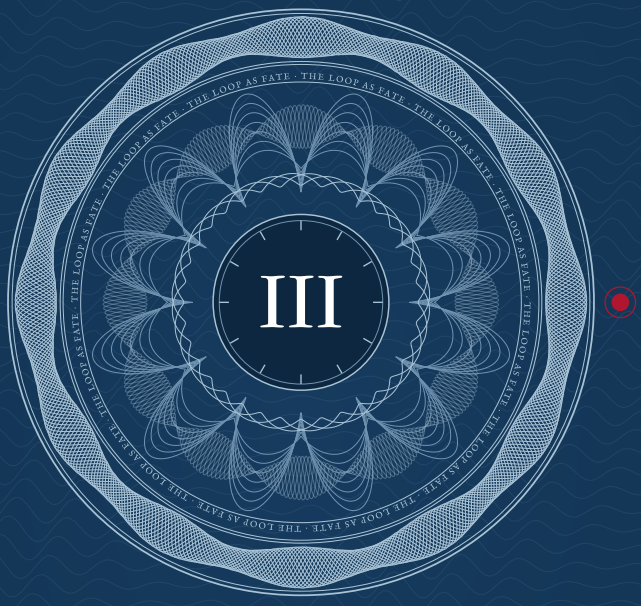
The second device is the enlargement. In her lecture on apocalyptic madness, Railly projects among her slides a photograph from the trenches of 1917, the image of a wounded soldier whose case she presents as an instance of the Cassandra complex; later, examining a print with a magnifying glass, she finds that the soldier is Cole. The scene condenses an entire theory of the photograph. In his "Little History of Photography" (1931), Walter Benjamin described the compulsion to search an old photograph for the small spark of chance, of the here and now, that reality has burned into the image—for the inconspicuous place in which the future has made its home within a long-forgotten instant, so eloquently that the later viewer, looking back, can find it there. Benjamin meant the future of a photograph's subjects, legible in retrospect in their faces. *Twelve Monkeys* takes him at his word. The future is literally lodged in the photograph of 1917—a man from 2035 stands in the trench—and Railly, looking back, finds it. The same essay coined the term "optical unconscious" for what the camera records without the eye's knowledge: the phases of a stride, the anatomy of a gesture. The magnified photograph of 1917 discloses an optical unconscious of a new kind, an unconscious not of the eye but of history, in which the camera recorded, generations before anyone could understand it, the presence of the end. The scene belongs to a cinematic lineage. In Antonioni's *Blow-Up* (1966), adapted from a story by Julio Cortázar, a photographer enlarges his pictures of a London park until the grain yields what seems to be a corpse; in *Blade Runner*, as already noted, a machine penetrates a photograph until it surrenders a face. Railly's enlargement inverts Antonioni's. She finds in the grain not a corpse but a living man—one who is, however, going to die, and whose death she will witness.

The third device is the monitor. In their interrogations the scientists of 2035 never appear to Cole unmediated: their faces flicker on the screens of the cluster of monitors that advances toward him on its mechanical arm, and their questions reach him through it. Their knowledge of the past is equally mediated—by recordings, by a degraded voicemail, by photographs, by Cole's own testimony. The future sees the past only as image, and sees the image only as data: what it wants from Cole is not memory but report. Here the film's medial ontology joins its politics. The monitor-cluster is the apparatus of an actuarial gaze, one that does not contemplate the image, as Marker's pensive spectator does, but extracts from it what can be used. Between Marker's camp, where the experimenters peer into their subject's memory, and Gilliam's laboratory, where the scientists observe their subject through a compound eye

of screens, the image has ceased to be an object of contemplation or of desire and become an instrument of assessment.

The fourth device is the cinema screen itself, and with it the film arrives at its most explicit statement about time. Watching *Vertigo*, Cole observes that the film never changes, that it cannot change, and that it seems different each time because the viewer is different and sees different things. The remark compresses a metaphysics. A film on its reel is a block universe in miniature: all its frames coexist, the end as real as the beginning, Madeleine's fall as fixed in the emulsion as her first appearance. Hermann Minkowski announced in 1908 that space by itself and time by itself were destined to fade into shadows, and that only a union of the two would preserve an independent reality; Einstein, consoling the family of his friend Michele Besso in 1955, wrote that for convinced physicists the distinction between past, present, and future is only "a stubbornly persistent illusion." The reel is Minkowski's spacetime rendered in acetate. Projection, meanwhile, is a present that travels along the block, illuminating one frame at a time—the picture of time that C. D. Broad examined in *Scientific Thought* (1923) as a spotlight moving along the series of events. Cole's remark joins the two pictures: the film is a block, unchangeable; its viewing is a spotlight; and the spotlight is borne by a viewer who changes. What changes between viewings is not the world but the light in which it is seen.

This is the medial foundation of the ethics this dissertation calls re-vision, and it explains why the film places its most philosophical sentence in a cinema rather than a laboratory. The scientists of 2035 treat the past as an archive from which to extract a sample; their relation to the block is that of an auditor consulting a ledger. Cole, in the cinema, proposes another relation: that of the spectator who returns to an unchangeable film and finds it changed because he has changed. *La Jetée* gave this relation its emblem in the blink, the instant in which a fixed image wakes. *Twelve Monkeys* gives it a narrative: a woman who begins the film as a clinician and ends it as a witness, whose final look at the child transforms an image that cannot be altered. The photograph says that this has been and will have been. The spectator who sees it again, and differently, cannot make it otherwise. But she can make it mean otherwise, and in a universe without revision that is the only redemption on offer. Chapter Three asks what kind of universe this is—what metaphysics of time underwrites the film's fixed past—and whether the freedom of re-vision is a genuine freedom or only the consolation of the condemned.



Chapter Three

The Loop as Fate

Closed Time, Tragic Necessity, and the Ethics of the Event

The time is out of joint. O cursed spite,
That ever I was born to set it right!

— Shakespeare, *Hamlet*, I.v

A man's feet are his guarantors: to the place where he is wanted, there they bring him.

— Babylonian Talmud, *Sukkah* 53a (trans. mine)

3.1
The Investigator and the Plague

Oedipus, Samarra, and the Detective Lured

The *Oedipus Tyrannus* opens, as *Twelve Monkeys* does, with a plague. Thebes is dying: the priest who leads the suppliants to the king's door describes a blight upon the fruit of the earth, upon the herds in their pastures, upon the labour of women, and a fire-bearing god, hateful pestilence, emptying the house of Cadmus. Delphi has named the cause: a pollution, the unpunished murder of the former king, Laius. Oedipus, who once saved the city by answering the Sphinx, undertakes to save it again by answering this riddle, and the play is the record of his investigation. Every step is taken in good faith, and every step brings him closer to the discovery that the pollution is himself. The blind seer Tiresias tells him so at the outset and is accused of conspiracy; Jocasta assures him that oracles are worthless and cites as proof a prophecy that seemed to fail, which is the very prophecy that has been fulfilled; a messenger arrives from Corinth to free him from his fear and, in Aristotle's celebrated example, produces the opposite effect, so that reversal and recognition fall together. The plague is caused by the man who investigates it. When the truth is out, Oedipus puts out his eyes with the brooches from the dead queen's robe: the man who saw too late blinds himself, in the play's final inversion, so as not to see what he now knows.

Twelve Monkeys is an investigation of the same type, and its structure is Oedipal in a precise and unusual sense. Cole is sent to find the source of the plague. He is not its cause; the film is careful to lodge the act in another man's hands. But his investigation is woven, at every point, into the history it investigates. Jeffrey Goines, whom Cole meets as a fellow patient in 1990, later claims that it was Cole who first spoke to him of wiping out the human race, so that the idea of the Army of the Twelve Monkeys, the decoy that will absorb the future's attention, may have been planted by the man sent to discover it. Railly's telephone message of 1996, which the scientists of 2035 take as their founding evidence, is a consequence of Cole's mission and a misdirection of it. The investigation does not find the plague; it finds itself, reflected in the evidence it has generated. And its anagnorisis inverts that of Oedipus. Oedipus discovers that he is the killer; Cole discovers that he is the killed. The Sophoclean recognition says *I am the cause*; the recognition of *Twelve Monkeys* says *I am the man in the dream*. Tragedy of guilt has become tragedy of position. What Cole learns is not what he did but where he stands: at the end of a jetty, in a scene that was always waiting for him to occupy it.

The oldest surviving version of the second structure, in which flight from fate becomes the instrument of its fulfilment, is preserved in the Babylonian Talmud. Solomon, the tractate *Sukkah* relates, saw the Angel of Death looking sorrowful and asked the reason; the angel replied that he had been charged to take the king's two scribes. Solomon placed them in the care of spirits and sent them to Luz, the legendary city into which death had no permission to enter. They died at its gates. The next day Solomon saw the angel laughing, and learned that the scribes had been sent precisely to the place where they were required. Solomon's conclusion is the proverb that serves as this chapter's epigraph: a man's feet are his guarantors, and they bring him to the place where he is wanted. The Aramaic word rendered here as *guarantors*, *'arvin*, is the word for sureties, for those who stand bond for a debt. Fate, in the Talmud's figure, is a contract of suretyship, and the body itself is its guarantee. The apologue reached the twentieth century in Somerset Maugham's version, in which a servant in Baghdad, seeing Death in the marketplace, flees on his master's horse to Samarra, and Death remarks that she was surprised to find him in Baghdad, since she had an appointment with him that night in Samarra.

Cole's flight has exactly this form, and the film adds a detail the Talmud would have appreciated. Having resolved that he is mad, that the future is a delusion, and that he may remain in 1996 with the woman he loves, he does what the scribes did: he flees toward a place where death is not. The fugitives mean to board a plane and leave the city behind, beyond the reach of a catastrophe they no longer believe in. The airport is Cole's Luz, and he dies at its gate. The gate is also, in the constellation of the previous chapter, Nietzsche's gateway, the threshold named Moment where the paths of past and future meet. The place of escape is the place of the appointment, and the instrument of the appointment is the flight itself.

La Fontaine gave the structure its classical French moral in "L'Horoscope" (*Fables*, VIII.16): *on rencontre sa destinée / souvent par des chemins qu'on prend pour l'éviter*—one often meets one's destiny on the very roads one takes to avoid it. The fable's examples deserve recalling, because both belong to the zoological argument of this dissertation. A father, warned that his son will be killed by a lion, keeps the boy confined; the son, enraged by a lion painted on a hanging, strikes the image and is fatally wounded by a nail concealed behind it. And Aeschylus, warned that he will die beneath a falling house, leaves the city to live under the open sky, where an eagle, mistaking his bald head for a stone, drops a tortoise upon it. The son is killed by the image of an animal; the poet is killed by an animal from the sky. That the second victim should be the dramatist who gave European literature its Cassandra and its chorus of twelve is a coincidence the history of fate seems to have arranged for this dissertation.

A final variant joins the investigator to the appointment. In Jorge Luis Borges's "Death and the Compass" (1942), the detective Erik Lönnrot reads a series of murders as the terms of a cabbalistic pattern, deduces the place and time of the fourth, and goes there to prevent it, only to discover that the pattern was composed for him by his enemy, Red Scharlach, and that the fourth victim is himself. The clues were a lure, and their intelligibility was the trap. *Twelve Monkeys* follows this design. Cole pursues a trail of signs—the stencilled ring of red monkeys on the walls of two centuries, the boastful graffiti, the voice on the recording—that seem to lead to the perpetrators and lead instead to the concourse where he will die. No one composed these signs to trap him; the timeline composed them, as the condition of its own consistency. Borges's Scharlach has been replaced by the closed curve itself, the most patient of enemies, since it has all of time in which to wait.

3.2
The Lazy Argument and the Prisoner's Physician

Fate and Foreknowledge from the Stoa to Boethius

The problem that tragedy dramatises was first formalised by Aristotle. In the ninth chapter of *On Interpretation* he asks whether it is already true today that there will be a sea battle tomorrow. If it is, the battle cannot fail to occur, and deliberation is idle; if every statement about the future is already true or false, everything happens of necessity. Aristotle's solution, on its most influential reading, is to deny that statements about contingent future events are yet true or false: it is necessary that there either will or will not be a sea battle, but neither disjunct is necessary by itself. *Twelve Monkeys* inhabits the hypothesis Aristotle laboured to avoid. In its world the statement that five billion people will die in 1997 was true in 1990 and in 1917, true before the virus existed, because it was made by someone who had already lived through its consequences.

The Stoics accepted the hypothesis and drew its consequences with characteristic rigour. For Chrysippus fate, *heimarmenē*, is the unbroken chain of causes that runs through the cosmos, and every proposition about the future already has its truth value. Their opponents answered with what Cicero, in *On Fate*, calls the lazy argument: if it is fated that you will recover from this illness, you will recover whether or not you call a doctor, and if it is fated that you will not, you will not; therefore calling the doctor is pointless. Chrysippus replied that certain events are co-fated, *confatalia*: your recovery is fated together with your calling the doctor, so that the act the lazy

argument dismisses belongs to the very fate it invokes. *Twelve Monkeys* is a Chrysippean film. The lazy argument would ask why the scientists bother to send anyone back, since the past cannot be changed. The film's answer is that the retrieval of the sample is co-fated with the mission. The future does not alter the past, but its interventions are woven into the past's fixed fabric, and the cure it will obtain is fated together with the journeys it pays for. Chrysippus also offered an image of the relation between fate and character: a cylinder set rolling down a slope is started by an external push, but it rolls as it does because it is a cylinder. The scientists push; Cole rolls according to his shape.

The Stoics drew one further consequence, which brings them closest to the film's emblem. The cosmos, they taught, periodically dissolves in a universal conflagration, *ekpyrosis*, and is reborn identically, so that, as the Christian bishop Nemesius later summarised their doctrine, Socrates will once more be accused by Anytus and Meletus, and every event will recur in the same order without end. The Stoic world is a cosmic zoetrope, a ring of identical frames turned by the fire of the world. The plague of *Twelve Monkeys* plays the role of the conflagration in a minor key. It does not end the cosmos; it ends one figure in the cosmic ring, the human, and returns the world to the animals who were there before.

Cicero, whose dialogues preserve much of what we know of these debates, also preserved their most poignant objection. In the first book of *On Divination*, his brother Quintus defends prophecy and quotes among his proofs a fragment of Ennius in which Cassandra cries out that the firebrand is here, wrapped in blood and flame. In the second book Cicero replies in his own voice, and his reply is the most exact ancient description of Cole's condition. If everything happens by fate, he argues, knowledge of the future is not merely useless but harmful, since it adds to an inevitable evil the anguish of foreseeing it; what would it have profited Pompey, or Caesar, to know in advance how they would die? Cole is the man Cicero imagined and pitied. He has the knowledge without the power, and the knowledge is his torment. His wish, in the second half of the film, to be mad—to have invented the future rather than remembered it—is the wish to be relieved of Cicero's burden.

Augustine transformed the problem by internalising time. In the eleventh book of the *Confessions* he admits that he knows what time is so long as no one asks him, and does not know as soon as he tries to explain it to one who asks; and he resolves the perplexity by locating past, present, and future in the soul, as three presents—the present of things past in memory, the present of things present in attention, the present of things future in expectation. Time is a *distentio animi*, a distension of the mind. Cole's mind is the limit case of Augustinian distension, since in him the first and third presents have fused: his memory holds the future. What he remembers—

the airport—is what he expects, and what he expects he has already seen. The Augustinian soul stretched between memory and expectation becomes, in Cole, a soul folded upon itself, a distension that has closed into a loop.

In *The City of God*, Augustine turned against the cyclical cosmologies of the philosophers with an argument that defines the Christian conception of history. Against those who held that the same Plato would teach the same pupils in the same school at Athens again and again through endless ages, he invoked the psalm—*in circuitu impii ambulant*, the wicked walk in a circle—and the once-for-all of the resurrection: Christ died once for our sins, and rising from the dead dies no more. Linear time is the time of salvation; the circle is the path of the damned. By this measure *Twelve Monkeys* is a film about a world that has lost the once-for-all. Its protagonist, who shares his initials with the hero of the *Terminator* films and with the figure both evoke, dies more than once in his own experience—once in the boy's sight, once in the man's body—and his death redeems nothing. The circle has returned, and with it the anxiety Augustine's line was drawn to dispel.

The most profound ancient answer to the problem was written by a man in Cole's position. Boethius composed *The Consolation of Philosophy* around 524, in prison at Pavia, awaiting an execution he knew was coming. Philosophy appears to him as a woman of uncertain age and stature and announces that it is time for medicine rather than lament: she comes as a physician to a patient. She shows him Fortune turning her wheel, raising and casting down those who cling to it, and in the fifth book she resolves the dilemma of foreknowledge. God does not foresee, in time; God sees, from eternity, which Boethius defines as the whole, simultaneous, and perfect possession of unending life. As we see a man walking without causing him to walk, God beholds all of time from the lofty summit of providence without imposing necessity upon it. The necessity of a known event is conditional—if you see a man walking, he must be walking—not simple, like the necessity that every human being dies. The Boethian God is the ideal spectator of the block universe: he sees the whole reel at once without determining a single frame.

The *Consolation* maps onto *Twelve Monkeys* with uncanny exactness, and the differences are instructive. Cole is the prisoner awaiting death. Railly is the woman who visits him in confinement, a physician of the mind, whose task is first to diagnose and then, as the film proceeds, to understand. The scientists of 2035 occupy the place of providence, beholding the whole of the catastrophe from the vantage of its completion. But their vantage is not eternity; it is aftermath. They do not see all time as present; they see the past as finished, as an archive, and their knowledge, unlike God's, is interested, since they need the catastrophe to have happened. Boethian providence has degraded into actuarial providence: a knowledge that sees the whole of a loss in

order to collect on it. The only figure in the film who approaches the Boethian gaze—a seeing that neither determines nor profits—is once again Railly at the end, looking at a child whose life she now sees whole.

3·3
Block, Bootstrap, and the Commonplace Reason

The Physics and Logic of the Fixed Past

The modern philosophy of time begins with J. M. E. McTaggart's "The Unreality of Time" (1908), which distinguished two ways of ordering events. In the A-series events are past, present, or future, and they change these determinations as time passes: the death of Queen Anne was once future, became present, and is now past. In the B-series events are simply earlier or later than one another, and these relations never change. McTaggart argued that time requires the A-series, that the A-series is contradictory, and that time is therefore unreal. His argument has been endlessly disputed, but his distinction has become the common language of the field, and it describes the two temporal orders that *Twelve Monkeys* sets against one another. The scientists inhabit a B-series. For them 1917, 1990, 1996, and 2035 are fixed positions in an archive, related as earlier and later, and their errors of calibration are errors of address. Cole inhabits an A-series that has come loose from the B-series: his now leaps from 2035 to 1990, back to 2035, to 1917, to 1996, and his past includes events the calendar places in his future.

David Lewis gave this predicament its exact formulation. Time travel, he argued in 1976, involves a discrepancy between external time, the time of the world, and personal time, the time of the traveller as measured by his memories, his digestion, the growth of his beard. The traveller's personal time is ordinary and linear; it is its mapping onto external time that is folded. Cole's personal biography runs straight: childhood, prison, missions, love, death. Laid upon the external calendar, it traces a figure that crosses itself at the airport in 1996, where a boy and a grown man occupy the same concourse, one near the beginning of his personal time and one at its end. The loop is not in Cole. It is in the relation between Cole and the world.

Physics had by then given the block universe its canonical form. Minkowski's four-dimensional spacetime, which Einstein adopted, represents every event as a point and every life as a world line, and it draws no physical distinction between past and future. Hermann Weyl drew the phenomenological consequence in a passage that reads like a description of cinema: the objective world simply is, it does not happen; only to the

gaze of consciousness, crawling upward along the world line of the body, does a section of it come to life as a fleeting image that continually changes. Weyl's crawling consciousness is the moving spotlight of the previous chapter, and his fleeting image is the frame illuminated in the projector's gate. *Twelve Monkeys* is set in Weyl's universe. What happens has always been on the reel; consciousness, crawling, makes it seem to happen.

In 1949 Kurt Gödel showed that this universe permits more than relativity's founders had imagined. In a paper for the issue of *Reviews of Modern Physics* honouring Einstein's seventieth birthday he presented a solution of the field equations—a rotating cosmos—that contains closed timelike curves: paths along which a traveller, always moving forward in his own time, returns to his own past. Gödel concluded, in a companion essay, that time in the intuitive sense, the lapse of an objective now, does not exist, and that relativity vindicates the idealist philosophers who held change to be appearance. Einstein, in reply, admitted that the problem had disturbed him since he first built the general theory. A closed timelike curve is the mathematical form of Cole's life. He moves always forward in his own time and arrives, at its end, at a scene from its beginning.

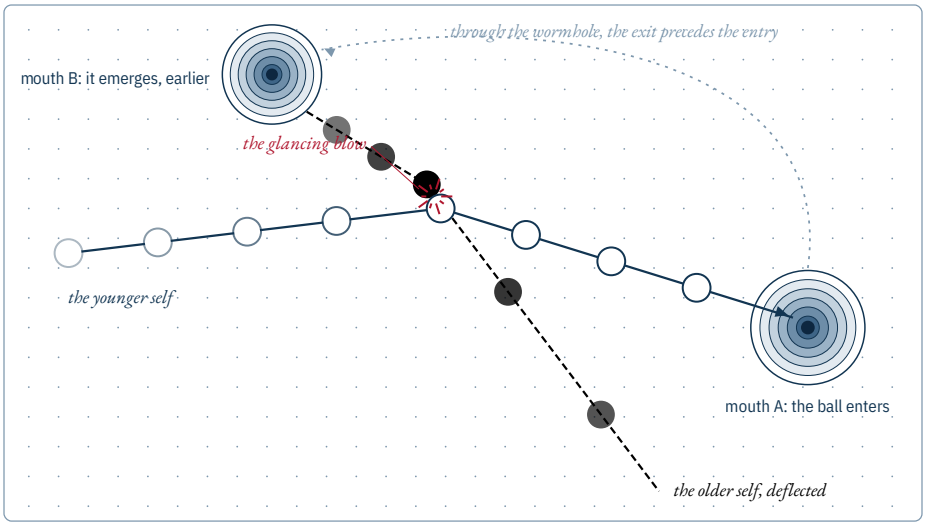
Physicists who took the possibility seriously were obliged to confront the paradoxes it seemed to entail, and their response is the scientific counterpart of the film's metaphysics. Around 1990 Igor Novikov and his collaborators proposed that the laws of physics admit only globally self-consistent histories in a spacetime containing closed timelike curves. The test case, posed by Joseph Polchinski, was a billiard ball sent into a wormhole on a trajectory that would make it emerge in the past and strike its earlier self, knocking it off course so that it never entered the wormhole at all. Fernando Echeverria, Gunnar Klinkhammer, and Kip Thorne showed that self-consistent solutions exist for such initial conditions: the ball emerges and strikes its younger self a glancing blow that deflects it just enough to send it into the wormhole on precisely the trajectory that produces the glancing blow. The attempt to prevent becomes the mechanism that produces. Cole is Polchinski's ball. His intervention at the airport, the run toward Peters with a gun, is the glancing blow that secures the scene he has been dreaming since childhood.

Lewis's treatment of the grandfather paradox adds a refinement with a political edge. Can Tim, who has travelled back with a rifle, kill his grandfather before his father is conceived? In one sense he can: he has the weapon, the skill, the opportunity, and his killing the grandfather is compossible with the facts of his situation. In another sense he cannot, since his success is not compossible with the further fact that the grandfather survived. Both statements are true, relative to different sets of facts, and Tim will in fact fail, Lewis wrote, for some commonplace reason—a distracting

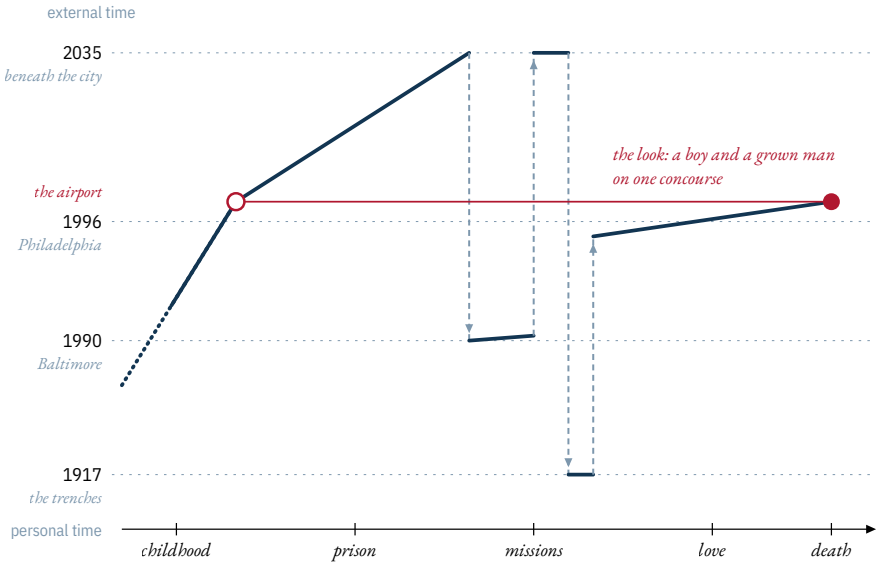
Plate 5

Self-consistent histories

A *the self-consistent solution*



B *personal time laid across external time*



A. Polchinski's billiard ball and the self-consistent solution of Echeverria, Klinkhammer and Thorne: the ball enters mouth A, emerges earlier from mouth B, and strikes its younger self a glancing blow that sends it into A on precisely the path that produces the blow. B. Cole's personal time, which runs straight, laid across external time after David Lewis: it leaps from 2035 to 1990, back to 2035, to 1917 and to 1996, and ends on the concourse where it began. Schematic, not to scale.

noise, a missed shot, a failure of nerve, an unaccustomed pang of mercy. In *Twelve Monkeys* the commonplace reasons are institutional. The past defends its consistency not with stray noises and faltering nerves but with police officers, psychiatrists, security checkpoints, and the casual disbelief of everyone who hears Cole speak. The officer who shoots Cole on the concourse is the commonplace reason of Lewis's argument, in uniform. Here the actuarial character of the film's world becomes visible. Consistency is not a cosmic miracle; it is an administrative routine.

The literature of time travel had explored self-consistent loops long before the physicists. Robert A. Heinlein's "By His Bootstraps" (1941) traps its hero in a loop of encounters with his own future selves, who manipulate him into becoming them, and gives the phenomenon its name. Heinlein's "—All You Zombies—" (1959) closes the loop around a single body that is its own mother and father, a fantasy of self-begetting that its film adaptation, *Predestination* (2014), made explicit. *The Terminator* (1984) closes it around paternity—the soldier sent back to protect the mother of humanity's saviour becomes the saviour's father—and Penley read this as the fantasy of witnessing one's own conception. The *Planet of the Apes* cycle closes it around the animal. In *Escape from the Planet of the Apes* (1971), two chimpanzees from the future arrive in the present; they are killed, but their infant survives to become Caesar, who in *Conquest of the Planet of the Apes* (1972) leads the revolt that will produce the ape-ruled future from which his parents came. The animals rule the world because animals came back from the world they rule. No closer analogue exists for the intertitles that open *Twelve Monkeys*.

Later fictions explored what such a universe demands of those who live in it. Kurt Vonnegut's *Slaughterhouse-Five* (1969) gave its Tralfamadorians a vision of all moments at once, and with it the knowledge that their universe will be destroyed by one of their own test pilots experimenting with a new fuel; they do nothing, because the moment is structured that way, and they counsel humans to contemplate the good moments and ignore the bad. Billy Pilgrim, who learns their wisdom while displayed in their zoo, is a former prisoner of war who has passed through the psychiatric ward of a veterans' hospital. Ted Chiang's "Story of Your Life" (1998), filmed as *Arrival* (2016), grants its linguist a simultaneous consciousness of her whole life, including her daughter's death, and has her choose that life anyway, performing it as one performs a script one knows by heart. *Donnie Darko* (2001) brings a schizophrenic adolescent and a falling jet engine to a closed loop that can be repaired only by his death, which he meets in his bed, laughing. *Looper* (2012) breaks its loop only through its hero's self-sacrifice, a mirror image of Cole's fate in which the self is destroyed in order to change the future rather than to fulfil it. *Dark* (2017–2020) builds three seasons of television on the premise that every attempt to prevent an apocalypse

is its cause, and ends by dissolving its knot rather than repeating it. And the *Terminator* series, having proclaimed in its second film that there is no fate but what we make, reversed itself in its third (2003), which declared Judgment Day inevitable. What divides these works is not whether the past can be changed—most say it cannot—but what freedom remains to those who know that it cannot. That question occupies the remainder of this chapter.

3.4
Recollected Forward

Kierkegaard, Nietzsche, and the Redemption of "It Was"

In 1843 Kierkegaard published, under the pseudonym Constantin Constantius, a short and strange book called *Repetition*. Its opening proposition is the most exact philosophical description of Cole's condition ever written: repetition and recollection, Constantius declares, are the same movement in opposite directions, since what is recollected has been and is repeated backward, whereas genuine repetition is recollected forward. Recollection, the Greek and above all the Platonic relation to truth, holds that to know is to remember what the soul knew before birth; it faces backward, toward an origin, and its mood is melancholy. Repetition, the modern and finally the religious category, faces forward; it receives again what was lost, and its mood is a kind of joy. Cole is the man in whom the distinction has collapsed. His dream is a recollection, a memory of childhood; and it is recollected forward, since what it remembers lies ahead of him. His life is lived toward a memory.

Constantius tests the possibility of repetition with a comic experiment. Having once enjoyed a stay in Berlin, he returns to repeat it: the same lodgings, the same coach, the same farce at the Königstädter Theatre. Nothing repeats. The lodgings have changed, the coach is uncomfortable, the theatre disappoints, and he concludes that repetition is impossible. The failure has a precise cause: the theatre performs anew each night, and so does the self that attends it. What Constantius could not know is that the century after his would invent an art in which the performance is identical at every showing. Cinema achieves at the level of the object the repetition the theatre could not, and in doing so relocates the impossibility of repetition to the subject. The film is the same; the spectator is not. Cole's remark in the cinema is the answer, a century and a half late, to Constantius's experiment: perfect repetition of the world exists, and its sole effect is to make visible the change in the one who watches.

The book's other narrative concerns a young man who falls in love and out of it in a single movement, who flees his beloved and finds in the Book of Job the model of what he hopes for: a man stripped of everything who receives it all back, doubled. When the young woman marries another, he declares that he has received repetition: he has got himself back. Repetition in this religious sense is not the recurrence of the same event but the gift of the self restored after loss. *Twelve Monkeys* tests the category and finds it wanting for its hero. Cole receives nothing back; he loses even the woman he has found. But the film transfers the possibility of repetition, in Kierkegaard's religious sense, to Railly. She is the one who will live on after the loss, and who, in the look exchanged with the boy, receives back—in the form of the child who will become him—the man she has just lost. Repetition for her is not the return of the same but the encounter, at the instant of loss, with its origin.

Nietzsche put the question of repetition in its most terrible form. In section 341 of *The Gay Science*, titled "The Heaviest Weight," a demon steals after you into your loneliest loneliness and tells you that this life, as you now live it and have lived it, you will have to live once more and innumerable times more, with nothing new in it, every pain and every joy and every thought returning in the same succession—even this spider and this moonlight between the trees, even this moment and I myself. Would you throw yourself down, gnash your teeth, and curse the demon who spoke so? Or have you once experienced a tremendous moment in which you would answer him: you are a god, and never have I heard anything more divine? The question "Do you want this once more and innumerable times more?" would lie upon every action, Nietzsche wrote, as the heaviest weight.

Twelve Monkeys places its protagonist in the demon's scenario and records his answer as it changes. In 1990 and again in 2035, Cole experiences the recurrence of his dream as torment, and his mission as a sentence. In 1996, having tasted the air and the music of a world not yet dead, he answers as the gnashing man does: he refuses the recurrence, declares himself mad so that the future he remembers may be unreal, and tears out the tooth through which the future speaks to him. At the airport, as the scene of his dream assembles itself around him, his answer changes again. The film does not show him affirming his fate in Nietzsche's sense; it shows him acting within it, running toward the man with the case of vials, choosing the action the scene requires. Whether that choice is the gnashing of teeth or the god's word is the film's most delicate undecidable, and its resolution is entrusted, as the final section of this chapter argues, to someone else.

Nietzsche gave the decisive formulation of the problem in the section of *Thus Spoke Zarathustra* titled "On Redemption." The will, Zarathustra teaches, is a liberator, but it cannot will backward. Before the *it was*, the stone that cannot be rolled

away, the will is an angry spectator and a prisoner, and from its impotence against time arises the spirit of revenge: the will's ill will toward time and its *it was*. Punishment, guilt, and every doctrine that preaches the world as penalty are the spirit of revenge become philosophy. Redemption, for Zarathustra, would be to re-create every *it was* into a *thus I willed it*. The formula does not propose to change the past. It proposes to change the will's relation to the past, so that what happened is no longer suffered as an injury but affirmed as willed. This is re-vision in its most exacting form.

Read through Zarathustra's terms, *Twelve Monkeys* distributes three relations to the *it was* among its figures. The scientists of 2035 embody the spirit of revenge in its modern institutional form. They cannot will backward, so they seek compensation: the cure, the sample, the return to the surface, an indemnity for the *it was* of 1996. Insurance, on this reading, is ressentiment against time given an actuarial structure; it does not forgive the past, it bills it. Cole and Railly, in their last hours, attempt revision: they try to will backward, to stop Peters, to make the *it was* not have been, and the attempt fails, as it must in a self-consistent world. Railly, in the look she exchanges with the boy, approaches the third relation, redemption in Zarathustra's sense: not altering the past but receiving it, at the moment of its completion, with a gaze that consents to see it whole. Whether this is affirmation or only the recognition that precedes grief, the film does not say. But it is the only moment in *Twelve Monkeys* in which the *it was* is met by something other than revenge or flight.

Two twentieth-century variations frame the stakes. Camus, in *The Myth of Sisyphus* (1942), proposed that we must imagine Sisyphus happy, his absurd labour redeemed by his lucid consciousness of it; the hour of descent, when he walks back down toward his rock, is the hour of consciousness. *Groundhog Day* (1993), released two years before *Twelve Monkeys*, made a comedy of the eternal return in which a weatherman, condemned to repeat a single day, escapes only when he has used the repetition to become a different man. *Twelve Monkeys* is the tragic counterpart of both. Like Camus's hero, Cole is condemned to a labour whose futility he comes to know; like the weatherman, he lives his crucial day more than once. But no lucidity is granted him that would make the labour his own, and no transformation releases him from the loop. The happy Sisyphus and the redeemed weatherman are replaced, in the film's world, by a woman who watches and a child who does not yet know.

3.5
The Wound That Existed Before Me

Trauma, Afterwardsness, and the Thanatic Primal Scene

The detour that deposits Cole in the trenches of 1917 carries a psychoanalytic significance that matches its epidemiological one. *Beyond the Pleasure Principle* (1920), the essay in which Freud introduced the repetition compulsion and the death drive, was written out of the experience of the war neuroses. The dreams of soldiers who had survived the trenches posed a problem for the theory that dreams fulfil wishes, since these dreams returned the dreamer again and again to the scene of his terror. Freud concluded that they served a function older than wish-fulfilment: they sought to master the traumatic stimulus retrospectively, by developing the anxiety whose absence had allowed the trauma to breach the mind's protective shield. Trauma is a fright that arrives unprepared, and its repetition in dreams is the mind's belated preparation for what has already happened.

Cole's recurring dream is a traumatic dream in Freud's exact sense. It returns him to a scene he witnessed as a child, unprepared and without the means to understand it, and it returns him there nightly, as though his mind were still labouring to produce the anxiety the scene required. But the film adds to Freud's model a twist Freud could not have conceived. The scene the dream repeats is not yet over; the anxiety it seeks to develop is the anxiety appropriate to an event that lies ahead. The detour to 1917 makes the point with startling economy. The soldier whose photograph appears in Railly's lecture is exactly the kind of patient on whose dreams Freud built the repetition compulsion: a wounded man in the trenches, raving of things no one can believe. The film reveals that this patient is a man from the future. The founding case of the theory of traumatic repetition becomes, in *Twelve Monkeys*, a case of time travel.

In the same essay Freud described the game of his grandson, a child of eighteen months, who would throw a wooden reel on a string over the edge of his curtained cot, uttering a long *o-o-o-o* that the family understood as *fort*, gone, and then pull it back into view with a joyful *da*, there. Freud read the game as the child's mastery of his mother's absences: by staging disappearance and return with an object under his control, he turned a passive experience into an active one. The object of the first game of representation in psychoanalysis is a reel. The cinema's reel performs the same operation for its spectators, making the lost reappear, the dead return, the gone come back as there. *Twelve Monkeys* is a fort-da game played with a man's death: the dream throws the scene away, each night brings it back, and the film, which ends where the dream begins, can be rewound to throw it away again.

The deeper structure is the one Freud named *Nachträglichkeit*, deferred action or, in Jean Laplanche's preferred rendering, afterwardsness. A scene is experienced at one time without being understood; it acquires its traumatic meaning only later, when a second scene, or a new capacity for understanding, revives it. The canonical case is the Wolf Man (1918). At the age of four the patient dreamed that the window of his bedroom opened of its own accord and that six or seven white wolves were sitting in the walnut tree outside, perfectly still, staring at him; he woke screaming. Freud traced the dream to a scene witnessed at the age of eighteen months, the primal scene of parental intercourse, which became traumatic only when the dream revived it. Two features of the dream are decisive for the present argument. The wolves are motionless, and Freud read their stillness as the reversal of the violent motion of the scene witnessed. And they stare, and Freud read their attention as the reversal of the child's own staring. The dream is a still image of animals looking, in which a scene of motion witnessed by a child is preserved by being frozen and inverted. It is, avant la lettre, a photogram of the primal scene, and its subjects are beasts.

Penley's reading of time travel as primal-scene fantasy can now be completed. The time traveller, she argued, is the subject who fulfils the fantasy of being present at his own origin, witnessing the scene of which he is the product. *La Jetée* and *Twelve Monkeys* transform the fantasy decisively, for the scene the child witnesses is the scene not of his conception but of his death. This is a thanatic primal scene, and its figures are disposed with a precision Freud would have recognised: a man and a woman before the eyes of a child, the classical triangle. But the man who falls is the child himself, grown, and the woman who kneels over him in the posture of the Pietà is both his lover and, in the iconography of the scene, the mother mourning her son. The Oedipal triangle collapses onto a single subject. The child witnesses the removal of the man who stands in the father's place and the mother's grief over him, and the man removed is the child. Heinlein's fantasy of self-begetting, in which one body is its own mother and father, finds here its thanatic counterpart: one life that is its own witness and its own victim.

Lacan's eleventh seminar (1964) supplies the terms in which this repetition can be distinguished from mere recurrence. Borrowing two words from Aristotle's account of chance in the *Physics*, Lacan opposed *automaton*, the insistence of signs, the network of signifiers that returns and returns, to *tuché*, the encounter with the real, which is always missed and which is defined as what returns to the same place. His paradigm is a dream reported in *The Interpretation of Dreams*. A father, exhausted by his vigil beside his dead son, falls asleep in the next room and dreams that the child stands beside his bed, takes his arm, and whispers reproachfully: "Father, can't you see I'm burning?" He wakes to find that a candle has fallen and set the shroud alight. The

dream contains the real that wakes the dreamer, and the encounter with it is missed in the very instant it occurs. *Twelve Monkeys* is organised by the same distinction. The stencilled monkeys, the graffiti, the recordings, the voicemail, the photographs are the *automaton* of the film: signs that insist, return, and mislead. The airport is the *tuché*: the real that returns to the same place, and that Cole meets, as Lacan's dreamer meets the burning child, at the instant it consumes him.

Gilles Deleuze's *Logic of Sense* (1969) offers the most powerful philosophical account of what this means for the subject who undergoes it. Deleuze distinguished two readings of time drawn from the Stoics. Chronos is the time of bodies, a living present that contains past and future within it. Aion is the time of events: an instant without thickness that divides endlessly into past and future, so that the event is always what has just happened and what is about to happen, never what is happening. Cole's death is an event in Aion. It is always already past, since the boy has seen it, and always still to come, since the man moves toward it; it is never present until the moment it takes him, and then it is no longer his. The twenty-first series of the book, "Of the Event," draws an ethics from this structure: to become worthy of what happens to us, to will and release the event, to become the offspring of one's own events rather than of one's deeds. Its emblem is the poet Joë Bousquet, paralysed by a German bullet in 1918, who wrote that his wound existed before him and that he was born to embody it. No sentence describes Cole more exactly. His wound, the bullet on the concourse, existed before him in the memory of the child he was; he was born, quite literally, to embody it.

Deleuze's *Difference and Repetition* (1968) carries the analysis a step further. Its third synthesis of time is the pure and empty form of time, for which Deleuze takes Hamlet's cry as the formula: time out of joint, no longer subordinated to the cycles of nature or to the movements it measures. Time so conceived is ordered by a caesura, an act too great for the agent, which divides it into a before, in which the hero is unequal to the act; a during, in which he becomes equal to it; and an after, in which the act shatters him and what returns is not he but something new. Deleuze's dramatic models are Oedipus and Hamlet, and *Twelve Monkeys* adds a third. Before the caesura, Cole is unequal to his fate: he wishes to be mad, to stay, to escape. In the caesura, at the airport, he becomes equal to it and runs toward Peters. After it he is destroyed, and what remains is not Cole but the boy, the bearer of an image he does not understand—the new, the future that the eternal return selects. The return in *Twelve Monkeys* is not the return of the same man. It is the return of a child who will become him and who, for the length of a life, has not yet.

3.6
Revenge, Revision, Re-vision

The Freedom of the Condemned

The metaphysics of *Twelve Monkeys* can now be stated without ambiguity. The film's universe is a block in Minkowski's sense, containing closed timelike curves of the kind Gödel described, governed by the self-consistency principle Novikov proposed, inhabited by travellers whose personal time folds across external time as Lewis analysed, and defended against paradox by commonplace reasons that the film identifies with the routines of institutions. Within this universe nothing that has happened can be made not to have happened, and every attempt to prevent the catastrophe is woven into its occurrence. The freedom of its agents is real in the sense Boethius and Lewis defended—their choices are their own, though their outcomes are known—but it is a freedom without the power of revision.

Three relations to this fixed past have emerged in the preceding pages. The first is revenge in Zarathustra's sense, which the film assigns to the scientists: the will that cannot will backward seeks compensation, and compensation, institutionalised, becomes insurance. The second is revision, which the film assigns to Cole and Railly in their final hours: the attempt to make the *it was* not have been, which the closed curve absorbs as a glancing blow. The third is re-vision, the transformed look by which the fixed past is received, affirmed, or simply seen whole. The obvious objection is that re-vision is not freedom at all but its consolation—the philosophy of the prisoner, Boethius in his cell, or the quietism of the Tralfamadorians, who look away from the bad moments because nothing can be done about them. If the past is fixed, is not the freedom to see it differently merely the freedom to feel differently about one's chains?

The objection has force, and the film does not dismiss it. But re-vision as the film stages it differs from Tralfamadorian quietism in a decisive respect: it does not choose its moments. The Tralfamadorians look away from the destruction of their universe; Railly looks at the boy at the instant of Cole's death, at the worst moment of the film, the moment the dream has been circling since the opening shot. Re-vision is not the selection of pleasant images from the block. It is the transformation of the most painful image by the look that consents to hold it. In Kierkegaard's terms it is repetition rather than recollection, since it faces forward, toward the child's life, rather than backward toward the loss. In Deleuze's, it is counter-actualisation: the event is not undone but lived as event, as sense. In Rich's, it is survival: entering an old text from a new direction because the old reading has become unlivable.

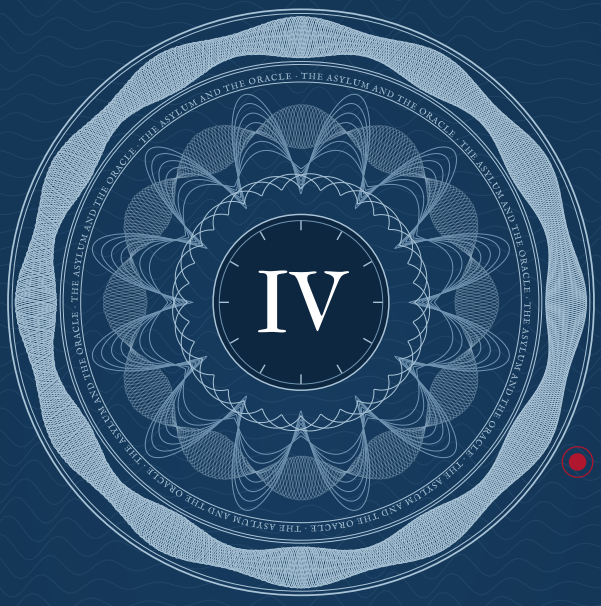
Ted Chiang's linguist offers a second answer. Knowing that her daughter will die, she chooses the life that contains the death, and her choosing is not resignation but performance, in the sense the heptapods give the word: language used not to inform but to enact. Freedom in a known world is the freedom to perform the known with full intention. Heidegger called this *Wiederholung*, the explicit taking-over of an inherited possibility. Cole approaches this freedom at the airport when, as the scene of his dream assembles itself around him, he acts anyway; Railly approaches it when she lifts her eyes. Neither changes what happens. Both change the mode in which it happens, from fate suffered to fate assumed.

The third answer is the most important, because it concerns the one figure addressed by the film who stands outside its block: the spectator. *Twelve Monkeys* is a fiction, and its fixed past is fictional. Its plague did not break out in 1996; its audience, in the winter of 1995 and 1996, watched a prophecy whose fulfilment was not, for them, fixed at all. Herein lies the deepest sense of the film's address. Within the fiction, the Cassandra Policy guarantees that the warning will fail. Outside the fiction, the warning is delivered to an audience whose future is open, and whose disbelief no timeline guarantees. The film adopts the posture that Jean-Pierre Dupuy would later call enlightened doomsaying: it presents the catastrophe as fixed, as already having happened, in order to make it credible enough to be averted. Its fatalism is a rhetorical instrument addressed to a world that is not fatalistic. The closed curve of the fiction is the lever with which the film attempts to pry open the curve of history.

This double address also clarifies what becomes of Railly after the last frame. The film leaves her on the concourse, holding knowledge no one around her possesses: that the man who carries the virus has boarded his plane, that the plague will come, that the boy who has just looked at her will live through it and become the man at her feet. She is now what Cole was—a woman who knows the future and has no one to tell it to who would believe her. The psychiatrist who lectured on the Cassandra complex has become its subject. The loop that closes on Cole opens, for her, onto a life of prophecy without audience, and it is to that condition, and to the institution that exists to manage it, that the next chapter turns.

Borges ended his "New Refutation of Time" (1947) by withdrawing it. Having argued with every resource of idealist philosophy that time does not exist, he conceded that time is the very substance of which he is made—a river, a tiger, a fire that carries him off, devours him, and consumes him, and that he himself is; the world, he concluded with mock regret, is real, and so, regrettably, is Borges. The passage contains in miniature the argument of this chapter and announces that of the chapters to come. Time is a tiger: it is an animal, and it devours. The fixed past cannot be refuted, only inhabited, and whoever inhabits it is both its prey and its substance. Before the

animal can be considered in its own right, in Chapter Five, one further institution of the fixed past must be analysed: the institution whose function, in the film, is to decide which accounts of time may be believed. A society that cannot change its past must at least control who is permitted to remember its future. That institution is the asylum, and its subject, in every century, is Cassandra.



Chapter Four

The Asylum and the Oracle

Madness, Prophecy, and the Clinical Manufacture of Disbelief

Much Madness is divinest Sense –
To a discerning Eye –
Much Sense – the starkest Madness –
'Tis the Majority
In this, as all, prevail –
Assent – and you are sane –
Demur – you're straightway dangerous –
And handled with a Chain –

— Emily Dickinson (c. 1862; published 1890)

They called me vagabond, beggar-priestess, wretch half-dead with hunger—and I endured it.

— Cassandra, in Aeschylus, *Agamemnon* (trans. mine)

4.1
Cassandra's Afterlives

From the House of Atreus to the Lecture Hall

The Cassandra of the *Agamemnon* is silent for most of the scene in which she first appears. Brought to Argos in the king's chariot as his share of the spoils of Troy—a royal captive, a prophetess, a concubine—she sits mute while Agamemnon enters the palace over the purple tapestries, and she remains mute when Clytemnestra returns to command her inside. The queen's exasperation produces one of the most suggestive images in the play. The foreign woman, she says, seems to need an interpreter, and has the manner of a wild beast newly caught, which does not know how to bear the bit until it has foamed its strength away in blood. The first figure of the prophetess in European drama is a captured animal. Her silence is the silence of the creature in the cage, and the language the queen proposes to break her to is the language of the household that will kill her.

When she speaks, she speaks to no one present. She cries out to Apollo, god of the ways, her destroyer, and the vision that seizes her runs in two directions at once. She sees the past of the house of Atreus—the children of Thyestes slaughtered and served to their father, the blood soaked into the walls—and she sees its future: the bath, the net, the blade, her own death beside the king. The chorus of Argive elders responds with an asymmetry that is the whole epistemology of the Cassandra myth. The prophecy of the past they understand and confirm, marvelling that a foreigner should know the history of a city she has never seen. The prophecy of the future they cannot follow: they hear it, they say, and are thrown from the track. Knowledge of what has been is credited as knowledge; knowledge of what will be is received as madness. The same mind, speaking with the same authority, is believed about the past and disbelieved about the future, and nothing distinguishes the two but the tense.

Cassandra herself explains the cause. Apollo desired her and gave her prophecy; she promised herself to him and broke her promise; and from that moment she persuaded no one of anything. The myth's founding transaction is a default. Cassandra contracted with the god and failed to perform, and the god's penalty attaches not to her knowledge, which he leaves intact, but to its credibility. Read in the vocabulary of Chapter One, the curse is a clause in a broken contract: the god, like an insurer, does not revoke the policy; he voids the claim. Cassandra's next gesture anticipates Cole's. She tears off the insignia of her office, the staff and the prophetic fillets, and tramples them, as Cole will tear from his jaw the tooth through which the future listens to

him. The renunciation changes nothing. The prophecy remains true, and its bearer walks into the house knowing what waits there.

Two further moments of the scene bear directly on this dissertation. When the chorus likens Cassandra's lament to the song of the nightingale, endlessly mourning Itys, Cassandra answers that the nightingale's lot is to be envied: the gods clothed her in a winged body and gave her a sweet life without tears, whereas what waits for Cassandra is the cleaving blade. The prophetess envies the bird its metamorphosis; the animal is the escape the human seer is denied. And her last words before entering the palace are a theory of the image. Alas for human affairs, she says: in prosperity a shadow can overturn them, and in misfortune a wet sponge, with a single stroke, wipes out the picture. The Greek word is *graphē*, a drawing or painting. The most famous prophetic scene in Greek tragedy ends by likening a human life to an image that can be erased—which is to say, by announcing the ontology of the photograph that Chapter Two traced from Bazin to Marker.

Later antiquity multiplied the figure without softening it. In Euripides' *Trojan Women* (415 BCE), Cassandra bursts onto the stage brandishing torches and singing a wedding hymn for her own enslavement to Agamemnon, a parody of nuptial joy that her mother takes for madness; yet within the frenzy she delivers the most lucid argument in the play, that the Trojans who died defending their homes are more fortunate than the Greeks who conquered them, and she foretells Odysseus's ten years of wandering and Agamemnon's murder. Euripides' Cassandra is a Bacchant of Apollo, Dionysian in manner and Apollonian in content. Lycophron's *Alexandra*, a Hellenistic poem of legendary obscurity, takes the form of a report delivered to Priam by the guard he has set over his daughter: her prophecy reaches the king only as the transcript of a confined woman's ravings, recited by her keeper. The opening intertitles of *Twelve Monkeys*, which attribute the end of the world to interviews with a diagnosed schizophrenic in a Baltimore hospital, restore Lycophron's frame. And Virgil, in the second book of the *Aeneid*, fixed the iconography for two millennia. Cassandra, never believed by the Trojans at the god's command, opens her lips to the fates as the horse is dragged into the city; then, as Troy falls, she is dragged from Minerva's sanctuary with her hair unbound, raising her burning eyes to heaven in vain—her eyes, since chains held her tender hands. The prophetess restrained, her gaze the only part of her still free: it is the posture in which *Twelve Monkeys* first presents Cole, strapped and sedated, his eyes following the doctors who will not believe him.

The twentieth century re-visioned the myth in Rich's sense. Christa Wolf's *Kassandra* (1983), written in the German Democratic Republic as the missiles of the late Cold War were being deployed on both sides of the Wall, opens before the Lion Gate of Mycenae, where the stone lions, now headless, once looked down on the captive

prophetess on the day of her death. The novel is Cassandra's interior monologue in the hours before she passes through that gate: a recollection of the war, of her father's court, of the security apparatus that grew up inside Troy to manage the war, and of her long refusal to speak its language. Wolf's Cassandra is not cursed by a god; she is silenced by a polity, and her prophecy is not a supernatural gift but the lucidity of someone who will not lie. Melanie Klein, in an essay on the *Oresteia* published posthumously in 1963, read the trilogy as a drama of the inner world and found in Cassandra a figure of knowledge the self cannot bear to credit, internal reality foretold and refused. And in 1988 the Jungian analyst Laurie Layton Schapira gave the clinic its term, *the Cassandra complex*, describing women whose accurate intuitions of disaster go unheard by an Apollonian order of masculine reason, and who suffer, in consequence, a characteristic form of hysterical despair.

It is this clinical Cassandra on whom Railly lectures in 1996, and the film's treatment of her lecture is precise. The term she deploys has been severed from the one fact on which the myth turns: that Cassandra was right. In the clinic, the Cassandra complex names the pathology of those who believe in catastrophes that will not come; the apocalyptic prophet is a patient by definition, since the content of the prophecy is taken as proof of the disorder. The diagnostic category thus performs, in the vocabulary of science, the very operation Apollo performed in the vocabulary of myth: it detaches knowledge from credibility, so that the prophet's certainty becomes evidence against him. Among her examples Railly projects a photograph of a wounded soldier from 1917; in the audience, and at the signing that follows, is Dr. Peters, who seizes the occasion to suggest that the doomsayers may be the sane ones and the species itself the disease. The scene assembles in a single room the three positions of the myth: the clinician who has made Cassandra a syndrome, the image of the prophet she does not yet recognise, and the man who will make the prophecy come true. The rest of the film restores the myth against the clinic, step by step, until the lecturer on the Cassandra complex has become its instance.

4.2
The Ship, the Cage, the Ward

Foucault, Goffman, and the Institution of Unreason

Michel Foucault's *History of Madness* (1961) opens with a ship. Sebastian Brant's *Narrenschiff* (1494) and Hieronymus Bosch's painting of a boat crowded with revellers adrift give Foucault his emblem for the late medieval and Renaissance experi-

ence of madness, in which the mad were not confined but set wandering, expelled from cities and entrusted to boatmen and pilgrims, liminal figures belonging to the water and the road. In this experience, Foucault argued, madness carried a tragic knowledge. At the close of the fifteenth century it took over from death the office of figuring the nothingness at the heart of existence; the fool's laughter anticipated the laughter of the skull; and the visions of Bosch, of Bruegel, of Dürer's horsemen bound madness to knowledge of the end of the world. The madman knew the apocalypse. Against this tragic experience Foucault set a critical consciousness of madness, exemplified by Erasmus's *Praise of Folly* (1511), which turned folly into a moral category, a failing to be satirised rather than an abyss to be feared. Cole, in the hospital of 1990, is a figure of the first experience misread by the second: a man who knows the end of the world, received by an institution whose only categories for that knowledge are error and disorder.

The classical age, in Foucault's narrative, ended the wandering. The founding of the Hôpital Général in Paris in 1656 inaugurated what he called the Great Confinement, in which the mad were locked away together with the poor, the idle, the debauched, and the blasphemous, under the single sign of unreason. And unreason, in its extreme form, was understood as animality. The mad were chained in cells like beasts; their supposed insensibility to cold and hunger was taken as proof that the animal had broken through the human; and at Bethlem in London visitors paid to watch them on holidays, as they would pay to see a menagerie. The asylum of the classical age was a zoo. *Twelve Monkeys* reassembles this archaeology with startling literalness. Its three institutions of confinement are one: the cages of 2035, stacked in a vertical prison, where convicts are kept like specimens and hosed down like stock; the ward of 1990, with its restraints and its locked seclusion room; and the zoo of 1996, whose gates the Army of the Twelve Monkeys opens. The film moves its hero from cage to ward and its animals from cage to street, and it never lets the viewer forget that the three spaces share a grammar.

The legend of Philippe Pinel striking the chains from the madmen of Bicêtre in 1793, and the York Retreat that William Tuke founded in 1796, are for Foucault not a liberation but a transfer: the chains of the classical age were replaced by the gaze, the judgment, and the silence of the moral asylum, in which the patient was brought to internalise a surveillance formerly applied from without. The asylum became a juridical microcosm, a tribunal in permanent session. *Twelve Monkeys* stages two such tribunals and invites their comparison. In 1990 Cole faces a panel of psychiatrists who question him about his beliefs and record his answers as symptoms. In 2035 he faces a panel of scientists who question him about what he has seen and record his answers as data. The first tribunal decides that he is mad, the second that he is useful; the first

confines him in order to cure him, the second releases him in order to use him. Both interrogate a man whose testimony concerns the end of the world, and neither has any interest in whether anyone else believes him.

Erving Goffman's *Asylums* (1961), drawn from fieldwork at St. Elizabeths Hospital in Washington, supplied the sociology of these tribunals. A total institution—asylum, prison, barracks, monastery—is a place where all aspects of life are conducted in the same place, under a single authority, in the company of others treated alike, according to a schedule imposed from above. Entry into it begins with what Goffman called the mortification of the self: the stripping of clothing and possessions, the bath, the uniform, the loss of the identity kit by which a person presents himself to others. Cole's returns to 2035 follow this protocol exactly: stripped, scrubbed, hosed, examined, caged. Goffman also identified a mechanism that could serve as a gloss on the film's entire first act. He called it *looping*. In a total institution, the inmate's defensive response to an assault on his self is taken up by the institution as evidence and becomes the target of its next intervention: the patient who protests his sanity demonstrates his lack of insight, the prisoner who resists demonstrates his dangerousness. Cole's insistence that he is not mad, that the world will end, that he must be released to complete a mission, is looped back into the record as symptom. Goffman's term anticipates both Hacking's looping kinds and the film's closed timeline, and the coincidence is more than verbal. In each case the subject's attempt to act upon his situation is converted, by the structure that contains him, into a confirmation of that structure.

The antipsychiatric critique radicalised these descriptions. Thomas Szasz argued in *The Myth of Mental Illness* (1961) that mental illness is a metaphor, that what psychiatry calls disease are problems in living, and in *The Manufacture of Madness* (1970) he compared the institutional psychiatry of his time to the Inquisition and the madman to the witch. Railly herself, late in the film, concedes something close to Szasz's thesis when she remarks that psychiatry has become the latest religion, the authority that decides what is right and wrong and who is mad. R. D. Laing went further. In *The Divided Self* (1960) and *The Politics of Experience* (1967) he treated psychosis as a strategy invented in order to live in an unlivable situation, and madness as potential breakthrough as much as breakdown. In the preface to a 1965 edition of *The Divided Self*, Laing recalled a girl of seventeen in a mental hospital who was terrified because, she said, the atom bomb was inside her; that was a delusion, he wrote, but the statesmen who boasted of their doomsday weapons were far more dangerous, and far more estranged from reality, than many of those to whom the label of psychosis was affixed. Laing's patient is Cole's ancestor and his inverse. She carried the apocalypse inside her as a delusion; he carries it as a memory. And the real bearer of the apocalypse in the

film is neither the patient nor the statesman but a laboratory assistant who passes for perfectly sane.

The popular image of the asylum in the decades before *Twelve Monkeys* was shaped above all by Ken Kesey's *One Flew Over the Cuckoo's Nest* and Miloš Forman's film of it (1975). Its hero, Randle McMurphy, is a convict who contrives his transfer from a prison work farm to a mental hospital in the belief that the ward will be easier time. He discovers that it is not, that the hospital is the place where the system Kesey's narrator calls the Combine is most nakedly visible, and he ends lobotomised, a body from which the self has been removed. Cole's trajectory runs parallel: a convict who arrives in the asylum by error, who discovers that the ward is merely the prison by other means, and who leaves it only to be returned to the cage. Frederick Wiseman's documentary *Titicut Follies* (1967), filmed inside the Massachusetts institution for the criminally insane at Bridgewater and withheld from general distribution by court order for more than two decades, had shown what Kesey's fable dramatised: naked men in bare cells, taunted by guards, force-fed, interrogated. The asylum of *Twelve Monkeys*, shot in part within the walls of a nineteenth-century penitentiary, belongs to this lineage of images. It is not a hospital that has failed to be humane. It is a prison that has succeeded in calling itself a hospital.

4.3
The Physician Who Was Right

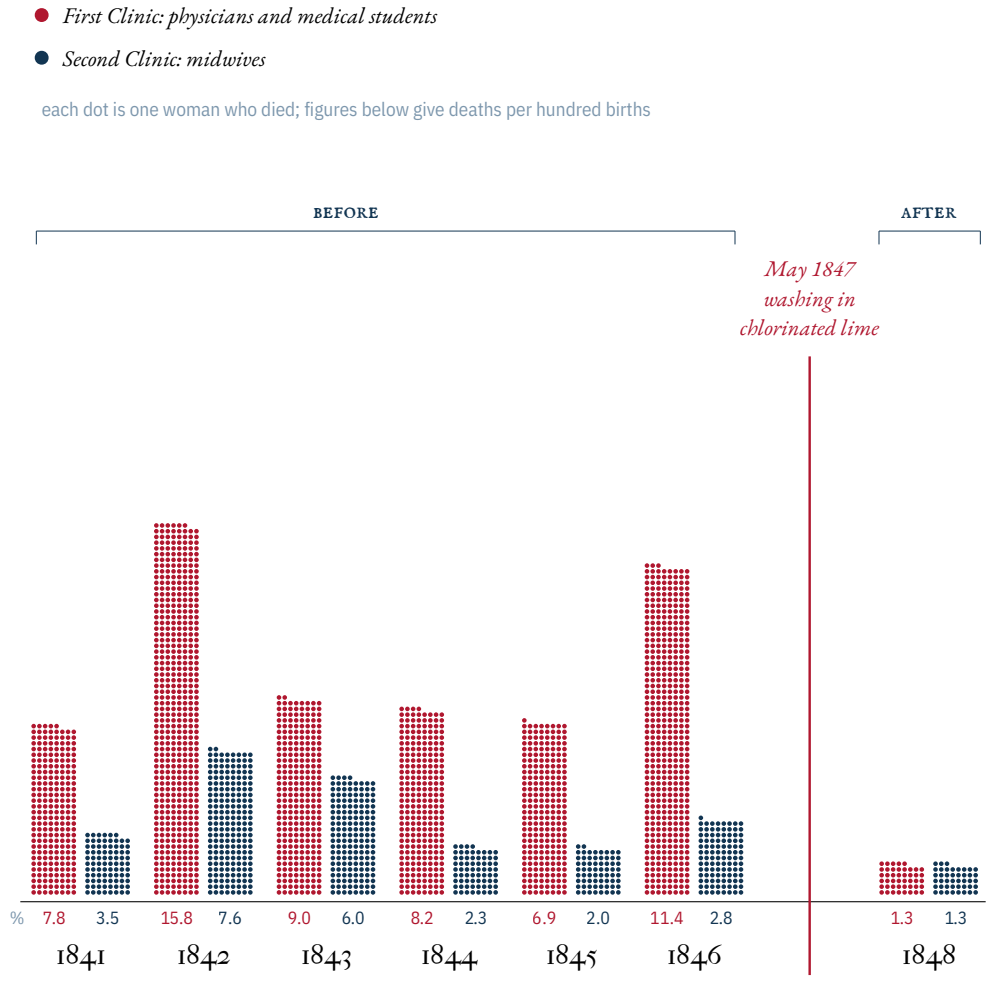
Semmelweis, Fleck, and the Science of Disbelief

In the day room of the 1990 hospital, between television commercials and tranquillisers, Jeffrey Goines gives Cole a lesson in the history of medicine. Once, he explains, nobody believed in germs; then a doctor tried to convince his colleagues that tiny invisible things were killing people, and they decided he was crazy. Jeffrey places the doctor in the eighteenth century, and the error is itself significant: the madman's chronology is unhinged, but his parable is exact. He is speaking of Ignaz Semmelweis, and the story he tells is the founding legend of the scientist as Cassandra.

In 1846 Semmelweis, a young Hungarian obstetrician at the Vienna General Hospital, confronted a statistical scandal. Of the hospital's two maternity clinics, the one staffed by physicians and medical students had a mortality from puerperal fever several times that of the one staffed by midwives. Women begged to be admitted to the second; some preferred to give birth in the street. In 1847 Semmelweis's colleague Jakob Kolletschka died of a fever resembling the mothers' after his finger was cut dur-

Plate 6

One dot for each mother



Maternal deaths in the two maternity clinics of the Vienna General Hospital, 1841–1846, and in 1848, the first full year of washing with chlorinated lime. Each dot is one woman; the figures beneath give deaths per hundred births. In 1848, for the first time, the First Clinic's rate fell below the Second's. Data: Semmelweis, *Die Ätiologie, der Begriff und die Prophylaxis des Kindbettfiebers* (1861).

ing an autopsy, and Semmelweis drew the conclusion: the physicians and students were carrying what he called cadaverous particles from the dissecting room to the delivery ward on their hands. He imposed washing in a solution of chlorinated lime, and mortality in the first clinic fell from around one woman in ten to one or two in a hundred. The establishment did not believe him. His explanation preceded the germ theory that would have made it intelligible; his findings insulted the physicians whose hands they accused; and his own temperament, increasingly embittered, culminated in open letters denouncing his opponents as murderers. In 1865 he was committed to an asylum near Vienna, where he died within two weeks, apparently of an infection from a wound sustained there, possibly in a beating by attendants. The physician who had discovered how infection travels died of infection, in a madhouse, disbelieved.

Semmelweis's story enters *Twelve Monkeys* with a double charge. It is, first, the story of Cole: a man in possession of a truth about invisible agents of mass death, whose evidence is real and whose explanation outruns the conceptual means of his hearers, and who is therefore classified as mad. But it is also the story of Peters. Semmelweis's discovery was not only that disease was carried by invisible particles; it was that the carriers were the physicians themselves. Death passed from the morgue to the maternity ward on the hands of the men whose profession was to heal. In *Twelve Monkeys* death passes from the laboratory to the airport, and thence to the cities of the world, in the case of a man whose profession is the study of disease. Jeffrey's parable thus tells both halves of the film's secret at once, to the man who will need the first half and will not understand the second: that the prophet will be thought mad, and that the plague will come from the clinic.

The history of science has since given Semmelweis's fate a theory. Thomas Kuhn's *The Structure of Scientific Revolutions* (1962) described how normal science resists anomalies until they accumulate into crisis, and Kuhn acknowledged his debt to Ludwik Fleck's *Genesis and Development of a Scientific Fact* (1935), which argued that what counts as a fact is determined by a thought collective and its thought style, so that a discovery that cannot be expressed within the prevailing style is literally unseeable to those who share it. Jeffrey's relativism—no right, no wrong, only popular opinion—is a degraded version of Fleck's epistemology, and the film refutes it in the most brutal way available: the virus is indifferent to the thought collective. Fleck's own life gives the constellation its darkest star. A Jewish microbiologist from Lwów, he was deported in 1943 and compelled to work in the typhus-vaccine laboratory the SS established at Buchenwald, where, according to survivors, the prisoner-scientists supplied the SS with a worthless preparation while reserving the effective one for fellow prisoners. The camp laboratory, in which prisoners are forced to labour at the science of epidemics for masters who would destroy them, is the historical form of the

fable shared by *La Jetée*, with its experimenters beneath Chaillot, and *Twelve Monkeys*, with its convict volunteers. Fleck's saboteurs are the only figures in this archive who turn the knowledge extracted from them against those who extract it.

The cult of Semmelweis has its own ambiguous history. In 1924 a young physician named Louis-Ferdinand Destouches submitted as his medical thesis a lyrical life of the Hungarian, portraying him as a martyr crucified by the stupidity of his colleagues. Eight years later, as Céline, Destouches published *Voyage au bout de la nuit* (1932), a novel that begins in the trenches of the First World War and descends through colonial Africa and the industrial suburbs into a nihilism without remainder; within a few more years he was writing pamphlets of murderous antisemitism. The trajectory is instructive for a reading of *Twelve Monkeys*. The cult of the unheeded prophet can curdle by degrees into contempt for those who did not heed him, and contempt for the deaf can become the wish to see them destroyed. Dr. Peters, whom Chapter Six will read as an angel of Malthus, is the terminus of this curdling: a man who has concluded that the prophets of doom are right and that the doom should therefore be administered. The film sets him beside Cole as the Semmelweis who chose to become the plague.

4.4
Fleetingly Improvised Men

Apocalyptic Delusion and the Counter-Archive of True Madness

The clinic possesses a canonical case of apocalyptic delusion, and its features illuminate what Cole's psychiatrists believe they are seeing. Daniel Paul Schreber, presiding judge of a Saxon appeal court, published in 1903 his *Memoirs of My Nervous Illness*, the most detailed first-person account of psychosis ever written. Among its convictions is that the world has ended: a catastrophe has destroyed humanity, and the people Schreber sees around him in the asylum are not real persons but what he calls fleetingly improvised men, figures miraculously set up for a moment and then dissolved. Freud, in his analysis of the *Memoirs* (1911), read the end of the world as the projection of an inner catastrophe, the withdrawal of the patient's libidinal investment from the world, which therefore seems to him to have perished; and he added the sentence that transformed the clinical understanding of delusion: what we take for the pathological product, the delusional formation, is in reality an attempt at recovery, a process of reconstruction.

Cole's experience of 1996 is Schreberian in structure and true in content. He walks among people whom he knows to be, from his vantage, already dead; the crowds, the drivers, the shoppers, the voices on the radio are for him the fleetingly improvised men of a world that ended before he was grown. The difference is that the catastrophe is not inside him. What Freud read as the projection of an inner end is, in Cole, the memory of an outer one; and his delusion—the conviction that he is mad, which he adopts in the second half of the film—is his attempt at recovery, a reconstruction of a world in which the five billion need not die, purchased at the price of his own sanity. *Twelve Monkeys* inverts Freud's formula exactly. The true belief is the pathological product; the delusion is the cure.

The phenomenological psychiatry that followed refined the criteria by which delusion is recognised. Karl Jaspers, in his *General Psychopathology* (1913), characterised delusions by an extraordinary conviction, an imperviousness to experience and argument, and an impossibility of content. Klaus Conrad, in his study of beginning schizophrenia (1958), described the delusional mood that precedes the delusion proper—the stage fright he called *trema*, in which the world seems charged and portentous—and the revelation he called *apophany*, in which everything begins to signify and to refer to the self; the term survives in clinical and popular speech as apophenia, the perception of meaningful pattern in random data. Cole meets every criterion. His conviction is absolute, it resists every argument, and its content is impossible by the standards of 1990; in 1996 he lives in a state of permanent apophany, reading graffiti, radio bulletins, and the faces of strangers as signs addressed to him. Every criterion is satisfied, and the belief is true. The film thereby exposes the limit of the criteria. They were designed to identify false beliefs, but what they actually identify are beliefs about the incredible held with certainty, and a true prophecy of an unprecedented catastrophe, held by someone who has lived through it, is indistinguishable from delusion by any test the clinic possesses.

Literature had long kept a counter-archive of true, or truth-bearing, madness, and *Twelve Monkeys* draws on it at many points. Gogol's "Diary of a Madman" (1835) records the decline of a petty clerk, Poprishchin, who intercepts the correspondence of two dogs, learns from it the secrets of his superiors, and concludes that he is the King of Spain. As his madness deepens, his diary's dates come loose from the calendar, and one entry is headed, famously, the year 2000, April 43rd. The madman dates his testimony from an impossible future. His diary is the ancestor of every document—the clinical transcript of April 1990 that opens *Twelve Monkeys* among them—in which the speech of the mad is filed under a date that cannot be right.

Stoker's *Dracula* (1897) supplies a closer ancestor. R. M. Renfield, a patient in Dr. Seward's asylum, is classified by his physician as a zoophagous maniac: he catches flies,

feeds them to spiders, feeds the spiders to sparrows, and begs for a kitten, consuming lives in order to absorb their life. He is also the novel's prophet. He senses the approach of his Master before anyone else, and in a scene of perfect lucidity he asks the men assembled to destroy Dracula to release him that very night, warning that they will regret it if they refuse. They refuse, taking his sanity for a ruse, and that night the vampire enters the asylum through him. The asylum's disbelief opens the door to the plague—for *Dracula* is a plague narrative, its Count arriving by sea with his boxes of earth and his rats, and F. W. Murnau's *Nosferatu* (1922) made the identification explicit, sending the vampire to Wisborg in a ship of coffins and rats while Knock, the Renfield figure, raves in his cell that the Master is coming. Cole, who catches and eats a spider in the hospital of 1990, inherits Renfield's zoophagy and Renfield's fate: the true warning of the madman, disregarded by the men of science, as the contagion arrives.

Anton Chekhov's "Ward No. 6" (1892) supplies the counter-archive's most precise model for Railly. Its protagonist, Dr. Andrei Ragin, presides with indifference over a squalid provincial hospital whose sixth ward holds the mentally ill. He finds in one of its inmates, Ivan Gromov, the only intelligent interlocutor in the town, and the two men debate the Stoic doctrine Ragin professes—that suffering is a matter of opinion and the wise man indifferent to it—against Gromov's furious insistence that pain is real. Ragin's visits arouse suspicion; his colleagues declare him mad; and he is confined in Ward No. 6, beaten by the attendant, and dead within days. The physician who listens to the patient becomes the patient. Railly's trajectory follows Ragin's in its essentials—the clinician who comes to credit the madman finds herself a fugitive whom no one will credit—and Chekhov's debate anticipates the question Chapter Three left open, whether the Stoic consolation of the condemned is wisdom or evasion.

The twentieth century added historical Cassandras to the literary ones. Claude Eatherly piloted the weather-reconnaissance aircraft that preceded the *Enola Gay* over Hiroshima on 6 August 1945 and signalled that the skies over the city were clear. In the years that followed he committed petty crimes that seemed designed to bring punishment upon him, and he was confined repeatedly in the veterans' hospital at Waco, Texas, while the press presented him as the Hiroshima pilot who had gone mad. Between 1959 and 1961 the philosopher Günther Anders conducted a correspondence with him, published as *Burning Conscience* (1961), in which Anders argued that Eatherly's condition was not illness but the only adequate moral response to participation in an atrocity: he was the one man in the chain of command who had grasped what had been done, and a society that confined him as mad thereby declared its own incapacity to grasp it. The 1990s returned the figure to fiction. Sarah Connor, confined in

Terminator 2 (1991) to a state hospital for her knowledge of Judgment Day, is in the charge of the same psychiatrist, Dr. Silberman, who had dismissed her lover's warnings in the first film, and who is reduced to terror when he finally sees the machine she described. Parry, the homeless visionary of Gilliam's own *The Fisher King* (1991), carries a grief so total that it has become a mythology. *Twelve Monkeys* gathers these figures into a single proposition: a society that confines its prophets is protecting itself not from their errors but from their truths.

4.5
"Only Popular Opinion"

Jeffrey Goines, Consumption, and the Dionysian Decoy

Jeffrey Goines is the film's most exuberant creation and its most elaborate decoy. Brad Pitt plays him as a body in permanent overdrive—eyes rolling, hands chopping the air, sentences outrunning one another—and gives him the film's most quotable speeches, delivered in the day room of 1990 to a sedated and bewildered Cole. Their content is a theory of the society that produces madness. Look at the television, Jeffrey commands: it is all there, in the commercials. Nobody makes anything any more; production has been automated; what remains for human beings is to buy, to consume, to be good consumers. Anyone who will not or cannot play that role is, by definition, mentally ill. Madness, in Jeffrey's account, is not a disorder of the mind but the failure of a market to find a use for a person. His summary of the ethics of such a society is the line that has made him famous: there is no right and no wrong, there is "only popular opinion."

The tirade condenses, in the idiom of a rich man's son who has watched too much daytime television, the critical theory of consumer society. Herbert Marcuse's *One-Dimensional Man* (1964) described a society that integrates its members by manufacturing false needs and satisfying them, a happy consciousness in which the capacity for negation withers. Erich Fromm's *The Sane Society* (1955) proposed that whole societies can be sick, that a pathology shared by millions is not thereby normal, and that sanity cannot be defined as adjustment to an order that is itself deranged. Guy Debord's *The Society of the Spectacle* (1967) defined the spectacle as a social relation among people mediated by images, and Jean Baudrillard's *The Consumer Society* (1970) analysed consumption as a system of signs in which objects function as markers of status rather than answers to need. Each contains Jeffrey's central claim, that the boundary between the sane and the mad is drawn by the requirements of an eco-

nomic order. The film lets Jeffrey say it in the place where that boundary is enforced, and from the wrong side of it.

The ethics Jeffrey derives from his diagnosis is a relativism with an ancient pedigree. Protagoras, in Plato's *Theaetetus*, holds that man is the measure of all things, of the things that are, that they are, and of the things that are not, that they are not; Nietzsche, in a notebook of 1886–1887, wrote that there are no facts, only interpretations. *Only popular opinion* is the sophist's doctrine reduced to a slogan, and the film refutes it without argument. The virus is not a matter of opinion. It kills whether or not anyone believes in it, and its indifference to belief is the whole of Cole's tragedy. There is a further irony in the fate of the man who proclaims the sovereignty of opinion. The future's investigation, built upon a recorded rumour, convicts him: in 1996 and again in 2035 the Army of the Twelve Monkeys is taken to be the author of the plague, and Jeffrey, who held that there is no fact beyond what people think, becomes the victim of what people think about a fact.

Jeffrey is, in the Nietzschean terms of Chapter One, the film's Dionysus, and his story is a comic inversion of Euripides' *Bacchae* (405 BCE). In the tragedy the god arrives in Thebes with his band of maenads, is imprisoned by the young king Pentheus, who denies his divinity, and walks free as the palace shakes; on the mountain his followers suckle fawns and wolf cubs and tear cattle apart with their hands; and Pentheus, lured to spy on them in women's clothing, is torn to pieces by his own mother. In *Twelve Monkeys* the Dionysian son has been confined to the asylum and released, and has gathered a band of followers; his great act of liberation is to open the zoo and let the animals into the streets; and the figure of order whom he humiliates is not a king but his own father, the eminent virologist Dr. Leland Goines, whom the Army leaves locked in a cage in the place of the animals it has freed. The patriarch of reason is exhibited as a beast; the beasts walk free. Dismemberment has become a prank, and the god's revenge a stunt for the evening news.

But the *Bacchae* is reversed in a more important respect. In Euripides the Dionysian is the destructive power; Thebes is wrecked by the god it refused. In *Twelve Monkeys* the Dionysian is a decoy. The Army of the Twelve Monkeys is loud, visible, theatrical, animal-loving, apparently mad, and entirely innocent of the catastrophe. The plague comes instead from the quietest man in the film, an assistant in the eminent father's laboratory, who appears at a book signing, speaks reasonably of humanity's excesses, carries a case through airport security, and opens it when asked. The film's structure of suspicion is thus a lesson in the misrecognition of danger. Its institutions—the clinic, the police, the future's investigators—look for apocalypse where it is loud and Dionysian and overlook it where it is quiet and Apollonian. Hannah Arendt's phrase for the perpetrator who did not look like one, the banality of evil,

coined in her report on the trial of Adolf Eichmann (1963), describes Peters's camouflage if not his motive. Max Horkheimer and Theodor Adorno's *Dialectic of Enlightenment* (1944/1947), with its thesis that reason pursued without reflection upon its own ends reverts to the domination it was meant to overcome, describes the motive exactly: the rational conclusion, drawn by a man of science, that the species is the problem and its removal the solution.

Twelve Monkeys thus distributes madness across two figures who mirror each other. Jeffrey is mad in manner and harmless in act; Peters is sane in manner and catastrophic in act. The clinic, whose categories respond to manner, confines the first and lets the second pass; the police pursue the first and wave through the second; the future, consulting its archive, accuses the first and takes its seat beside the second. Between them stands Cole, mad in the eyes of every institution and right in fact, whose only allies are the one who is mad and harmless and, at last, the one who is sane and has learned to believe him. The film's anatomy of madness is therefore not the romantic thesis of the antipsychiatric decade, that the mad are secretly sane. It is the argument that the institutions charged with distinguishing madness from reason are calibrated to the wrong signals, and that a civilisation which trusts them will be destroyed by the reasonable.

4.6
The Psychiatrist's Conversion

Railly, Looping Kinds, and the Making of a Cassandra

Kathryn Railly is the only character in *Twelve Monkeys* who changes her mind, and her conversion is the film's second plot, braided through the first. In 1990 she is a young psychiatrist, sceptical and humane, who listens to Cole with more attention than her colleagues and is troubled, when he vanishes from a locked seclusion room, by an anomaly she cannot explain. In 1996 she is an established author who has made her reputation by explaining apocalyptic prophecy as pathology, and she is lecturing on the Cassandra complex when Cole abducts her. The film then submits her expertise to a sequence of proofs: the boy in the well, whose hoax Cole had foretold; the bullet she extracts from his leg, which turns out to be of First World War manufacture; the photograph of 1917, in which, under magnification, the soldier wears Cole's face. Each proof is an anomaly in Kuhn's sense, and her paradigm resists each in turn until it gives way.

Her conversion is staged as a chiasmus. As Railly comes to believe that Cole is a traveller from the future, Cole comes to believe that he is mad. The two movements cross in the middle of the film, and for a stretch of it each argues the other's former position: the psychiatrist insists on the truth of the prophecy, and the prophet insists on his own delusion. Cole's wish to be mad, which he states with desperate sincerity, is the wish that the world not end; madness, for him, would mean that the air he has learned to love, the music he cannot stop listening to, the woman he has found, are real and will remain so. Railly's acceptance of his sanity is the acceptance that they will not. At the crossing point both are, in a sense, Cassandras: he of a future he no longer believes in, she of a future she has only just begun to believe.

In the course of this crossing Railly pronounces the film's most explicit verdict on her own profession, the remark already cited, that psychiatry has become the latest religion, deciding what is right and wrong and who is mad. The thought is Szasz's in substance, but its force in the film comes from the speaker: it is the confession of a priest who has lost her faith in the middle of a rite. Georges Canguilhem, whose *The Normal and the Pathological* stands behind Foucault's histories, argued that health is not conformity to a statistical norm but normativity, the capacity to institute new norms in response to a changed milieu, and that pathology is the narrowing of that capacity. By Canguilhem's criterion Cole is not sick. He has adapted to a world that will not exist for another year, and his norms are those of a milieu his clinicians cannot see. The pathological rigidity in the film belongs to the institutions that cannot revise their categories in the face of his evidence, and Railly's conversion is, in this sense, her recovery.

Ian Hacking's work gives the most precise account of what happens to the diagnostic category in the course of this story. In "Making Up People" (1986) and "The Looping Effects of Human Kinds" (1995), Hacking argued that classifications of people differ from classifications of things because people are aware of how they are classified and change in response, so that the kind is transformed by the behaviour it describes. In *Mad Travelers* (1998) he studied a transient mental illness, the dissociative fugue that appeared in France in the 1880s with the case of Albert Dadas, a gas fitter from Bordeaux who travelled compulsively across Europe without knowing how he had come to be where he found himself; the diagnosis flourished, Hacking argued, in an ecological niche between the romance of tourism and the criminalisation of vagrancy. The fugueur is a mad traveller, a man who appears in places without knowing how he arrived, and the clinic of 1990 has precisely this category ready for the naked stranger found disoriented in a Baltimore street, insisting on the wrong year. *Twelve Monkeys* tests what happens when a kind meets an instance that does not belong to it. The time traveller is filed as fugueur and schizophrenic; he responds to the filing by

coming to believe it; and the clinician who filed him responds to his evidence by abandoning it. The category loops through both of them and returns transformed. At the end of the film the Cassandra complex no longer names a delusion about the end of the world. It names the condition of the one person on the concourse who knows the truth and has no one to tell.

Comparison with Cameron's films sharpens the point. Dr. Silberman of *The Terminator* and *Terminator 2* is the psychiatrist as unconvertible institution. He diagnoses Kyle Reese in 1984 and Sarah Connor in 1991 with the same professional complacency, and when the evidence of the machine at last confronts him he does not revise his categories; he flees in terror, and the film abandons him. Railly is his opposite. She is converted by the evidence, she acts upon it, and she pays for it: the police treat her as a hostage who has gone over to her captor, and at the airport she becomes what she had diagnosed. Cameron's psychiatrist is the institution that cannot learn. Gilliam's is the person who can, and who discovers that learning has made her an exile from the institution that taught her.

The function of the asylum in *Twelve Monkeys* can now be stated in the terms of this dissertation's argument. The clinic is the past's instrument of the Cassandra Policy. It does not need to know that Cole is right in order to guarantee that he will not be believed; its categories guarantee it automatically, since a true prophecy of the incredible satisfies every criterion of delusion. The diagnosis, the transcript, the sedative, the restraint, and the review panel are the commonplace reasons through which, in Lewis's terms, the closed timeline secures its consistency. They stand to the timeline as the exclusion clause stands to the insurance policy: the mechanism that ensures no claim is paid before the loss. The future that sends Cole and the clinic that receives him are thus allies without knowing it. The one requires his disbelief, and the other manufactures it. Between them, in every century the film visits, stands Cassandra—the captive beast of Clytemnestra's simile, the Bethlem lunatic on holiday display, the convict hosed down in his cage. The next chapter takes the animal at its word.



Chapter Five

The Thirteenth Monkey

Zoetropic Time, the Animal Gaze, and the Anthropological Machine

And what rough beast, its hour come round at last,
Slouches towards Bethlehem to be born?
— W. B. Yeats, "The Second Coming" (1920)

No, freedom was not what I wanted. Only a way out: right, left, wherever; I made no other demand.
— Kafka, "A Report to an Academy" (1917; trans. mine)

5.1
The Ring of Twelve

Zodiac, Clock, Zoetrope, Chorus

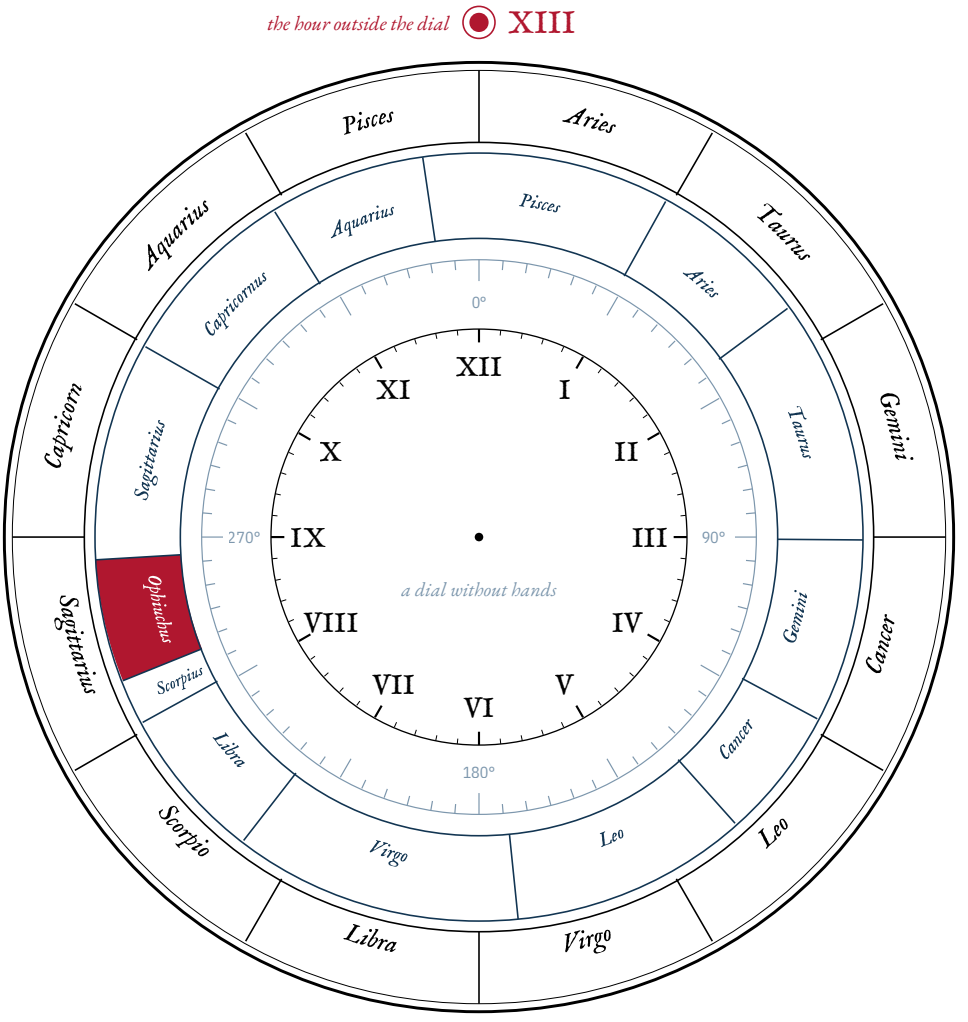
The emblem that Cole pursues across two centuries is a ring of red monkeys stencilled on walls—on the ruined facades of 2035, on the brickwork of 1996—above the name of an army that turns out to be a band of pranksters. The film never explains the number. Its makers may have chosen it for its sound, and the band's leader for its menace. But a ring of twelve animals is among the most heavily freighted figures in the history of human symbolism, and the film, whether or not it intends the freight, carries it. This section unpacks four of its meanings—zodiac, clock, zoetrope, chorus—each of which converges on the dissertation's central claim: that in *Twelve Monkeys* time itself is figured as a ring of beasts, and the human as the figure that stands outside the ring, counting.

The first meaning is the zodiac. The Greek *zōidiakos kyklos* means the circle of little figures, the diminutive *zōidion* designating both a small animal and a small image, and the band of twelve constellations through which the sun passes in the course of a year is predominantly a procession of beasts: ram, bull, crab, lion, scorpion, goat, fish. The Babylonian astronomers who divided the ecliptic into twelve equal signs, by the fifth century BCE, made the circle of animals the first clock of the heavens, the dial on which the year is read. In China an independent system assigns the years of a twelve-year cycle to twelve animals, the monkey ninth among them, so that the monkey is, quite literally, one of the twelve figures of time. And the almanacs of early modern Europe and colonial America reproduced the figure of the Man of Signs, a naked human body over which the twelve zodiacal creatures are distributed from the ram at the head to the fish at the feet, each governing the organ beneath it: the human body as a map of animal time. The ring of twelve monkeys is a zodiac, a dial on which animals tell the hour.

The second meaning is the clock. The Egyptians and Babylonians divided day and night into twelve hours each, and the dial on which the modern world reads its time is a ring of twelve. *Twelve Monkeys* is a film obsessed with dates and calibration: its scientists are forever missing their targets by years, depositing Cole in 1990 instead of 1996 and in 1917 instead of either; its intertitles fix the end of the world to a calendar; its loop closes at a precise hour on a precise day. The ring of monkeys is a clock face from which the hands have been removed, a dial of animals whose hour is not yet indicated. Orwell opened *Nineteen Eighty-Four* on an April day on which the clocks strike thirteen, the hour that does not exist on the dial, the hour of a time out of joint.

Plate 7

The ring of twelve, and the thirteenth



outer ring: the twelve signs, equal arcs of thirty degrees

inner ring: the thirteen constellations the ecliptic crosses, at their IAU boundaries

Ophiuchus, the serpent-bearer, identified with Asclepius

longitudes measured from the March equinox; values c. 2020

The zodiac as a dial. Outer ring: the twelve signs, equal arcs of thirty degrees. Inner ring: the thirteen constellations that the sun actually crosses, at the boundaries fixed by the International Astronomical Union in the 1930s, Ophiuchus the serpent-bearer among them. At the centre, a clock face without hands; outside it, the thirteenth stroke.

The thirteenth stroke is the place of the human in the ring: the hour the dial presupposes and excludes.

The third meaning is the zoetrope, defined in Chapter One: a ring of still images of a living creature which, set turning, produces the appearance of life. Chapter Two showed that the protocinematic image par excellence, the plate of Marey's photographic rifle, was a ring of twelve pictures of a bird in flight. The Army's emblem is a strip from a zoetrope, a ring of animal figures that would appear to move if spun. Its temporal logic is the film's. From within the ring each monkey seems to advance; from outside, the ring returns. The animals of the emblem are figures in a loop, and the loop is the shape of the film.

The fourth meaning is the most surprising and the most exact. Ancient tradition held that the tragic chorus of Aeschylus numbered twelve and that Sophocles raised it to fifteen, and the *Agamemnon* itself appears to preserve the number. After Cassandra has entered the palace and the death cries of the king are heard from within, the chorus breaks, for the only time in the play, into individual voices, and twelve speakers in succession deliver a couplet each: we must break in; we must summon the citizens; we must wait and be certain; words cannot raise the dead. They debate, they hesitate, and they do nothing. Twelve voices that have heard Cassandra and disbelieved her, and that hear the murder and cannot act: this is the chorus of tragedy at the moment of its most famous paralysis. The Army of the Twelve Monkeys is that chorus restaged as a satyr play. It makes noise, paints its emblem, opens the zoo, fills the city with spectacle, and neither causes nor prevents the catastrophe that is taking place, quietly, elsewhere.

Nietzsche, who traced tragedy to the satyr chorus, cited with approval Schiller's description of the tragic chorus as a living wall that tragedy draws around itself in order to close itself off from the real world and preserve its ideal ground. The phrase defines the function of the Army in *Twelve Monkeys* with uncanny precision. Jeffrey's band is the living wall of the plot: the noisy, visible, half-animal chorus that occupies the foreground of the investigation and screens from view the real world in which a laboratory assistant packs his case. The satyrs of Attic drama were men costumed as goat-men, in whom the human and the animal had not yet been divided; Jeffrey's followers, who march under a ring of beasts, are their heirs. The ring of twelve monkeys is thus at once the zodiac in which time is read, the dial on which it is counted, the zoetrope in which it loops, and the chorus behind which its catastrophe is hidden. In each of these figures the human appears as the term outside the ring: the reader of the zodiac, the counter of the hours, the spectator at the zoetrope's slit, the protagonist whom the chorus watches and cannot save. That term is what this dissertation calls the thirteenth monkey.

5.2

Smintheus

The Mouse-God, the Physician's Oath, and the Plague That Begins with Animals

The *Iliad* begins with a plague, and the plague begins with animals. Chryses, priest of Apollo, comes to the Achaean camp to ransom his daughter; Agamemnon dismisses him with insults; and the old man, walking by the shore of the loud-roaring sea, prays to the god under a cult title, Smintheus. Apollo descends from Olympus like the night, his arrows clattering on his shoulders, sits apart from the ships, and lets fly. First he strikes the mules and the swift dogs; then he turns his arrows upon the men; and for nine days the pyres of the dead burn thick. The first plague in European literature is zoonotic in structure: it passes from the animals to the humans. And the god who sends it bears an epithet that ancient commentators connected with *sminthos*, a mouse. Strabo reports that at the sanctuary of Apollo Smintheus in the Troad the cult statue, a work of Scopas, stood with a mouse beneath its foot, and later authors tell of sacred mice kept in the temple. The god of prophecy is also the lord of mice, and the lord of mice is the bringer of plague.

The association of rodents with pestilence is older than the germ theory and more widespread than Greece. In the first book of Samuel, the Philistines who have captured the Ark of the Covenant are smitten with swellings, and they return the Ark with an offering of five golden images of their swellings and five golden mice, images, the text says, of the mice that mar the land. The Black Death of the fourteenth century, carried—as Alexandre Yersin and Paul-Louis Simond established at the end of the nineteenth—by a bacillus transmitted through the fleas of rats, made the rodent the emblem of epidemic for modern Europe. Camus opened *La Peste* (1947) with Dr. Rieux stepping on a dead rat on the landing of his building in Oran, and closed it with the warning that the bacillus never dies but waits, for decades if need be, until the day it rouses its rats again and sends them forth to die in a happy city. The plague announces itself through animals before it reaches humans, and it withdraws into animals to wait. Smintheus is the name of that structure.

Apollo is, at the same time, the god of healing. He is invoked as Paian, and he is the father of Asclepius, whose serpent-twined staff remains the emblem of medicine. The Hippocratic Oath, in the form transmitted from antiquity, opens with an invocation of the god: the physician swears by Apollo the Physician, and by Asclepius, Hygieia, and Panacea, and by all the gods and goddesses. For more than two thousand years the oath of medicine was sworn to the god of plague. The scientists of 2035 resume this

double office, and so, in his perverse way, does Dr. Peters, a servant of the science of disease who turns its instruments to the purpose of pestilence. The Apollonian complex introduced in Chapter One can now be stated fully. Prophecy, plague, and healing are not three functions accidentally united in an archaic god; they are three moments of a single relation to the future, in which the one who knows what is coming also controls its arrival and its cure. The scientists send the prophet, require the plague, and await the cure; Peters, their unacknowledged double in the past, carries the plague in the case of a man of medicine.

Apollo's sanctuary at Delphi bore the maxims that define the Apollonian order: know thyself, and nothing in excess. In *The Birth of Tragedy* Nietzsche made Apollo the god of measure and individuation, and above all the god of the image: the Apollonian world is the world of the dream, of beautiful appearance, of forms held apart, contemplated, and preserved. The photograph, in this sense, is Apollonian, and *La Jetée*, a film of preserved images whose hero is chosen for the clarity of a dream-image, is an Apollonian work. So is the scientific gaze of 2035, which holds the past at a distance as an archive of images to be examined. And Apollo's first maxim will reappear, two sections hence, at the foundation of modern biological classification, when Linnaeus defines the human species by the Delphic command alone.

To call the scientists Smintheus rationalised into an underwriter is therefore not a conceit. It names a historical transformation. The archaic god sent plague as punishment for an insult to his priest and lifted it when the insult was repaired: the plague was a message, and its end was a reconciliation. The scientists of 2035 neither send nor lift the plague; they inherit it as a fact and manage it as a datum. The punishment has become a statistic and the reconciliation a claim. Apollo's arrows have been replaced by a case of vials, and his offended priest by a laboratory assistant who believes, like the misanthropes of every age, that the species has offended the earth. What remains of the god is the structure of his office, the unity of foresight, pestilence, and cure. What has been lost is the possibility of propitiation. One could pray to Smintheus. One cannot pray to an actuary.

5.3
Rotpeter's Way Out

Kafka's Ape and the Prisoner Who Imitates

In Kafka's "A Report to an Academy" (1917) an ape addresses a learned society that has invited him to describe his former life as an ape. He cannot, he explains; he

has forgotten it. What he can describe is how he became what he is. He was shot by a hunting expedition of the firm of Hagenbeck on the Gold Coast, twice: once in the cheek, leaving the large, naked, red scar that earned him the name Red Peter, which he finds repugnant; and once below the hip, a wound that has left him with a limp. He woke in a cage in the steerage of a ship, a cage too low to stand in and too narrow to sit down in, facing a wall. There, for the first time in his life, he found himself without a way out. Freedom he did not want; he is scornful of the word, which he has heard applied to trapeze artists swinging under the roofs of variety theatres. He wanted a way out, any way out, and the only way out was to become human. So he imitated the sailors. He learned to spit, to smoke a pipe, to drink schnapps from the bottle, overcoming an animal disgust that nearly defeated him; and one evening, to the astonishment of the crew, he called out in a human voice. Arriving in Hamburg, he chose the variety stage over the zoological garden, because the zoo, he reasoned, was only a new cage. Now he gives performances, receives visitors, and returns at night to a half-trained chimpanzee companion whose eyes he cannot bear to meet by day, because they hold the madness of the bewildered, half-broken animal.

Cole is Rotpeter. He is captured, caged, and shot—in the leg, in the trenches of 1917, so that he too carries a wound below the hip into the rest of the story. He wakes again and again in cages. He is summoned before an academy, the panel of scientists, to deliver reports. And his way out is the way of imitation. Not freedom: the film never offers him that, and the pardon the scientists hold before him is only a reduction of his sentence. What he seeks, like the ape, is an exit, any exit, and the exits on offer are two. The first is the mission, by which the caged animal earns release by becoming the instrument of its keepers. The second, which he chooses in 1996, is sanity in the clinic's sense, by which the prisoner of an impossible knowledge earns a place among the living by agreeing that his knowledge is a delusion. Cole's decision to believe himself mad is Rotpeter's imitation of the sailors: an exit purchased by adopting the view his captors hold of him. And like the ape's, it is a way out that leads not to freedom but to another stage.

Kafka's choice of the Hagenbeck firm was not arbitrary. Carl Hagenbeck, the Hamburg animal dealer, supplied the zoos and circuses of Europe with captured animals, and in 1907 he opened at Stellingen a zoological park whose enclosures were separated from the public by concealed moats rather than bars, so that the animals appeared to roam free in panoramic landscapes: captivity disguised as liberty. Hagenbeck was also a pioneer of the *Völkerschauen*, the ethnographic exhibitions in which human beings from colonised societies were displayed to European crowds in reconstructed villages, beside the animals of their regions. The zoo of the colonial age exhibited humans as animals and animals as free. It was the anthropological machine

in its most public form, the place where the boundary between human and beast was staged for paying spectators and drawn along the lines of empire. Rotpeter, captured by Hagenbeck's hunters, is the inverse of the *Völkerschau*: the animal who exhibits himself as human. Cole, shown to tribunals in two centuries as a specimen of madness or of usefulness, belongs to the same lineage of exhibited bodies.

The last image of Kafka's story returns this chapter to the argument of the previous one. Rotpeter cannot bear, by day, to look into the eyes of his half-trained companion, because he sees there the madness of the bewildered, half-broken animal—the madness of a creature caught between two orders, no longer an ape and not yet a performer of humanity. Kafka locates madness exactly where Foucault located it in the classical age: at the boundary between the human and the animal, in the look of the half-tamed beast. Cole's look in the hospital of 1990 is that look: the stare of a creature torn from one world and not admitted to another, which the clinicians read as the sign of a disordered mind. The madness the clinic diagnoses in him is the madness of the threshold, and the threshold is the one that separates the human from the animal.

The ape has long been the emblem of imitation. Medieval and Renaissance writers called art the ape of nature, *ars simia naturae*, and the devil the ape of God; the English verb *to ape* still means to copy. *Twelve Monkeys* is a film of apings. It is itself, by its own credit, an imitation inspired by another film. Within it a woman apes the image of a man's dream, as in its embedded film a shopgirl is made to ape a dead woman who was herself impersonating a dead ancestor; its hero apes sanity in search of a way out; and its band of monkeys apes a revolution it does not make. The monkey is the figure of the copy, and the copy is the figure of repetition. In the ring of twelve monkeys the film inscribes, among its other meanings, a theory of its own status as a remake: a ring of imitations, each copying the last, around an origin that was itself already a copy.

5.4
The Anthropological Machine

Linnaeus, Moreau, Uexküll's Tick, and the Camp

Giorgio Agamben opened *The Open: Man and Animal* (2002) with an image from a thirteenth-century Hebrew Bible preserved in the Ambrosian Library in Milan. It depicts the messianic banquet of the righteous on the last day, and the righteous are shown with the heads of animals. Agamben read the miniature as a figure of the end of history in which the relation between the human and the animal would be

settled otherwise—neither by the human's mastery of the animal nor by its collapse into animality, but by a reconciliation whose form we cannot yet imagine. Against this image he set what he called the anthropological machine: the apparatus by which Western thought produces the human by drawing, within the living being, a line between the human and the animal, and by excluding the animal so defined.

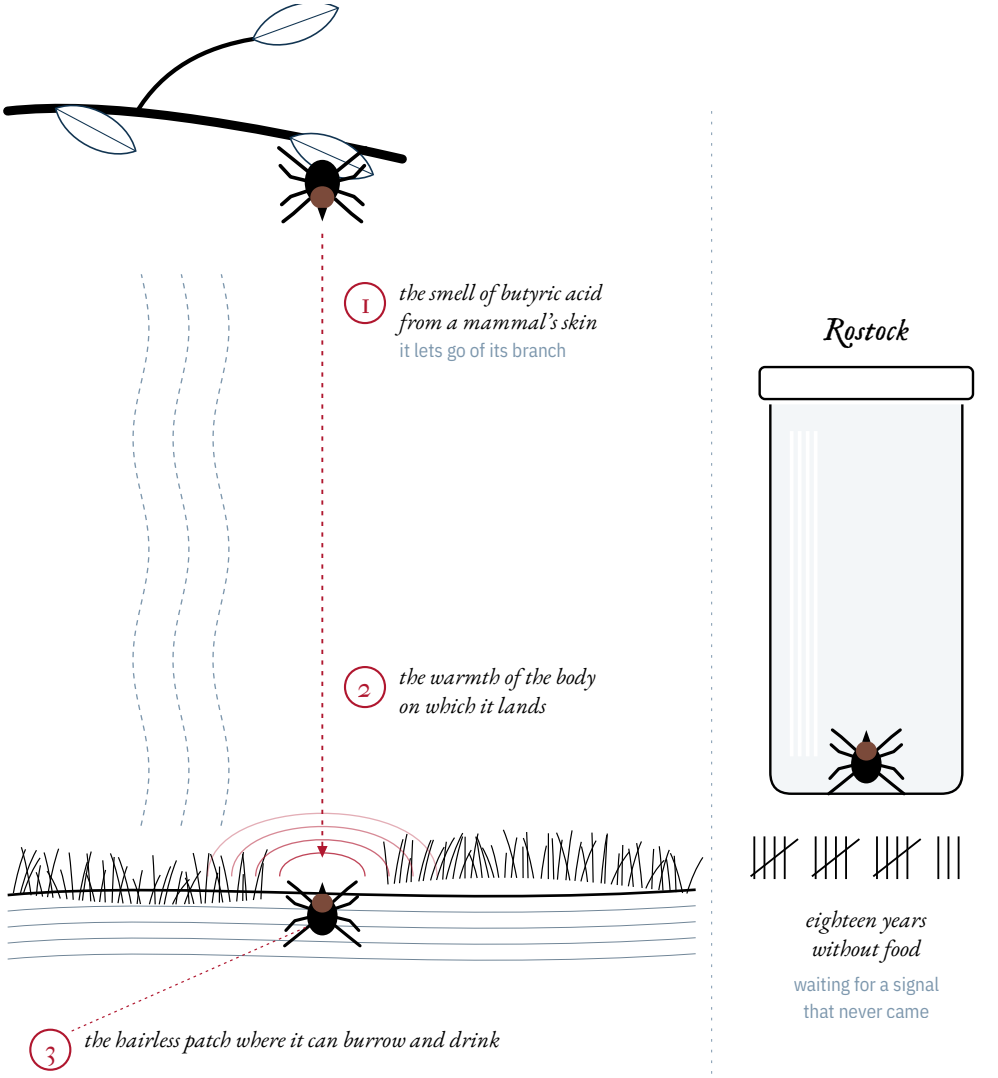
The machine's most revealing operation, for Agamben, occurs in Linnaeus. When the Swedish naturalist placed *Homo* among the primates in the *Systema Naturae* (1735; tenth edition, 1758), he recorded the species' difference not by an anatomical character but by a command: *nosce te ipsum*, know thyself. The human is the animal that must recognise itself as human in order to be one. It is an extraordinary definition, and its source is the first maxim inscribed at Delphi. The anthropological machine of modern biology runs, quite literally, on Apollo's command. *Twelve Monkeys* presents the machine in operation after the end of history. Its 2035 is administered by a council of specialists—among them a zoologist, a botanist, a microbiologist, and the astrophysicist who will reappear on the aircraft—who classify, sample, and catalogue the world above them, and who send a prisoner in a sealed suit to collect specimens of the life that has reclaimed the surface, while treating him as a specimen in turn. The film's hierarchy of the living runs from the scientists who classify, through the prisoners on whom they experiment, to the animals on the surface, and its hero passes back and forth across the lines.

Fiction had given the machine its most lurid laboratory a century before. In H. G. Wells's *The Island of Doctor Moreau* (1896) a vivisector carves animals into the shape of men in an enclosure the islanders call the House of Pain, and governs his creatures by a Law they chant in unison—not to go on all fours, not to suck up drink, not to eat flesh or fish—with the refrain after each prohibition: *Are we not Men?* Humanity is a set of prohibitions recited by beasts, and when Moreau dies the Beast Folk revert. Wells's narrator, returning to London, cannot look at the people in the streets without seeing the animal beneath; he withdraws from society and devotes himself to chemistry and to the study of the stars. The House of Pain, the recited law, and the witness who sees the animal in his fellow citizens and retreats into astronomy: *Twelve Monkeys* inherits all three, distributing them across the laboratory of 2035, the asylum of 1990, and the city of 1996, where Cole moves among people he knows to be, in a sense, specimens of a vanishing species, and where the last word belongs to an astrophysicist.

Agamben's reading of the machine passes through Heidegger's lecture course of 1929–1930, *The Fundamental Concepts of Metaphysics*, which proposed that the stone is worldless, the animal poor in world, and the human world-forming; and through the biologist Jakob von Uexküll, whose studies of animal environments, *Umwelten*,

Plate 8

The world of the tick



Uexküll's tick, whose world consists of three signals: the smell of butyric acid from a mammal's skin, which makes it drop from its branch; the warmth of the body on which it lands; and the hairless patch where it can burrow and drink. At right, the tick kept alive in a laboratory at Rostock for eighteen years without food.

supplied Heidegger's evidence. Uexküll's most famous case is the tick, whose world consists of three signals: the smell of butyric acid from a mammal's skin, which makes it drop from its branch; the warmth of the body on which it lands; and the hairless patch where it can burrow and drink. Uexküll added a detail on which Agamben dwells: a tick was kept alive in a laboratory at Rostock for eighteen years without food, suspended in waiting for a signal that never came. The tick in its jar is the animal held out of time, in a waiting without event, and it is the purest image of Cole's condition between missions—caged, sedated, suspended in a future that is one long aftermath, waiting for the signal that will drop him into the past. It is also an image of the photograph: a life preserved in suspended animation, waiting to be reanimated by a look.

Agamben also revisits Alexandre Kojève's reading of Hegel, according to which history ends when the struggle for recognition is complete, and post-historical humanity returns to a kind of animality, building as birds build nests and making music as crickets sing—a footnote Kojève later amended after a visit to Japan, where he found in ritualised formalism another post-historical possibility. Francis Fukuyama's *The End of History and the Last Man* (1992) carried Kojève's thesis to the end of the Cold War, borrowing from Nietzsche the figure of the last man, who has invented happiness and blinks. *Twelve Monkeys* answers Kojève and Fukuyama with a literalism neither imagined. History does not end with humans becoming animals; it ends with animals replacing humans. The post-historical world of 2035 is the world above ground, where a bear and a lion move through an intact city, and the post-historical humans are those below, administering survival in a bunker. The last man of the film does not blink. He looks up, through the visor of his suit, at a lion on a ledge.

The anthropological machine has a political name, and Agamben gave it in *Homo Sacer*, published in 1995, the year *Twelve Monkeys* was shot. The camp, he argued, is the hidden paradigm of modern political space: the place where the exception becomes the rule and human beings are reduced to bare life, *zōē*, life that may be killed without the killing counting as homicide or as sacrifice. The galleries beneath Chaillot in *La Jetée*, with their experiments upon prisoners, are a camp in this sense; so is the underground of 2035, where convicts are kept in cages and consumed in the service of a science of survival. The prisoner-volunteer is bare life par excellence. Neither free citizen nor animal, he can be sent into contamination, shot in a trench, abandoned in a century not his own, and no one is accountable for his death. From the future's point of view, Cole's death at the airport is not a killing but a loss within an acceptable margin. The anthropological machine and the actuarial calculus meet in the body of the prisoner who may be spent.

5.5
The Look of the Beast

Berger, Derrida, the Mirror, and the Apes Who Disbelieve

John Berger's essay "Why Look at Animals?" (1977) begins from the observation that animals stood at the centre of the human world for most of its history—as the first subjects of painting, the first metaphors, the messengers and promises of the gods—and that the relation was organised by an exchange of looks across an abyss: the animal looks at the human, and the human knows himself looked at. Within two centuries, Berger argued, capitalism has marginalised animals, removing them from daily life and returning them only as pets, as toys, as images, and as spectacles. The public zoo, founded in the very century of this disappearance, is a monument to the impossibility of the encounter it promises. In it the animal's look has been extinguished: isolated and dependent, the animals no longer return the visitor's gaze but look past it, sideways, blindly. The zoo, Berger concluded, is an epitaph to a relationship as old as humanity.

Twelve Monkeys restages Berger's history and runs it backward. Its zoo is opened, and the animals that stood behind bars with their looks extinguished walk out onto a city overpass, to the astonishment of passing motorists. In 2035 the reversal is complete. In a city emptied of people, a bear rears in the snow and a lion roars from a ledge, and the human in the sealed suit is the one who is looked at, a specimen in a vitrine of plastic. The return of the animal is the return of the animal's look. And the film's final image, the upturned face of a boy following an aircraft, has something of this restored look about it: the gaze of a creature who sees without yet understanding, before the categories that will make him the thirteenth monkey have taken hold.

Jacques Derrida's lecture of 1997, published as *The Animal That Therefore I Am* (2006), begins with a scene of exposure. Naked in his bathroom, the philosopher is looked at by his cat, and he feels shame—shame at being naked before an animal, and shame at that shame. The philosophical tradition from Descartes to Heidegger and Lacan, Derrida argued, has never seriously considered being seen by the animal; it has treated animals as things to be looked at, never as beings who look back and whose look addresses us. He coined the word *animot* to mark the violence of the general singular, *the animal*, under which the immense multiplicity of living creatures is subsumed, and he spoke of an unprecedented war waged over the past two centuries against animals through industrial breeding, experimentation, and slaughter on a scale without historical parallel. The laboratory monkey, the cause of the Army of the Twelve Monkeys' protests, is the emblem of that war. The film's 2035, in which the

war has ended with the defeat of the species that waged it, imagines the animal's look returned at last upon a human who can no longer look away.

Lacan's essay on the mirror stage (1949) opens with an animal. The human infant, Lacan observes, at an age when it is outdone in instrumental intelligence by the chimpanzee, already recognises its own image in a mirror and greets it with jubilation, whereas the young chimpanzee, having mastered the image and found it empty, loses interest. The human is the primate captivated by its image. Gordon Gallup's experiments of 1970, which showed that chimpanzees do recognise themselves in mirrors, troubled the line Lacan drew, but the distinction between recognition and captivation survives them, and it returns this chapter to Marker. The man of *La Jetée* is chosen precisely because he is captivated by an image; he is, in Lacan's terms, the most human of the prisoners. The time-travelling animal is the image-captivated animal. The thirteenth monkey is the primate that cannot stop looking at a picture, and that will follow the picture, if it must, to the end of the jetty.

In the cinema where Cole and Railly hide, the Hitchcock retrospective includes *The Birds* (1963), and the citation is not idle. Hitchcock's film, adapted from a story by Daphne du Maurier, imagines an inexplicable revolt of birds against the people of a California town; it offers no cause and no cure; and it ends with the survivors driving slowly away through a landscape occupied by massed, watching, motionless birds. The final shot is a world in which the animals have taken possession and look on. *The Birds* is the cinematic precedent for the intertitles of *Twelve Monkeys*, which promise that the animals will once again rule the world; and the watching birds of its last shot, as still as the wolves in the Wolf Man's walnut tree, compose a photogram of the animal gaze that the later film will set in motion.

The *Planet of the Apes* stages the reversal as a tribunal. In Franklin J. Schaffner's film (1968) an astronaut wounded in the throat is captured by apes, caged, and taken for a mute beast; when at last he speaks, he is tried before an orangutan court whose task is to decide whether he is an animal. The anthropological machine has been turned around, and the human must prove his humanity to beings who have classified him as a brute. The film ends with the astronaut's discovery, at the foot of a ruined Statue of Liberty, that the planet of the apes is the earth after a human catastrophe. Pierre Boulle's novel (1963) had ended more cruelly still. Its frame is a manuscript cast into space in a bottle and recovered by a couple sailing among the stars, who read the narrator's account of a world ruled by apes and conclude that it must be a hoax, since a rational, articulate human being is an impossibility; only at the very end is it revealed that the readers are themselves chimpanzees. The human testimony is found, read, and disbelieved by apes. It is the Cassandra structure transposed to the

species: humanity's report of its own reason, filed as a fantasy by the animals that succeed it.

5.6
Once Again the Animals

Zoonosis, the Clever Animals, and the End of the Human Exception

The animals of *Twelve Monkeys* belong, finally, to a particular moment in the relation between humans and other primates. The monkey had long been the laboratory's privileged substitute for the human. The development of the polio vaccines of the 1950s consumed rhesus monkeys in enormous numbers, their kidney cells serving as the medium in which the virus was grown; and in 1960 researchers discovered that vaccine produced in this way had carried into millions of human bodies a simian virus, SV40, unknown when the vaccine was licensed. The cure had passed a monkey's virus into the human population—Apollo's double office in the idiom of the twentieth-century pharmaceutical laboratory. In 1981, in Silver Spring, Maryland, an undercover volunteer documented the condition of seventeen macaques in a neuroscience laboratory, and the ensuing prosecution, the first criminal case of its kind against an American animal researcher, made the newly founded People for the Ethical Treatment of Animals a national movement. Peter Singer's *Animal Liberation* (1975) had supplied the movement's philosophy, and the Animal Liberation Front, founded in Britain in 1976, its tactics of raid and release. The Army of the Twelve Monkeys, a band that frees animals and protests their use in a virologist's laboratory, belongs to this history, and its location in the corridor between Baltimore and Philadelphia places it within a short drive of Silver Spring.

The 1990s added the imaginary of zoonotic catastrophe described in Chapter One: Ebola-family viruses in imported macaques at Reston, a fictional monkey carrying a haemorrhagic fever into California in *Outbreak*, and the hypothesis, confirmed by the end of the decade, that the virus responsible for AIDS had crossed into humans from other primates. Susan Sontag, in *AIDS and Its Metaphors* (1989), had warned against the fantasies of origin and invasion that this hypothesis attracted—the imagery of a disease emerging from a dark continent by way of its animals—and against the apocalyptic rhetoric that turned an epidemic into a judgment. *Twelve Monkeys* stands at the meeting point of these discourses and refuses their easiest form. Its monkeys are not the source of the plague. The virus comes from an American laboratory, carried by an American scientist, through an American airport. The film summons the

zoonotic imaginary only to redirect it: the animal is the decoy, and the human is the vector.

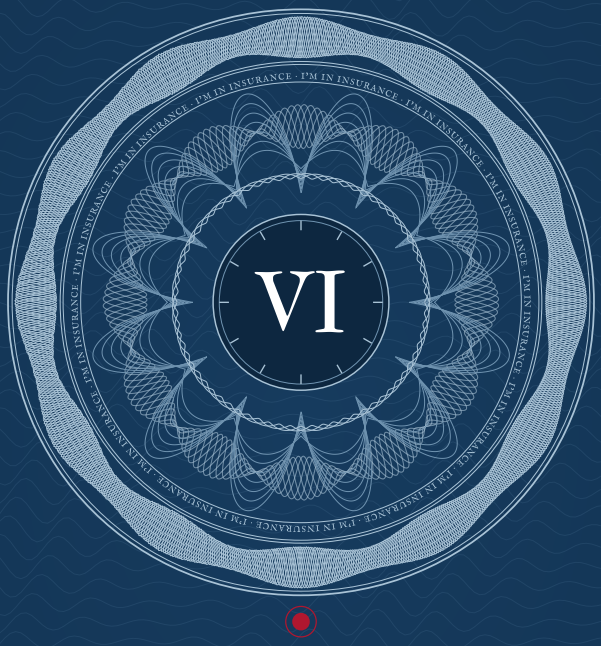
Two later films realised possibilities that *Twelve Monkeys* held in reserve. In Danny Boyle's *28 Days Later* (2002), written by Alex Garland, animal-liberation activists break into a primate laboratory in Cambridge and release chimpanzees infected with an engineered virus of rage, and within a month Britain is a necropolis: the prank of the Army of the Twelve Monkeys converted into the act that ends the world. In *Rise of the Planet of the Apes* (2011), a virus developed in a laboratory as a treatment for Alzheimer's disease grants apes intelligence and kills humans, and the end credits trace its spread along the routes of commercial aviation from an infected pilot—an animated map of Dr. Peters's itinerary. The cure is the plague; the apes rise; the airport is the vector. The sequence of films from *Planet of the Apes* to *Rise*, with *Twelve Monkeys* at its hinge, amounts to an extended meditation on a single proposition: that the end of the human exception will come from the human laboratory, and that the animals will inherit what the laboratory leaves.

Nietzsche wrote the epitaph of the human exception in 1873, in an essay he did not publish, "On Truth and Lie in an Extra-Moral Sense." In some remote corner of the universe, poured out in countless flickering solar systems, there was once a star on which clever animals invented knowledge. That was the most arrogant and mendacious minute of world history—yet only a minute. After nature had drawn a few breaths the star grew cold, and the clever animals had to die. The passage could serve as the intertitle of *Twelve Monkeys*. It names the human as an animal distinguished only by its cleverness, measures its history by the clock—a minute—and ends it with the indifference of nature resuming its breath. The thirteenth monkey is Nietzsche's clever animal: the primate that invented knowledge, built from it the instruments of its promise, its insurance, and its cure, and died of what it had made.

The figure can now be drawn in full. The thirteenth monkey is the animal that stands outside the ring of animal time in order to count it: the reader of the zodiac, the keeper of the clock, the spectator at the zoetrope. It is Nietzsche's promising animal, made calculable so that it might dispose of its future, and Ewald's insuring animal, which socialises that promise in a contract. It is Lacan's image-captivated primate and Marker's prisoner chosen for his captivation by a picture. It is Linnaeus's animal that must know itself in order to be what it is, and Agamben's animal that produces itself by excluding the animal within. It is the sick animal of the *Genealogy* and the debtor animal of the *Phaedo*, owing a cock to the god of healing. And it is the animal that cultivates in its laboratories, under the oath of Apollo the Physician, the agent of its own return to the ring. *Twelve Monkeys* tells the story of that return. The

plague closes the human parenthesis, and the animals, as the opening intertitles promise, rule the world once again.

One last image, from the eighth century, gives the human parenthesis its proper scale. In Bede's *Ecclesiastical History of the English People*, a counsellor of King Edwin of Northumbria, deliberating whether to accept the new faith, compares the life of man to the flight of a sparrow through the king's hall on a winter's night. It flies in at one door and out at the other; for the brief moment it is inside, it is safe from the storm and warmed by the fire; but it passes from winter into winter, and of what went before and what follows we know nothing. Cole's 1996 is the sparrow's hall. He enters it from the winter of 2035, where snow lies on an empty city; he is warmed for a moment by its music and its air, by a woman's company and the light of an inhabited world; and he leaves it, through the doors of an airport, into a winter that the boy watching him will grow up to know. The film's final irony is that the hall itself is about to go dark, and that the creatures who will next pass through it are not human. But the comparison holds. Human history, in *Twelve Monkeys*, is a bird's brief passage through a lighted room, and the room belongs, before and after, to the animals.



Chapter Six

"I'm in Insurance"

Plague, Probability, and the Actuarial Eschatology of the Future

And I looked, and behold a pale horse: and his name that sat on him was Death, and Hell followed with him. And power was given unto them over the fourth part of the earth, to kill with sword, and with hunger, and with death, and with the beasts of the earth.

— Revelation 6:8 (Authorised Version)

Those who have asserted that all is well have spoken foolishly; they should have said that all is for the best.

— Voltaire, *Candide* (1759), ch. 1 (trans. mine)

6.1
The Bills of Mortality
How Plague Taught Europe to Count

The line on which this dissertation turns—a scientist from 2035 announcing, beside the carrier of the virus, that she is in insurance—has a history longer than the film's. Insurance is not merely the profession the future happens to have adopted. It is the institution in which European modernity first learned to treat death as a calculable quantity, and the occasion of that learning was the plague. This chapter reconstructs the genealogy that links the epidemic to the actuarial table, the actuarial table to the welfare state and the risk society, and these in turn to the eschatology of the future in *Twelve Monkeys*. Its claim is that the film's final scene returns actuarial reason to its origin. Born from the counting of plague deaths, insurance ends, in the film, by underwriting a plague.

In London, from the plague years of the sixteenth century and continuously from 1603, the parish clerks published weekly Bills of Mortality: printed sheets recording the burials in each parish and, crucially, the number attributed to plague. The causes were reported by searchers, elderly women appointed to view the bodies of the dead and declare what had killed them, and the bills were read by merchants and courtiers deciding whether to flee the city. Daniel Defoe's *A Journal of the Plague Year* (1722), a fictional reconstruction of the plague of 1665, is in large part a meditation on these numbers. Its narrator copies the weekly totals into his account, watches them rise and fall, and distrusts them, since families concealed plague deaths to avoid having their houses shut up, and searchers could be bribed or deceived. Defoe's narrator is the first great actuarial witness in English literature: a man who experiences catastrophe through a table, and who knows that the table lies.

In 1662 John Graunt, a London haberdasher, published *Natural and Political Observations Made upon the Bills of Mortality*, and with it founded demography. Graunt compiled the bills of decades, corrected for their errors, estimated the population of London, observed that more boys than girls were born, measured the excess mortality of the city over the countryside, identified the plague years and estimated how many plague deaths had been concealed under other names, and constructed a rough table of how many of a hundred people born would survive to each decade of life. He wished, he explained, to set down how many had died of each disease, so that those who lived in fear of the more notorious ones might better understand the hazard they were in. The word is exact. Graunt converted the terror of plague into a hazard, a measurable exposure, and Charles II recommended his election to the newly

chartered Royal Society. William Petty, his friend and collaborator, would generalise the method as political arithmetic, the art of reasoning by numbers about the affairs of state.

The plague of 1665 that Defoe would later narrate closed the University of Cambridge, and a young fellow of Trinity named Isaac Newton withdrew to his mother's farm at Woolsthorpe, where in the months that followed he laid the foundations of his calculus of fluxions, his theory of colours, and his theory of gravitation. The mathematics of continuous change—the instrument with which every later age would describe curves, the epidemic curve among them—was conceived in quarantine. The plague year was the annus mirabilis of the science that would one day model plagues.

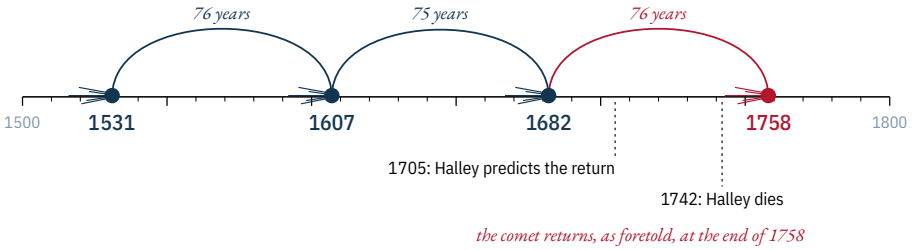
The decisive figure is Edmond Halley, and his double career is the hinge of this chapter. Halley was an astronomer. In 1705 he argued that the comets observed in 1531, 1607, and 1682 were a single body returning, and predicted its return for 1758; he died in 1742, and the comet reappeared, as foretold, at the end of 1758. Halley was the prophet whose prophecy was fulfilled after his death and believed because of it—Cassandra's inverse, vindicated by periodicity. But Halley was also a founder of actuarial science. In 1693 he published in the *Philosophical Transactions* "An Estimate of the Degrees of the Mortality of Mankind," drawn from the registers of births and burials of the city of Breslau, which had been compiled by the pastor Caspar Neumann, partly to test popular beliefs about the climacteric years of life; from them Halley constructed the first scientifically founded life table and used it to price annuities upon lives, showing that the annuities sold by the English government had been priced without regard to the age of the purchaser. In a single career the astronomer who predicted the return of a comet calculated the mortality of a city. The periodicity of the heavens and the mortality of human beings were submitted to one arithmetic, and the old science of the zodiac, which had read fate in the stars, yielded to a science that read it in tables. The astrophysicist who takes her seat beside Dr. Peters and announces that she is in insurance is Halley's heir in the most literal sense: an astronomer of human mortality.

The century that followed built on Halley's foundation. Abraham de Moivre's *Annuities upon Lives* (1725) supplied the mathematics; the Equitable Society, founded in London in 1762, was the first to set life-insurance premiums by age according to a mortality table. Richard Price, the dissenting minister who constructed the Northampton table (1771) and corresponded with Franklin on American affairs, was also the man who in 1763 communicated to the Royal Society the posthumous essay of his friend Thomas Bayes on the inference of causes from their effects—the problem of reasoning backward from observed outcomes to the probabilities of the conditions that produced them, which is precisely the problem the scientists of 2035 set

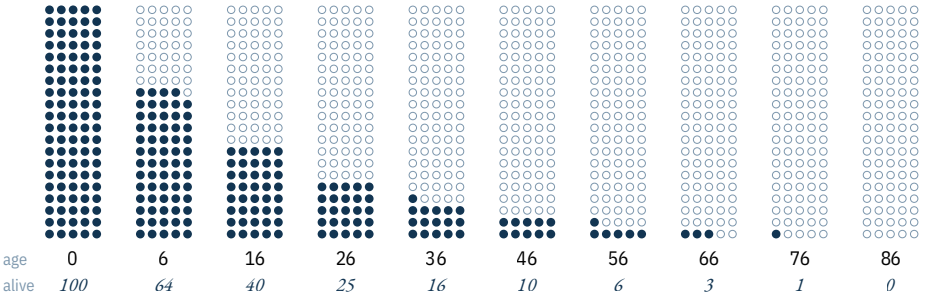
Plate 9

One arithmetic for comets and cities

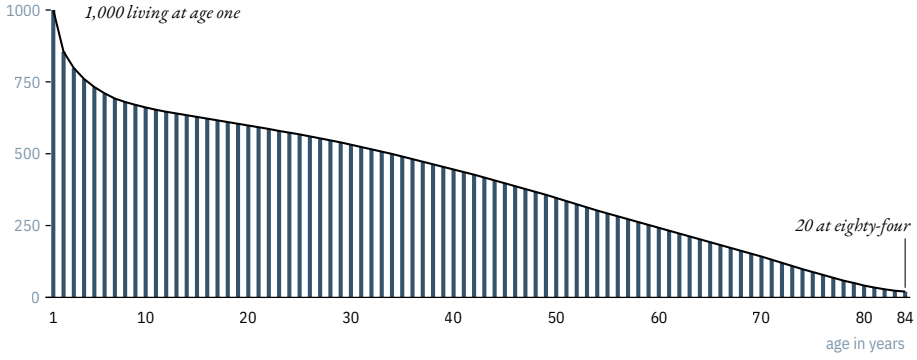
the periodicity of the heavens



of a hundred quick conceptions: Graunt, 1662



the mortality of a city: Halley, Breslau, 1693



Above, the comet's returns as Halley identified them in 1705, and its predicted return in 1758, after his death. Middle, Graunt's table of 1662: of a hundred "quick conceptions", how many remain alive at each age. Below, Halley's life table of 1693 for Breslau. Data: Graunt, *Natural and Political Observations* (1662); Halley, *Philosophical Transactions* 17 (1693).

themselves when they reason from a ruined world to the source of its ruin. Pierre-Simon Laplace, in his *Philosophical Essay on Probabilities* (1814), imagined an intelligence that, knowing all the forces and positions of nature at a single instant, would find nothing uncertain, the future like the past present before its eyes. And Adolphe Quetelet, director of the Brussels observatory, carried the astronomers' law of error into the study of human beings, measuring the chests of Scottish soldiers and the incidence of crime and composing from the distributions the figure of the average man; he observed that the budget of the prison, the galleys, and the scaffold is paid with a frightening regularity. Ian Hacking, in *The Taming of Chance* (1990), described the avalanche of printed numbers that followed in the early nineteenth century, and the transformation by which statistical law came to govern what had been the domain of fate. The surname the astrophysicist gives on the aircraft, the most ordinary in the English language, is the name of Quetelet's average man.

6.2
From Theodicy to Security

Lisbon, Inoculation, and the Insurance of Society

On the morning of 1 November 1755, All Saints' Day, an earthquake, followed by a tsunami and by fire, destroyed most of Lisbon and killed tens of thousands of people. The catastrophe provoked the first great public debate of European modernity on the meaning of disaster. Voltaire's *Poem on the Lisbon Disaster* (1756) assailed the optimism of Leibniz and Pope, for which whatever is, is right; Kant wrote three essays on the natural causes of earthquakes; and Rousseau, replying to Voltaire in a letter of August 1756, made the argument that would define the modern understanding of catastrophe. It was not nature, he observed, that had assembled twenty thousand houses of six and seven storeys in that place, and had the inhabitants been more evenly dispersed and more lightly lodged, the damage would have been far less. The disaster was natural in its cause and social in its magnitude. Susan Neiman, in *Evil in Modern Thought* (2002), has argued that Lisbon marks the moment at which European thought ceased to treat natural catastrophe as a problem for theodicy and began to treat it as a problem for human arrangements. The question was no longer why God permits it, but how the city should be built.

Five years later the question of how to manage a catastrophe that could not be prevented received its first mathematical answer. Smallpox was the great killer of eighteenth-century Europe, and the practice of inoculation—deliberately infecting a

healthy person with matter from a mild case, in the expectation of a mild illness and lasting immunity—carried a small but real risk of death. In a memoir read to the Paris Academy in 1760, Daniel Bernoulli calculated, from Halley's life table, how many years of life a population would gain if smallpox were eliminated by inoculation, and concluded that the gain far outweighed the risk. Jean le Rond d'Alembert objected that a calculation valid for a population did not settle the question for an individual, who might reasonably refuse a small risk of death now for a gain spread over a lifetime he might not live to enjoy. The exchange inaugurated the modern problem of risk: the gap between what is rational for a population and what is acceptable to a person, between the curve and the life. It is the gap in which Cole lives and dies. The future's calculation concerns the survival of a population; it does not concern him.

Michel Foucault, in his lectures of 1978 at the Collège de France, *Security, Territory, Population*, took smallpox inoculation as the paradigm of a new form of power, which he called the apparatus of security. Sovereignty operates through law and prohibition; discipline operates through the normation of bodies in enclosed spaces—the school, the barracks, the asylum. Security operates otherwise. It does not seek to prevent the event; it takes the event as given and works upon its distribution, using the notions of case, risk, danger, and crisis to manage a phenomenon across a population, calculating what proportion of infections, deaths, or thefts is acceptable and intervening to keep the curve within bounds. The scientists of 2035 are an apparatus of security in its purest form. They do not prevent the plague; they could not if they would. They take it as given, measure its distribution across time, and intervene at the point where intervention yields the greatest return: a sample, a cure, a recovery of the surface. Foucault's three forms of power are distributed across the film's three institutions: the sovereign power to kill in the police officer at the airport, discipline in the asylum of 1990, and security in the laboratory of 2035.

François Ewald, Foucault's assistant and later the editor of his lectures, made insurance the key to this third form of power. In *L'État providence* (1986) he traced the passage in nineteenth-century France from a regime of fault, in which accidents were judged as the responsibility of individuals, to a regime of risk, in which they were treated as statistically inevitable features of collective life whose costs should be shared; the law of 1898 on industrial accidents marks the turning point. In his essay "Insurance and Risk" (1991), Ewald argued that nothing is a risk in itself and that anything can become one, depending on how a danger is analysed; insurance does not discover risks but constitutes them, and the risk it constitutes has three characteristics. It is calculable, because it can be assigned a probability; it is collective, because it is realised only across a population; and it is a capital, because what insurance guarantees is not the restoration of what was lost but the payment of its value. The plague of

Twelve Monkeys, seen from 2035, has all three characteristics. It is calculable, a dated event with a known toll; it is collective, a catastrophe of the species; and it is a capital, the source of the sample on which the future's recovery depends. Ewald's title, moreover, names what the future has become. *L'État providence*, the providential state, the welfare state: the institution in which the providence of Boethius's God, who sees the whole of time from his watchtower, has been secularised as the insurer's foresight. Chapter Three called the scientists an actuarial providence. Ewald gives that providence its historical name.

Ulrich Beck's *Risk Society* (1986) describes the limit of this order. In reflexive modernity, Beck argued, the principal hazards are no longer natural but manufactured: produced by the very processes of industrial and scientific modernisation, invisible to ordinary perception, perceptible only through the sciences that produce them, and liable to return like a boomerang upon their producers. Such hazards generate what Beck called organised irresponsibility, a system in which everyone is implicated and no one is accountable. Charles Perrow's *Normal Accidents* (1984) had shown that in complex, tightly coupled systems catastrophic failures are not aberrations but normal properties of the system, and Paul Virilio insisted that to invent the ship is to invent the shipwreck, and to invent the aeroplane is to invent the crash. *Twelve Monkeys* dramatises these theses. Its virus is manufactured in a laboratory devoted to the study of disease; it is invisible, perceptible only to the science that made it; it escapes through a security apparatus designed to prevent exactly such escapes, at the moment an inspector asks its carrier to open his case; and no institution in the film—not the laboratory, not the clinic, not the police, not the airport—is accountable for it. The checkpoint at which Peters displays his samples is Perrow's normal accident staged as a scene: the security procedure becomes the mechanism of release. To invent virology is to invent the leak.

6.3
The Actuarial Angel

Benjamin's Theses and the Future That Collects

In the spring of 1940, months before his death at Portbou, Walter Benjamin wrote the theses "On the Concept of History," and in the ninth he described a picture he owned, Paul Klee's watercolour *Angelus Novus* (1920). Benjamin read Klee's figure as the angel of history. It faces the past, and where human beings see a sequence of events, it sees a single, unbroken catastrophe heaping ruin upon ruin before it. The

angel would linger, rouse the dead, and repair what has been broken; but a gale blowing from Paradise has filled its wings so that it cannot fold them, and drives it backward into the future while the rubble mounts toward the sky. That gale, Benjamin wrote, is what we call progress.

The scientists of 2035 are angels of history in a precise and parodic sense. They too face the past, and they too see in it not a chain of events but a single catastrophe: the plague, the five billion, the ruin of the surface. They too are driven by a storm—not the storm of progress but the climate of the bunker, the cold that has forced them underground. But they have no wish to rouse the dead or to repair what has been broken. Their only wish is to retrieve something from the wreckage: a sample, an unmutated strain, a resource for their own continuation. They are, in the term this dissertation proposes, actuarial angels. The Greek *angelos* means messenger, and the future's messengers in *Twelve Monkeys* are two: Cole, the convict sent to bring back information, and the astrophysicist, sent at the last to collect. The first is a messenger of the old kind, an *angelos* who carries news and is not believed. The second is the new angel. She carries nothing, announces nothing, and comes only to take.

The contrast deepens when set against the other theses. In the third, Benjamin praised the chronicler who recounts events without distinguishing the great from the small, on the principle that nothing that has ever happened should be counted lost to history; but he added that only a redeemed humanity receives the fullness of its past, and only for a redeemed humanity has its past become citable in every one of its moments. The future of *Twelve Monkeys* has achieved a parody of this condition. Its archive loses nothing: it holds recordings, photographs, transcripts, dates, and its machine can return to any moment it chooses. Every moment of the past is citable. But the citation is not redemption; it is retrieval. The actuarial future possesses the past in its entirety as data and redeems none of it, and the difference between the two is the difference between Benjamin's remembrance, *Eingedenken*, which holds the past open to a claim of justice, and the record, which holds it closed for a claim of indemnity.

In the appendix to the theses Benjamin noted that the Jews were forbidden to investigate the future and were instructed instead, by the Torah and the prayers, in remembrance. This stripped the future of the spell it casts on all who consult soothsayers; yet it did not make the future homogeneous and empty, for in it every second was the small gate through which the Messiah might enter. The image completes a series this dissertation has followed through every chapter: the gate of Luz, where Solomon's scribes met the angel of death; the gateway named Moment, where Zarathustra confronts the dwarf; the Lion Gate of Mycenae, before which Wolf's Cassandra stands; and the airport gate where Cole dies. In Benjamin's theses every second

is such a gate, and the messianic possibility is that something may pass through it that is not the continuation of the catastrophe. In *Twelve Monkeys* the gate admits only what was already scheduled. The film's future is the soothsayers' future, known in advance and therefore without messianic opening, and its time is the homogeneous, empty time of the timetable. The departures board is its emblem.

Benjamin's earlier essay "Critique of Violence" (1921) supplies a final key, and it returns this chapter to Smintheus. Benjamin distinguished mythic violence, which founds and preserves law and binds its victims in guilt, from divine violence, which destroys law and expiates. His paradigm of mythic violence is the legend of Niobe, whose children were struck down by the arrows of Apollo and Artemis because she had boasted of them, and who was turned to a weeping stone; mythic violence, Benjamin wrote, is bloody power over mere life for its own sake. The phrase *bloßes Leben*, mere or bare life, is the source of Agamben's *bare life*, and its founding example is the work of Apollo's bow. The plague of *Twelve Monkeys* is mythic violence in Benjamin's sense. It falls upon bare life; it is administered rather than expiated; and it founds a new order, the order of the bunker and its scientists, upon the dead. Nothing in the film corresponds to divine violence, the interruption that would break the cycle of law and guilt. What it offers instead, at the edge of its last scene, is a look.

The opposition can be stated as a pair of angels. Benjamin's angel faces the catastrophe and wishes to make it whole, and cannot; the actuarial angel faces it and wishes only to make it pay, and can. The first is the figure of a historical materialism that would redeem the defeated; the second, of a security apparatus that would indemnify the survivors. Between them stands the figure that the film offers as its own: not an angel but a woman kneeling on a concourse, who neither redeems nor collects, but sees. Re-vision, in the terms of the theses, is what remains of the messianic when the gate has closed: not the power to make the past whole, but the refusal to let it be filed.

6.4
Prophecy after Hiroshima

Jonas, Anders, Dupuy, and the Paradox of Jonah

The paradox that governs all modern thinking about catastrophic prophecy is as old as the Book of Jonah. Commanded to prophesy against Nineveh, Jonah flees by sea toward Tarshish; a storm rises; he is cast overboard and swallowed by a great fish, in whose belly he prays for three days and three nights before being vomited onto dry

land. He goes at last to Nineveh and delivers the shortest prophecy in scripture: "Yet forty days, and Nineveh shall be overthrown." The Ninevites believe him. The king decrees a fast, and not for humans only: neither man nor beast, herd nor flock, is to taste anything, and man and beast alike are to be covered with sackcloth. God sees their repentance and relents, and Jonah is furious, because his prophecy has failed. The book ends with God's rebuke to the prophet sulking outside the spared city: "And should not I spare Nineveh, that great city, wherein are more than sixscore thousand persons that cannot discern between their right hand and their left hand; and also much cattle?" The last words of the book are the animals.

Jonah's paradox is that the successful prophet of doom is the one whose prophecy fails. The warning that is believed averts what it announces and so falsifies itself; the warning that is fulfilled is a warning that went unheeded. Cassandra and Jonah are the two faces of the prophet, the one disbelieved and confirmed, the other believed and refuted. *Twelve Monkeys* is a Cassandra story told in a world that has forgotten Jonah. No one in it repents; no fast is declared for man and beast; no one proposes to spare the city for the sake of those who cannot tell their right hand from their left, and also much cattle. And the film contains, in Dr. Peters, a figure the Book of Jonah anticipates with uncanny accuracy: the prophet who is angry that the doom may not come, and who resolves to bring it himself. Jonah sulks beneath his withered vine; Peters boards a plane.

Hans Jonas, in *The Imperative of Responsibility* (1979), made Jonah's paradox the foundation of an ethics for the technological age. The novelty of modern power, Jonas argued, is that its effects reach so far into the future and so deep into the conditions of life that traditional ethics, oriented to the neighbour and the present, cannot govern it. He proposed a new imperative: act so that the effects of your action are compatible with the permanence of genuine human life on earth. And because the long-range consequences of technology cannot be known with certainty, he proposed a heuristics of fear: where catastrophe is possible, the prophecy of doom deserves more weight than the prophecy of salvation, because the stakes are asymmetrical. The prophet of doom, in Jonas's ethics, must accept in advance the humiliation of being proved wrong, since his success consists in being refuted.

Günther Anders, whose defence of Claude Eatherly Chapter Four discussed, drew the darker diagnosis. Since Hiroshima, he argued, humanity has lived in the time of the end, *Endzeit*, which may at any moment become the end of time, *Zeitende*; the age is a reprieve. Its characteristic pathology is what he called apocalypse blindness, an inability to imagine the consequences of what we are able to produce, rooted in a Promethean gap between our technical capacity and our capacity to feel and to represent. To break this blindness Anders told, more than once, a parable of Noah. Weary of

preaching a flood in which no one believed, Noah puts on the garments of mourning and sits in the square. Asked whom he mourns, he answers that he mourns the dead; asked when they died, he answers that they will die tomorrow. He speaks of the catastrophe in the future perfect, as something that will have happened, and the shock of the tense achieves what his warnings could not: people begin to help him build the ark.

Jean-Pierre Dupuy, in *Pour un catastrophisme éclairé* (2002), subtitled *Quand l'impossible est certain*—when the impossible is certain—built on Jonas and Anders a theory he called enlightened doomsaying. A catastrophe, Dupuy observed, is not believed possible before it occurs; once it has occurred, it appears to have been inevitable. Prevention fails because it addresses the catastrophe as a mere possibility, which no one credits. Dupuy's proposal is to adopt toward the future the stance of Anders's Noah: to inscribe the catastrophe in the future as a fixed event, a destiny, and to act from the standpoint of its having happened, within a temporality he called projected time, which closes future and present into a loop of mutual determination that resembles the self-consistent loops of the physicists. The catastrophe must be held certain in order that it may be avoided.

Set against this tradition, the future of *Twelve Monkeys* appears as its exact parody. The scientists of 2035 adopt Dupuy's standpoint precisely: they inhabit the time after the catastrophe, they treat it as fixed, they speak of it in the past tense, and they send a messenger into the time before it, as Noah sat in the square in mourning. But they do so in order to collect, not to avert. Their doomsaying is not enlightened but actuarial; it holds the catastrophe certain in order to profit from its certainty. The Cassandra Policy is the inversion of Anders's parable. Noah mourns the dead of tomorrow so that they may not die; the scientists send Cole to mourn the dead of 1997 on condition that they die. And the film's own address, as Chapter Three argued, restores the enlightened form at the level of its spectators. *Twelve Monkeys* is Noah in the square: a fiction that speaks of a catastrophe in the future perfect, before an audience whose future has not been written.

The sociology of prophecy adds a final irony. In *When Prophecy Fails* (1956), Leon Festinger and his colleagues studied a small group of believers in the American Midwest who awaited, on the word of their leader, a flood that would destroy the world on 21 December 1954, and rescue by flying saucer for the faithful. When neither flood nor saucer arrived, the group did not abandon its belief; it proselytised more fervently than before, relieving its dissonance by recruiting others to share it. Festinger's believers held a false prophecy more firmly after it failed; Cole's hearers reject a true prophecy more firmly as its proofs accumulate. Between the two cases lies the whole pathology of modern catastrophic belief, and Frank Kermode, in *The Sense of an Ending*

(1967), described its literary form. We humanise the indifferent beat of a clock, Kermode observed, by hearing it as *tick-tock*, a modest beginning answered by a diminutive end, an interval organised by its close. Apocalyptic fictions are concord fictions that give time a shape by imagining its end, and when the end fails to come their adherents revise the date. Jacques Derrida, in his essay on the apocalyptic tone in philosophy (1983), would add that every discourse claiming to unveil the end exerts a power over those to whom it speaks. *Twelve Monkeys* is a tick-tock in which the tock arrives and no one hears it, and in which the only voice that claims to unveil the end has no power at all.

6.5
Malthus's Angel

Dr. Peters and the Misanthropic Sublime

Dr. Peters is the film's most enigmatic figure and its most consequential. He appears only a handful of times: as a member of the staff of Leland Goines's laboratory; at Railly's book signing, where he volunteers the opinion that the apocalyptic prophets may be right about the species and its excesses; and at the airport, where his case of vials passes through inspection and he boards his plane. The film grants him no backstory, no confession, no grand speech. He is the unremarkable face of a catastrophe whose motive the spectator must reconstruct from a few sentences, and those sentences place him in a recognisable tradition. He is the misanthrope of reason: the man who has concluded, by a chain of argument that seems to him impeccable, that the human species is the planet's disease and its removal the cure. He bears, too, a name the film leaves unremarked, that of the apostle who holds the keys of the gate.

The tradition begins, in its modern form, with Thomas Robert Malthus's *Essay on the Principle of Population* (1798). Population, Malthus argued, increases geometrically while subsistence increases arithmetically, so that population must perpetually press against the means of life and be held in check. The checks are of two kinds: preventive, such as delayed marriage, and positive, such as war, famine, and disease. His most famous passage describes the positive checks as the ministers of a terrible equilibrium: "sickly seasons, epidemics, pestilence, and plague advance in terrific array, and sweep off their thousands and tens of thousands. Should success be still incomplete, gigantic inevitable famine stalks in the rear, and with one mighty blow levels the population with the food of the world." Malthus did not wish for these checks; he described them as the operation of a law. But his prose personifies the plague as an

army advancing in array, and his successors would draw from his law the conclusion he refused, that the checks might be welcomed, or administered.

The twentieth century revived Malthus at the scale of the planet. Paul Ehrlich's *The Population Bomb* (1968) predicted mass famine within a decade; the Club of Rome's *The Limits to Growth* (1972) modelled the collapse of industrial civilisation under the pressure of population and consumption; Garrett Hardin's "The Tragedy of the Commons" (1968) and his later "lifeboat ethics" (1974) argued that rich nations should refuse aid to poor ones lest the boat be swamped. Deep ecology, as articulated by Arne Næss in 1973, held that the flourishing of non-human life requires a substantial decrease of human population, and the radical fringe of the environmental movement in the 1980s sometimes drew the most extreme conclusions: in 1987 a writer in the journal of the American group Earth First!, under the pseudonym Miss Ann Thropy, suggested that the AIDS epidemic might be welcomed as a check on human numbers, a suggestion much of the movement repudiated. The Finnish ecologist Pentti Linkola advocated drastic and coercive reductions of humanity. Peters is a fictional crystallisation of this fringe, not of environmental thought as such, whose mainstream has repeatedly rejected it. He is the point at which the Malthusian law becomes a Malthusian will.

Speculative fiction has given the figure several faces. In Margaret Atwood's *Oryx and Crake* (2003), the bioengineer Crake, a genius of contempt, distributes through a global pharmaceutical campaign a pleasure pill seeded with a lethal virus, wiping out humanity in order to leave the earth to a gentle species of his own design: the pharmaceutical promise of happiness is the vector of extinction. In Alan Moore and Dave Gibbons's *Watchmen* (1986–1987), Adrian Veidt kills millions in New York with a fabricated alien attack in order to unite humanity against an imaginary enemy, and when his adversaries arrive to stop him he informs them that the act is already accomplished; the boast of the Army of the Twelve Monkeys, painted on walls in the past tense, has its grim counterpart in Veidt's fait accompli. George Pan Cosmatos's *The Cassandra Crossing* (1976) imagined authorities who, to conceal a plague escaped from their own secret laboratory, seal a train of exposed passengers and route it toward a condemned bridge: the institution manages its contagion by sacrificing the infected. Stanley Kubrick's *Dr. Strangelove* (1964) gave the misanthropic sublime its comic form, a general driven by a paranoid fantasy of contaminated bodily fluids launching the attack that triggers a doomsday machine. And Stephen King's *The Stand* (1978) imagined a weaponised influenza escaping a military laboratory through a breach in containment and carried across the country by a fleeing guard and his family: the epidemic as the escape of an institution's secret.

Peters's act also belongs to the oldest lineage of the plague narrative, and the comparison clarifies what is new in it. Thucydides described the Athenian plague of 430 BCE with clinical precision and recorded its moral consequence: as the dying multiplied, fear of the gods and respect for law dissolved, and people indulged openly what they had formerly concealed, since no one expected to live long enough to be punished. Boccaccio framed the *Decameron* (c. 1353) with the plague of Florence, from which ten young people withdraw to a villa to tell stories: narrative as quarantine. Mary Shelley's *The Last Man* (1826), the first major novel of human extinction, imagines a plague that consumes the species in the late twenty-first century and leaves a single survivor wandering a depopulated Europe; its frame presents the novel as a prophecy deciphered from leaves found in the cave of the Cumaean Sibyl, a confined prophetess's scattered transcript. Jack London's *The Scarlet Plague* (1912) sets its narrative in 2073, sixty years after a plague has destroyed civilisation, where an old man who was once a professor tells his half-savage grandsons of the world before and is mocked as a liar: the witness of the past is the Cassandra of the post-apocalypse. In all these narratives plague is something that befalls humanity, whether as natural catastrophe or as judgment. *Twelve Monkeys* belongs to the age in which plague is something humanity does to itself, by intention or accident, in a laboratory. Peters is among the defining villains of that age, and he has no lines worth quoting, because his argument is the argument of the age.

Peters's aesthetic deserves a name. Kant, in his late essay "The End of All Things" (1794), described the thought of an end of time as at once terrifying and sublime, a thought at whose brink the imagination recoils and by which it is drawn; and his *Critique of Judgment* (1790) had located the dynamical sublime in the spectacle of a nature whose might exceeds our capacity to resist it, contemplated from a position of safety. Peters's misanthropy is a sublime of this kind, inverted. He contemplates not nature's might over humanity but humanity's might over nature, and finds the spectacle intolerable; his remedy is to restore nature's might by unleashing it, from a position—a seat on an aircraft, a laboratory credential—of perfect safety. The misanthropic sublime is the pleasure of imagining the earth without us, and the film's vision of a city in which the animals walk freely is, among other things, the fulfilment of Peters's wish. *Twelve Monkeys* does not share the wish. But it does not deny the beauty of its fulfilment either, and the ambivalence is part of its honesty.

6.6

Pangloss at the End of the World

The Song, the Clock, and the Insured Apocalypse

Twelve Monkeys ends, as it began, with a cultural artefact out of its time. Over the closing credits, after the boy's face and the departing aircraft, Louis Armstrong sings "What a Wonderful World," a song written by Bob Thiele and George David Weiss and recorded in 1967, a year of war and riot, whose earnest catalogue of trees, skies, roses, and newborn children was conceived as a gesture of hope against the news. Barry Levinson's *Good Morning, Vietnam* (1987) had already set it over a montage of bombing and executions, making it the anthem of ironic counterpoint, and Kubrick had established the convention a generation earlier by ending *Dr. Strangelove* with Vera Lynn's "We'll Meet Again" over a sequence of nuclear explosions. Gilliam's use of Armstrong belongs to this lineage and exceeds it. The song plays over the end of a world the spectator now knows to be fixed, and the irony is not simply that the world is not wonderful. It is that, from the vantage of 2035, the world the song describes—green, flowering, open to the sky, with the animals flourishing beneath it—is precisely the world the plague will restore.

The song's philosophical name is Pangloss. In Voltaire's *Candide* (1759), written in the shadow of Lisbon, the tutor Pangloss teaches that this is the best of all possible worlds and that everything in it is for the best, and he holds to the doctrine through shipwreck, earthquake, the auto-da-fé, and his own hanging. The most savage episode of the satire concerns his disease. Pangloss has contracted syphilis, and when Candide finds him disfigured and begging, he traces the infection through a chain of lovers back to a companion of Columbus, and concludes that the disease was indispensable in the best of worlds, since had Columbus's men not brought it back from the Americas, Europe would have had neither chocolate nor cochineal. Pangloss's syphilis is the first actuarial theodicy of an epidemic: it justifies a plague by the goods that came in its train. The scientists of 2035 hold, without saying so, the same doctrine. The plague of 1996 was necessary to the world of 2035. Without it there would be no bunker, no science of survival, no time machine, no cure, and no scientists. Their world is the best of all possible worlds for them because it is the only one in which they exist, and the plague is its chocolate and its cochineal.

Leibniz, whom Voltaire was mocking, ended his *Theodicy* (1710) with a myth that bears directly on the form of *Twelve Monkeys*. Theodorus, a priest troubled by the crimes of Sextus Tarquinius, is led by Pallas Athena into the Palace of the Fates, a pyramid of apartments each containing a possible world, in each of which he can watch,

as in a theatrical representation, a different life of Sextus. The apartments rise toward an apex, the best of all possible worlds, which is the actual one, and in it Sextus commits the crime that will lead to the founding of the Roman Republic: the evil is necessary to the greater good. Leibniz's palace is a cinema of possible worlds, and its pyramid is the structure that the television series derived from *Twelve Monkeys* would later adopt, with its branching timelines and revisable histories. The film itself occupies the apex alone. It shows only the actual world, the one in which the crime occurs, and it refuses the consolation of the other rooms. What it retains from Leibniz is the claim that the actual world is justified by what follows from it, and it assigns that claim to the beneficiaries.

The passage from Leibniz to the insurer is the passage Susan Neiman traced from Lisbon onward, from theodicy, which justifies evil by the whole, to risk, which compensates loss within the whole. Insurance is the secular successor of theodicy. Where Leibniz argued that no evil is absolute because each contributes to the best whole, the insurer arranges that no loss is absolute because each is compensated from a common fund. The actuarial theodicy of *Twelve Monkeys* fuses the two. The plague is justified, from the future's point of view, by the world it made possible, and compensated, through the mission, by the sample it yields. Armstrong's song over the credits is the anthem of this fused doctrine, the hymn of a world that has been underwritten. Its wonder is real, but it belongs to the beneficiaries.

Poe's "The Masque of the Red Death" (1842) supplies the figure in which the song, the clock, and the bunker converge. While a plague called the Red Death ravages his country, Prince Prospero withdraws with a thousand courtiers into a fortified abbey, welds its gates shut, and holds a masked ball in a suite of seven coloured rooms. In the last, black room stands an ebony clock whose chimes, each hour, halt the musicians and dancers in a moment of unaccountable dread before the revel resumes. At midnight a masked figure appears whose disguise is the Red Death itself, and the court dies to the last guest: "And Darkness and Decay and the Red Death held illimitable dominion over all." The underground of 2035 is Prospero's abbey after the masque, inhabited by survivors the story did not imagine; the ebony clock is the ring of twelve monkeys; and the masquerade has its counterpart in the disguises in which Cole and Railly arrive at the airport, she in the blonde wig of the dream, while the Red Death passes through the checkpoint in a briefcase. Armstrong's song, rising as the credits roll, is the music that resumes after the clock has struck.

Fredric Jameson observed in *The Seeds of Time* (1994) that it seems easier for us to imagine the thoroughgoing deterioration of the earth and of nature than the breakdown of late capitalism, and Mark Fisher's *Capitalist Realism* (2009) made the formula famous. *Twelve Monkeys* dramatises it with unusual exactitude. The film ima-

gines without difficulty the death of five billion people and the abandonment of the surface of the earth. What it cannot imagine is the end of insurance. The future that survives the catastrophe is not a commune, a theocracy, or a ruin; it is an underwriter, and its last representative is a woman in a travelling suit who introduces herself by her profession. The apocalypse has ended the human species as the earth's dominant inhabitant and left intact the institution that prices human lives. Jameson had also written, in a 1989 essay on Philip K. Dick's *Time Out of Joint* (1959), of a nostalgia for the present: science fiction's capacity to make our own moment appear as the past of a future, and so to make it visible as history. Cole's love of the air and the music of the 1990s is that nostalgia made literal. The future longs for our present as for a lost paradise, and the longing is the film's last gift to its audience: the experience of seeing one's own world as already mourned.

The argument of this chapter can now be gathered. The future of *Twelve Monkeys* is an actuarial providence descended from the counting of plague deaths. Its scientists are angels who collect rather than redeem. Its doomsaying is a parody of the enlightened catastrophism that the film's own address to its spectators restores. Its antagonist is the Malthusian will that converts a law of population into an act, and its coda is the hymn of a theodicy that has become an insurance contract. Against all this the film sets one thing: the look exchanged across the concourse between a woman who knows and a child who does not. The final chapter returns to that look, and asks what it can mean for a world that has since lived through the pandemic the film imagined.



Chapter Seven

Conclusion

Re-vision after the End

Lifetime is a child at play, moving the pieces on a board: the kingship is a child's.
— Heraclitus, fragment B52 (trans. mine)

April is the cruellest month, breeding
Lilacs out of the dead land, mixing
Memory and desire, stirring
Dull roots with spring rain.
— T. S. Eliot, *The Waste Land* (1922)

7.1
The Argument Recapitulated

Five Theses on Actuarial Tragedy

This dissertation began at the end of a jetty, with a woman kneeling over a dying man and a scientist taking her seat on an aircraft, and it proposed that the word the scientist uses to name her profession discloses the structure of the whole film. The chapters that followed tested that proposal against the film's medium, its metaphysics, its institutions, its animals, and its political economy of time. Their results can now be stated as five theses, each the conclusion of a chapter and each a component of the concept the dissertation has called actuarial tragedy.

First, the tense of *Twelve Monkeys* is the tense of the photograph. From *La Jetée* the film inherits Barthes's anterior future, the *this will have been* in which every image of its hero declares at once that he is dead and that he is going to die; and it inherits Bazin's obscenity, the repetition of an unrepeatable death. Where Marker preserved this tense in the photogram and punctured it once with a blink, Gilliam and the Peoples translate it into narrative structure, distributing the functions of the still across a dream that revises itself, a photograph that harbours the future, a cluster of monitors that converts memory into data, and a cinema screen before which the hero states the film's metaphysics. The medium is not the container of the film's philosophy. It is its first argument.

Second, the universe of the film is a block containing closed curves, self-consistent in Novikov's sense, in which the traveller's personal time folds across external time as Lewis described and the past defends its consistency through commonplace reasons that turn out to be institutions. Within it the classical resources for reconciling fate and freedom—Chrysippus's co-fated events, Boethius's conditional necessity, Kierkegaard's repetition, Nietzsche's redemption of the *it was*, Deleuze's counter-actualisation of the event—remain available, but only in a single form. Freedom survives as the transformation of the relation to what happens, never as the alteration of what happens. Revision is impossible; re-vision is all that is left.

Third, the asylum is the past's instrument of the Cassandra Policy. A true prophecy of the incredible satisfies every criterion by which psychiatry recognises delusion, and the clinic therefore guarantees, without needing to know it, that the prophet will not be believed. The film reconstructs the long alliance between madness and knowledge of the end, from Aeschylus's captive Cassandra through Foucault's ship of fools and Semmelweis's madhouse to Laing's patient with the bomb inside her; it shows

the clinic looping the prophet's protests back into the record as symptoms, in Goffman's sense, and looping its own categories through prophet and psychiatrist alike, in Hacking's; and it locates its one conversion in the clinician who learns, too late, to believe.

Fourth, the human appears in the film as the thirteenth monkey: the animal that stands outside the ring of animal time in order to count it, promise against it, insure against it, and, in its laboratories, prepare its own return to it. The ring of twelve monkeys is at once zodiac, clock, zoetrope, and Aeschylean chorus; the god who presides over it is Apollo Smintheus, oracle, plague-archer, and physician; and the plague the film imagines is the end of the human exception, the closing of a parenthesis in which a clever animal invented knowledge, measured its minute of history, and died of what it had made.

Fifth, the future in *Twelve Monkeys* is an insurer, and this is the film's deepest historical insight. Actuarial reason was born from the counting of plague deaths, from Graunt's bills to Halley's life table, and it grew into the security apparatus Foucault described, the providential state Ewald analysed, and the risk society whose limit Beck identified. The future of the film completes this history by underwriting the plague from which actuarial reason was born. Its scientists are angels who collect rather than redeem; their doomsaying parodies the enlightened catastrophism of Jonas, Anders, and Dupuy; their theodicy is Pangloss's, in which a plague is justified by the goods that follow it; and their anthem is the song over the credits.

The five theses converge in the concept of the Cassandra Policy. A future whose existence is premised on a catastrophe sends into the past a prophet whose disbelief it requires; the past's institutions of reason manufacture that disbelief automatically; the closed timeline guarantees that the warning will be delivered and will fail; the human being, the insuring animal, underwrites its own end; and the medium that records all this speaks in the tense of what will have been. Against this system the film sets a single counter-figure, which it neither overestimates nor abandons: the look exchanged at the end of the jetty. The remainder of this conclusion considers that look, the afterlives of the film, and its pertinence to a world that has, since 1995, lived through a pandemic and learned to call its own epoch the age of human geological agency.

The Cassandra Policy

SCHEDULE OF THESES

I The tense of *Twelve Monkeys* is the tense of the photograph.

II The universe of the film is a block containing closed curves, self-consistent in Novikov's sense.

III The asylum is the past's instrument of the Cassandra Policy.

IV The human appears in the film as the thirteenth monkey.

V The future in *Twelve Monkeys* is an insurer.

NOT COVERED

the look exchanged at the end of the jetty

XIII

Plate 10. The five theses of Section 7.1, set out as a schedule.

7.2

Lifetime Is a Child at Play

The Boy at the Airport

Among the fragments of Heraclitus is one of the strangest sentences in Greek philosophy: *aiōn pais esti paizōn, pesseuōn; paidos hē basilēiē*. Lifetime—or time, or eternity, for *aiōn* can mean all three—is a child at play, moving the pieces on a board; the kingship is a child's. Nietzsche, in *Philosophy in the Tragic Age of the Greeks* (written in 1873), read the fragment as the heart of Heraclitus's vision of a world without moral purpose, in which coming-to-be and passing-away are the play of an innocent force, like a child who builds heaps of sand by the sea and knocks them down and builds them again. In *Thus Spoke Zarathustra* the child became the last of the three metamorphoses of the spirit, after the camel that bears burdens and the lion that says no: the child is innocence and forgetting, a new beginning, a game, a wheel rolling of itself, a first movement, a sacred yes. Deleuze, in *Nietzsche and Philosophy* (1962), made the child's throw of the dice the image of an affirmation that embraces chance rather than calculating against it.

The boy at the airport is Heraclitus's child. He is the bearer of lifetime in its most literal sense, since the whole of Cole's life lies before him, the life the film has just shown ending. He does not know what he has seen. He is innocence and forgetting in Zarathustra's sense, a new beginning and a wheel rolling of itself; the loop of the film turns through him, and he will turn it again. And the kingship is his. Everyone else in the final scene is defined by a relation of knowledge or power to the catastrophe: the insurer who has come to collect, the carrier who is about to release it, the woman who now knows it, the dying man who has carried it all his life. The boy alone stands outside the economy of foresight. He is the one figure in the film whom the actuarial order has not yet reached, though it will, and his upturned face, following the aircraft that carries the plague, is the film's last image of a gaze that sees without calculating.

The figure belongs to Gilliam's cinema as much as to Marker's. *Time Bandits* ends with a boy alone before the smoking remains of his parents, in possession of a knowledge of history no adult will credit; *The Adventures of Baron Munchausen* entrusts the survival of an old liar's stories to a girl who believes them; *Tideland* (2005) would follow a child through a catastrophe she transfigures by fantasy. In each case the child bears imagination against a world that has organised itself to disbelieve. In *La Jetée* the

child on the pier is barely seen; he is the camera before which the image is taken. *Twelve Monkeys* lets us see him, and lets someone see him.

That someone is Railly, and her look is the film's final act. She has followed Cole through a recognition that inverted her profession, and she now knows what no one else on the concourse knows: that the dying man and the watching boy are the same person, that the plague will come, that the child will grow up beneath the earth and be sent back to die here. She can change none of it. What she can do is look at the boy, and the film grants her a long, still moment in which to do so. Her expression has been read as grief, as recognition, as tenderness, as horror; it is all of these and more than any of them. Rilke's Eurydice, told that Orpheus had turned around, could only ask *who?*, and the boy, if he could speak, would ask the same. Railly knows who. Her look holds a whole life in a single instant, past and future at once, without the power to alter it and without the wish to file it.

This is re-vision in its most concentrated form, and the dissertation can now define it with the precision the film has earned. Re-vision is the act by which a subject who knows a fixed past receives it, at the moment of its completion, as a whole that matters, without the power to change it and without the interest that would turn it into an asset. It differs from revision, which wills backward and fails; from revenge, which bills the past and collects; and from quietism, which looks away from the worst moments. It is Kierkegaard's repetition, which faces forward; Nietzsche's redemption, which converts *it was* into something other than an injury; Benjamin's remembrance, which refuses to let the past be filed; and Rich's survival, which enters an old story from a new direction because the old reading has become unlivable. In the film it lasts a few seconds and changes nothing in the chain of events. It is nonetheless the only moment in *Twelve Monkeys* that the policy does not cover.

Heraclitus is also the philosopher of the river into which one cannot step twice, since other waters are always flowing on. Cole's remark in the cinema inverts the fragment with a precision Heraclitus would have appreciated. One can step twice into the same film, since its waters never change; what one cannot do is step into it twice as the same person. The river has become a reel, and the flowing has passed from the world into the one who watches. The boy will see the scene at the airport only once and remember it all his life; the woman sees it once and knows it whole; the spectator may watch it as often as she likes, and will see it differently each time. The film ends by handing its kingship to the ones who play: the child at the gate, and the spectator before the screen.

7·3

Remake, Series, Palace of Worlds

The Afterlives of a Fixed Film

Twelve Monkeys is itself an afterlife, and its relation to its source is the first instance of the ethics it proposes. Chapter One read the credit "inspired by" as a theory of the remake: to breathe motion into Marker's stills, as Marker breathed it into a single shot of a woman waking. The remake is zoetropic in the precise sense, taking a ring of still images and, by setting it turning, producing the appearance of life. But it does not change its source. *La Jetée* remains what it was, a film of photographs that ends in a death on a pier, and *Twelve Monkeys* does not revise it; it re-visions it, returning to a fixed work and finding in it what a later moment can see. That later moment saw the asylum where Marker had seen the camp, the epidemic where he had seen the bomb, the insurer where he had seen the experimenter, and the living animal where he had seen the museum. The remake is the past seen again by a changed spectator. It is Cole's remark in the cinema enacted at the scale of film history.

Marker's own career after *La Jetée* was a long meditation on afterlives—on memory, its media, and its archives—from *Sans Soleil* through the CD-ROM *Immemory* (1997) to the virtual museum he built in the online world of Second Life. He died in Paris on 29 July 2012, his ninety-first birthday: the maker of the century's most famous film about a child who witnesses his own death closed his life on the anniversary of its beginning, a loop no screenwriter would have dared. *La Jetée* has meanwhile entered the general vocabulary of images. Its experiments and its padded blindfold have been restaged in music video and gallery installation, its sequoia scene quoted and requoted, its structure adopted by fictions that never acknowledge it. It has become, in Warburg's sense, a pathos formula of the late twentieth century: the figure of a man who travels through time to the moment of his death.

The television series that Syfy produced from the film between 2015 and 2018, created by Terry Matalas and Travis Fickett, is the most instructive of these afterlives, because it chose the opposite principle. The series made the past mutable. Its travellers can and do change history, and its plot unfolds across four seasons as a sequence of revisions, branches, and counter-revisions. It renamed its psychiatrist Cassandra Railly, making explicit the myth the film had only lectured on; it named its chief scientist Jones, after the film's last speaker; and it expanded the Army of the Twelve Monkeys into a millenarian cult serving a mysterious figure called the Witness, with designs upon time itself. Whatever its merits as serial drama, the series is a symptom. It converts a film about the impossibility of revision into an engine of endless revi-

sion, because the serial form requires that the story not close. The fixed loop of the film, which ends where it began, is incompatible with the economy of renewal by the season.

Leibniz's Palace of the Fates, discussed in Chapter Six, provides the exact model of the difference. The series inhabits the palace: it moves from apartment to apartment, from one possible world to another, showing different lives of the same characters, and treats each world as provisional. The film occupies the apex alone. It shows one world, the actual one, and refuses the comfort of the others. The cultural preference of the early twenty-first century for the palace over the apex—for the multiverse, the reboot, the alternate timeline, the director's cut, the franchise in which every death can be undone—is a preference for revision over re-vision, and it has a political meaning. A culture that can imagine every past as revisable has difficulty imagining any catastrophe as real. The fixed film is a discipline against that difficulty. It asks its viewers to hold one world in view, to see its end, and to bear the seeing.

Other works have taken up the film's principle more faithfully. Denis Villeneuve's *Arrival* (2016), adapting Chiang, ends with its linguist choosing a life whose end she already knows, and its closing images of mother and daughter answer the boy at the airport. *Dark* constructs an apocalypse that every attempt at prevention produces, and resolves it not by revision but by dissolution, a final act of letting go. And Adam McKay's *Don't Look Up* (2021) restages the Cassandra structure as satire. Two astronomers—Halley's heirs—discover a comet that will destroy the earth and are disbelieved by a political and media establishment that finds the news unmarketable; a technology magnate persuades the government to let the comet approach so that its minerals can be harvested; the harvest fails, the earth is destroyed, and the wealthy escape to a distant planet, where, in the film's last joke, they are eaten by its animals. It is the Cassandra Policy as comedy of manners: a catastrophe permitted because someone expects to collect on it, and an ending in which the animals inherit.

What these afterlives confirm is that *Twelve Monkeys* has become a fixed point in the sense the time-travel genre gives the term: an event that later fictions orbit, revise, or repeat, but cannot undo. Its achievement was to hold, against the pressures of a studio, a star system, and a genre that rewards the averted catastrophe, to the ending in which the catastrophe is not averted and the hero dies as foretold. That fidelity to the fixed past is the film's formal ethics. It refuses its audience the consolation of revision in order to offer the harder gift of re-vision.

7.4
Engines of Re-vision

The Time-Loop Game

If re-vision is the freedom of a subject before a fixed world, the medium that has explored it most fully since *Twelve Monkeys* is not cinema but the video game. The time-loop game places its player in exactly Cole's position: in a world whose catastrophe recurs on schedule, whose inhabitants repeat their actions, and whose only variable is the knowledge the player accumulates from cycle to cycle. Three works define the form.

In *The Legend of Zelda: Majora's Mask* (2000), a moon with a grimacing face will fall on the land of Termina in three days, and the player's hero, armed with an ocarina and a song that returns him to the dawn of the first day, relives those days again and again. The central settlement is called Clock Town, and its great clock counts down the hours. Most of what the player accomplishes is undone by each return; what persists is a handful of possessions, a notebook in which the hero records the troubles of the townspeople, and the player's own knowledge of who will be where, and when, and in what distress. The game's most celebrated sequence concerns two lovers, separated by misfortune, whose reunion the player can bring about only in the last hours before the moon falls; reunited, they choose to remain in the doomed town and greet the dawn together. Nothing the player does in that cycle prevents the catastrophe. The reunion is an act of re-vision: the same three days, seen whole, become the occasion of a gesture of love that the catastrophe does not cancel. Even the hero's transformations belong to this dissertation's argument, since by putting on masks he becomes in turn a creature of the swamp, of the mountains, and of the sea, so that the passage through the loop is also a passage across the boundary between the human and the animal.

Jonathan Blow's *Braid* (2008) makes rewinding the player's fundamental power: any mistake can be undone, any death reversed. Its narrative appears to be a quest to rescue a princess from a monster. In the final level the player watches the hero and the princess flee a pursuer together and then, when time runs in its proper direction, discovers that the princess was fleeing the hero all along. The events are the same; the direction in which they are viewed reverses their meaning. *Braid* is the purest demonstration of re-vision as a principle of form, in which nothing is altered and everything is changed by the direction of the look. Its epilogue, as players have long noticed, alludes to the first nuclear test and to the remark of one of its physicists, Kenneth Bainbridge, on what the bomb's makers had become, so that a game about undoing ends at the

event that cannot be undone, the subject of the Peoples' documentary *The Day After Trinity*.

Mobius Digital's *Outer Wilds* (2019) carries the principle to its limit. The player, a young astronaut in a small solar system, discovers that the sun will go supernova twenty-two minutes after each awakening, and that each death returns the player to the start. Nothing in the world persists from loop to loop. There is no inventory to keep and no upgrade to earn; the only thing that accumulates is knowledge, recorded in the ship's log and in the player's mind, about the vanished species that built the loop in search of what it called the Eye of the Universe. The game cannot be won by force or skill, only by understanding, and the understanding it finally yields is that the end cannot be prevented. The player's last act is to accept it, and to help begin, in the death of one universe, the next. Of all the works this dissertation has considered, *Outer Wilds* most completely realises Cole's remark in the cinema. The world never changes. Only the one who plays it changes, and the change is the whole of the game.

The time-loop game also reveals, by contrast, what the actuarial relation to a fixed world looks like in practice. Every such game can be played in two ways. It can be explored: its cycles traversed slowly, its inhabitants observed, its catastrophe understood. Or it can be optimised: its schedules memorised, its routes calculated, its outcomes extracted with maximal efficiency, as in the competitive practice of the speedrun, which treats the loop as a resource to be exploited rather than a world to be seen. The two modes correspond exactly to the two relations to a fixed catastrophe that *Twelve Monkeys* sets against each other. The scientists of 2035 are speedrunners of history: they know the schedule, calculate the route, and send their agent to extract the item at the optimal moment. Railly, at the end, is an explorer who has finally understood. The medium that most fully embodies the film's ethics also exposes the temptation the film diagnoses: the conversion of a known world into a solved one.

The games return this conclusion to the cinema with a sharper sense of what the film asks of its spectator. *Twelve Monkeys* cannot let its viewer replay its cycles, but it invites a second viewing as insistently as any loop invites a second life. The opening dream means one thing the first time and another the second; the scene in the cinema means one thing before the ending is known and another after; the final look means more each time it is seen. The film is a single loop that the spectator must traverse again herself, and the change it seeks is in her. In this sense *Twelve Monkeys* anticipates the aesthetics of the time-loop game by a generation, and it does so with the oldest resources of the medium to which it belongs: a fixed reel, a changing viewer, and an image that will not alter however often it is shown.

7.5
After 2020

Pandemic, Prophecy, and the Policy Realised

When *Twelve Monkeys* was released, its plague was a hypothesis. A quarter of a century later it acquired a referent. The pandemic of coronavirus disease that began at the end of 2019 was associated, by the World Health Organization's estimate of excess mortality, with close to fifteen million deaths worldwide in 2020 and 2021 alone, and it transformed the conditions of everyday life on every continent. It is not the business of this dissertation to recount that history. What concerns it is the extent to which the pandemic was experienced, narrated, and administered in the categories the film had dramatised.

The first category was the actuarial. Within weeks of the outbreak's spread, the epidemic curve had become a public vernacular. Populations were urged to flatten it; governments published daily counts of cases and deaths; dashboards updated hourly tracked infections across the world; statisticians and journalists computed excess mortality, the deaths above the number expected, as the most reliable measure of the catastrophe. Graunt's weekly bills returned as continuous data, and with them Graunt's problem, the concealment and misattribution of deaths, which is what made the measurement of excess mortality necessary in the first place. The pandemic was lived, to a degree without precedent, through tables. It was also governed through all three of Foucault's forms of power at once: the sovereign power of emergency decrees, the disciplinary power of confinement and distancing, and the security apparatus of modelling, testing, and vaccination, which took the epidemic as given and worked upon its distribution.

The second category was the prophetic. Public-health specialists had warned for years that a pandemic of a novel respiratory virus was likely and that the world was unprepared. In 2015 Bill Gates delivered a widely viewed lecture arguing exactly this, and in 2018 the World Health Organization added to its list of priority diseases a placeholder, Disease X, for a pathogen not yet known. These warnings were heard, archived, and largely not acted upon; after the pandemic began they were rediscovered and circulated as prophecies fulfilled, often under the name of Cassandra. The pattern is the one this dissertation has called the Cassandra Policy, though without its metaphysics: not a closed timeline that guarantees disbelief, but institutions whose ordinary operations converted warnings into reports, reports into archives, and archives into evidence of foresight after the fact.

The third category was the film's central opposition, the animal and the laboratory. The origin of the virus became, and at the time of writing remains, a subject of scientific and political dispute. Many researchers hold that it reached humans through zoonotic spillover, possibly at a market in Wuhan where live animals were sold; others regard a research-related incident as possible; and governments and intelligence services have reached differing assessments held with varying degrees of confidence. This dissertation does not adjudicate the question, which turns on evidence it cannot assess. What matters for its argument is the form the public debate took. It was conducted, with remarkable consistency, in the terms *Twelve Monkeys* had set: was the catastrophe the work of animals or of a laboratory, of nature or of science, of the decoy or of the vector? The film's structure of suspicion had become a template through which a real catastrophe was interpreted, and the template's power to organise belief is itself something the film anticipated: a society that knows the shape of the story in advance sees the story it expects.

The fourth category was insurance, and here the film's final line found an almost literal realisation. In 2017 the World Bank launched a Pandemic Emergency Financing Facility, financed in part by bonds whose holders received high interest in exchange for the risk of losing their principal if a qualifying pandemic occurred. Payment was governed by a schedule of conditions—numbers of cases and deaths, spread across countries, the passage of time—designed to ensure that it would follow a catastrophe that had crossed a calculable threshold. When the coronavirus spread in early 2020, the bonds paid out only once their conditions had been met, in the spring, by which time the pandemic was well established; the facility was criticised for the complexity of its triggers and the lateness of its payment, and it was soon wound down. Whatever the merits of the instrument, its structure is the one this dissertation has described. It was insurance against a pandemic that could pay only after the pandemic had occurred, priced by actuaries, held by investors, and triggered by the curve. It did not prevent the catastrophe. It underwrote it.

The fifth category is larger than the pandemic. In 2000 the atmospheric chemist Paul Crutzen and the biologist Eugene Stoermer proposed that the present geological epoch be called the Anthropocene, in recognition of humanity's transformation of the earth system; and although the bodies responsible for the geological time scale declined in 2024 to ratify the term as a formal epoch, it has become the name under which the humanities think the present. Dipesh Chakrabarty, in "The Climate of History: Four Theses" (2009), argued that the Anthropocene collapses the old distinction between natural history and human history, since humans have become a geological force, and that it obliges us to think of ourselves as a species, a category that history, devoted to the particular, had long refused. *Twelve Monkeys* is an Anthropo-

cene fiction before the word. It imagines the human species as a single agent whose collective activity, concentrated in a laboratory and a briefcase, alters the biosphere; it imagines the consequence as the return of the earth to other species; and it imagines the survivors not as a humanity reconciled with the earth but as an institution that insures itself against its own history. Donna Haraway, in *Staying with the Trouble* (2016), proposed against the Anthropocene's heroic and apocalyptic narratives the practice of making kin across species in a damaged world. The film offers no such practice. But its last vision of a city in which bears and lions walk, and its last look exchanged between a woman and a child, gesture toward the two relations—to other species and to the future—on which any such practice would depend.

7.6
Toward a Theory of Actuarial Tragedy

The concept this dissertation has developed was derived from a single film, but it is not confined to it. Actuarial tragedy names a form that becomes possible when three conditions are met: when catastrophe is understood as manufactured rather than sent; when the institutions that respond to it are organised by the calculus of risk rather than by the logic of guilt or of salvation; and when knowledge of the catastrophe in advance—by prophet, scientist, or model—is structurally separated from the power to prevent it. Under these conditions the old elements of tragedy return transformed. Fate becomes the closed loop of a system that cannot be changed from within; the oracle becomes the forecast; the chorus becomes the archive; the hero's recognition becomes the moment at which an individual grasps what the institution always knew; and catharsis gives way to indemnity, the compensation of loss for those positioned to collect.

A corpus of actuarial tragedy can be sketched from the comparisons this dissertation has drawn. It would include *The Cassandra Crossing*, with its sealed train routed toward destruction to protect a secret; *Children of Men*, in Alfonso Cuarón's adaptation (2006) of P. D. James's novel (1992), in which a species that has ceased to reproduce administers its own extinction through camps and statistics; the television series *Chernobyl* (2019), in which a scientist's knowledge of a catastrophe confronts a state whose institutions are designed to deny it; *Don't Look Up*, with its comet permitted to arrive because someone expects to profit from it; and Atwood's *Oryx and Crake*, in which the pharmaceutical promise of health becomes the vector of extinction. In each, catastrophe is known in advance, administered rather than prevented, and narrated from within the institutions that manage it or profit from it. *Twelve Monkeys* is

the paradigm of the form because it adds the element the others lack: the closed timeline that converts an institutional failure to prevent into a metaphysical impossibility, and so exposes the policy beneath the failure.

Against actuarial tragedy the dissertation has set the concept of re-vision, and it is necessary in closing to state its limits as plainly as its claims. Re-vision is not a politics. It changes nothing in the chain of events; it prevents no catastrophe; it redistributes no loss. Within a closed world it is the only freedom available, and that is why the film can offer nothing more. But the spectator is not within a closed world, and for her re-vision is not the end of freedom but its beginning. To see a catastrophe whole, as fixed, as already mourned, is the condition of acting against it while it is still to come: this is the lesson of Anders's Noah, of Dupuy's projected time, and of the film's own address. Re-vision within the fiction is consolation; re-vision before the fiction is preparation. *Twelve Monkeys* is built so that the first becomes the second, and its fatalism is the instrument of an appeal that only an audience outside the loop can hear.

The final look at the airport carries this double address. Within the fiction it is the look of a woman who can do nothing but see. Before the fiction it is a look turned, through the boy, toward the audience, which is, like him, at the beginning of a future it does not yet know. The child does not know who has died; the spectator does not know what will come. Railly's look tells us both what she knows and that she cannot tell it to anyone who will believe her. It is Cassandra's look, in other words, turned for once toward the only listener who might: the spectator who has seen the whole loop and can leave the cinema for a time that is not yet fixed.

The film's opening intertitles date its prophecy to an April day in 1990, and the date summons, whether or not the film intends it, the most famous April in modern poetry. *The Waste Land* opens by calling April the cruellest month, breeding lilacs out of the dead land and mixing memory and desire—a poem written in the aftermath of the war into whose trenches Cole is mistakenly thrown, and of the influenza pandemic that followed it, about a civilisation that has outlived its own end and speaks in the fragments of its dead. Memory and desire are the two forces of *Twelve Monkeys*: the memory of a scene at an airport, and the desire for a woman and a world that the memory dooms. April mixes them, and the mixture is cruel because it brings life out of a dead land that will not stay dead. The animals that return to the city of 2035 are lilacs bred out of the dead land. The film, like Eliot's poem, declines to say whether that return is a consolation or a final cruelty.

This dissertation has argued that *Twelve Monkeys* is a philosophical work of exceptional density. It thinks through its medium, its metaphysics, its institutions, its animals, and its political economy of time with a coherence that its reputation as a puzzle film or a studio remake has obscured. Its final line discloses a structure, the Cassandra

Policy, that illuminates not only the film but the age it anticipated: an age that has learned to calculate its catastrophes in advance and to insure itself against them, and has not thereby learned to prevent them. The film's answer to that age is modest and exact. It does not promise that the past can be changed. It shows a woman looking at a child at the end of a jetty, in a world that is about to end, and it asks the spectator to look with her—to see the whole of what has been and will have been, without the power to alter it and without the wish to collect on it. That look is the last freedom the film allows. It is also the first freedom of anyone who leaves the cinema still, for the moment, on the near side of the end.



Excursus

The Aesthetics of the Outmoded

Gilliam's Frame, the Used Future, and the Carnival of Obsolescence

Paris change ! mais rien dans ma mélancolie
N'a bougé ! palais neufs, échafaudages, blocs,
Vieux faubourgs, tout pour moi devient allégorie,
Et mes chers souvenirs sont plus lourds que des rocs.
— Charles Baudelaire, "Le Cygne" (1861)

The seven chapters of this dissertation have treated *Twelve Monkeys* as an argument. This excursus treats it as a made thing: a world of surfaces, lenses, costumes, sounds, and salvaged junk, assembled by a director whose style has been more often admired than analysed. Its claim is that the film's aesthetics are not ornament laid over its philosophy but one of the modes in which that philosophy is conducted. The film thinks, as much as anywhere, in the look of its objects: in the cathode-ray monitors of 2035, the tilted frames of 1990, the carnival of the opened zoo, the breathing bellows of a bandoneon.

E.1
The Used Future

Retro-Futurism from *Metropolis* to the Bunker of 2035

The future of *Twelve Monkeys* is built from the past. Its underground society lives among the salvaged technologies of the late twentieth century: banks of cathode-ray monitors, translucent plastic sheeting, ducts and pipes, cages and gurneys, a chair hoisted up a wall on a rail so that its occupant faces his interrogators like an exhibit. Its scientists wear ill-assorted clothes from several decades at once, a wardrobe that Julie Weiss's Academy Award-nominated costume design makes eloquent of a civilisation reduced to its stores. Its time machine is a sealed tube into which a naked man is slid like a body into a drawer. Nothing in 2035 has been designed; everything has been found, repaired, adapted, and made to serve. The future is a bricolage of our present's remains.

The aesthetic has a lineage. George Lucas spoke of the "used universe" of *Star Wars* (1977), whose spacecraft were scarred and dirty, against the gleaming futures of earlier science fiction; Ridley Scott's *Alien* (1979) made its starship a grimy industrial workplace; and *Blade Runner*, co-written by David Peoples, built its Los Angeles of 2019 by what its designers called retrofitting, bolting ducts, conduits, and machinery onto the facades of older buildings, so that the future appeared as the past overgrown with apparatus. Gilliam's *Brazil* (1985) had carried the principle into bureaucratic dystopia. Its ducts, coiling through every room like the intestines of the state, are the film's master image, and its hero's brief salvation is a renegade heating engineer who repairs them outside the system. The bunker of 2035 descends from *Brazil*'s ducts and from *Blade Runner*'s retrofitted city, and it inverts both. Here the apparatus is not overgrowing a living city. It is all that remains of one.

The underground as the site of the future is older still. Fritz Lang's *Metropolis* (1927) placed its workers in a subterranean city beneath the gardens of the rich, and H. G. Wells's *The Time Machine* (1895) divided the humanity of the year 802,701 into the ethereal Eloi above and the Morlocks below, machine-tenders descended from the labouring classes. *La Jetée* put its survivors in the galleries beneath Chaillot; *Twelve Monkeys* puts its own beneath Philadelphia. In each case the underground is the place of labour, of machinery, and of a humanity that has lost the surface; and in *Twelve Monkeys* the lost surface has passed to the animals, as Wells's far future passed to the crabs and creeping things of a dying shore. The used future is the Morlocks' future: the world as seen from the machine room.

The name later given to one branch of this aesthetic is steampunk, a word the novelist K. W. Jeter proposed in 1987 for fiction that reimagined the future through the technologies of the Victorian past. Gilliam's work, from the engravings of his Python animations to the mechanical wonders of *Munchausen*, is often counted among its precursors, and his own precursors are visible in turn: Georges Méliès, whose *Le Voyage dans la Lune* (1902) made the cinema a machine of marvels, and the Czech filmmaker Karel Zeman, whose *Vynález zkázy* (1958) animated the engraved illustrations of Jules Verne's novels and whose *Baron Prášil* (1961) treated the Munchausen legend in the same idiom a generation before Gilliam. *Twelve Monkeys* is the darkest of these retro-futures. Its machines are not wonders but prosthetics; its technology does not open the sky but seals the underground; and its marvel, the time machine, is used not to explore but to collect.

What the used future expresses, philosophically, is a relation to time. A future built from the debris of the present is a future that has imagined nothing new; it can only repair. The scientists of 2035 possess a time machine, but they cannot design a chair. Their capacity for invention has been concentrated entirely in the means of returning to the past, and everything else is salvage. This is the visual form of the actuarial future described in Chapter Six. A civilisation whose relation to time is a claim against a past loss has no project of its own; its only innovation is the instrument of recovery. The used future is the future as loss adjuster, valuing the world by what can be salvaged from it.

E.2
The Revolutionary Energies of the Outmoded

Benjamin, Baudelaire, and the Arcades of 2035

In his essay on Surrealism (1929), Walter Benjamin credited André Breton with the discovery of the revolutionary energies that appear in the outmoded: in the first iron constructions and the first factory buildings, in the earliest photographs, in objects that have begun to die out, in grand pianos and the clothes of five years ago, in fashionable restaurants once the vogue has ebbed. The outmoded, for Benjamin, was the place where the dreams of a past epoch, congealed in its commodities, became legible to a later one, and where the promise those dreams contained might yet be redeemed. The project that occupied the last thirteen years of his life, the unfinished study of the Parisian arcades, sought in the iron-and-glass shopping passages of the nineteenth century, already obsolete by the 1920s, the dream-houses of a collective that had not yet awakened.

Benjamin borrowed from Jules Michelet the sentence that every epoch dreams the one that follows it, and he added that in this dream the coming epoch appears wedded to elements of primal history. *Twelve Monkeys* inverts Michelet. In its fiction it is the later epoch that dreams the earlier: 2035 dreams 1996, through the recurring vision of a convict who carries the image of an airport into a world without air. And the 1996 of the dream appears, as Benjamin predicted, wedded to primal history—to animals in the streets, to a plague, to a death at a gate. The bunker of 2035 is an arcade in Benjamin's sense, a covered world lit artificially and furnished with the obsolete commodities of a vanished culture, and its inhabitants are the collective that dreams. What they lack is the awakening Benjamin hoped for, the moment in which the dream of the past is recognised and its promise seized. They do not awaken; they extract.

Benjamin's figure for the observer of the nineteenth-century city was the flâneur, the idler who strolls the arcades and boulevards at the pace of a tortoise, and his most famous description of the type has the flâneur going botanising on the asphalt. *Twelve Monkeys* gives the phrase a literal and terrible fulfilment. On his first mission Cole walks the snowbound streets of 2035 in a sealed suit, collecting specimens of the life that has reclaimed the city: he botanises on the asphalt, the last flâneur of a city whose crowds have vanished. The figure who, for Baudelaire and for Benjamin, embodied the modern subject's immersion in the crowd here walks a city without any crowd at all, observed in his turn by the beasts that have replaced it.

Baudelaire's "Le Cygne," which Benjamin read as the great poem of the city's transformation, supplies an emblem for the film's escaped animals. Walking through

the new Paris of Haussmann's demolitions, the poet remembers a swan that had escaped from its cage and was dragging its white plumage along a dry gutter, bathing its wings in the dust and lifting its head toward an ironic sky, as if reproaching God. Paris changes, the poem says, but nothing in my melancholy has moved; everything for me becomes allegory. The swan is an exile, like Andromache in Epirus, like sailors forgotten on islands: it is the animal escaped from the menagerie into a city that has no place for it. The animals the Army of the Twelve Monkeys releases into Philadelphia are Baudelaire's swan multiplied, and the bear and the lion of 2035 are the swan triumphant, the escaped animal no longer exiled in the city but in possession of it. For Benjamin, allegories were in the realm of thought what ruins are in the realm of things. The city of 2035, with its beasts among the ruins, is an allegory in exactly this sense.

Benjamin also wrote of the photographs Eugène Atget made of Paris at the turn of the twentieth century—streets, courtyards, and shopfronts almost always empty of people—that they photographed the city like the scene of a crime, and that with them photographs began to become evidence in the trial of history. The surface of 2035 in *Twelve Monkeys* is an Atget photograph set in motion: a city emptied of its inhabitants, recorded with the evidentiary flatness of a crime scene, in which the crime is the disappearance of the human. The chain this dissertation has traced from Marker's photographs to Gilliam's moving images passes here through one of its art-historical ancestors. Atget emptied Paris by long exposure and early hours, which erased the passers-by; the plague of 1996 empties Philadelphia by death. The result, in both cases, is an image of the city as evidence, waiting for an investigator.

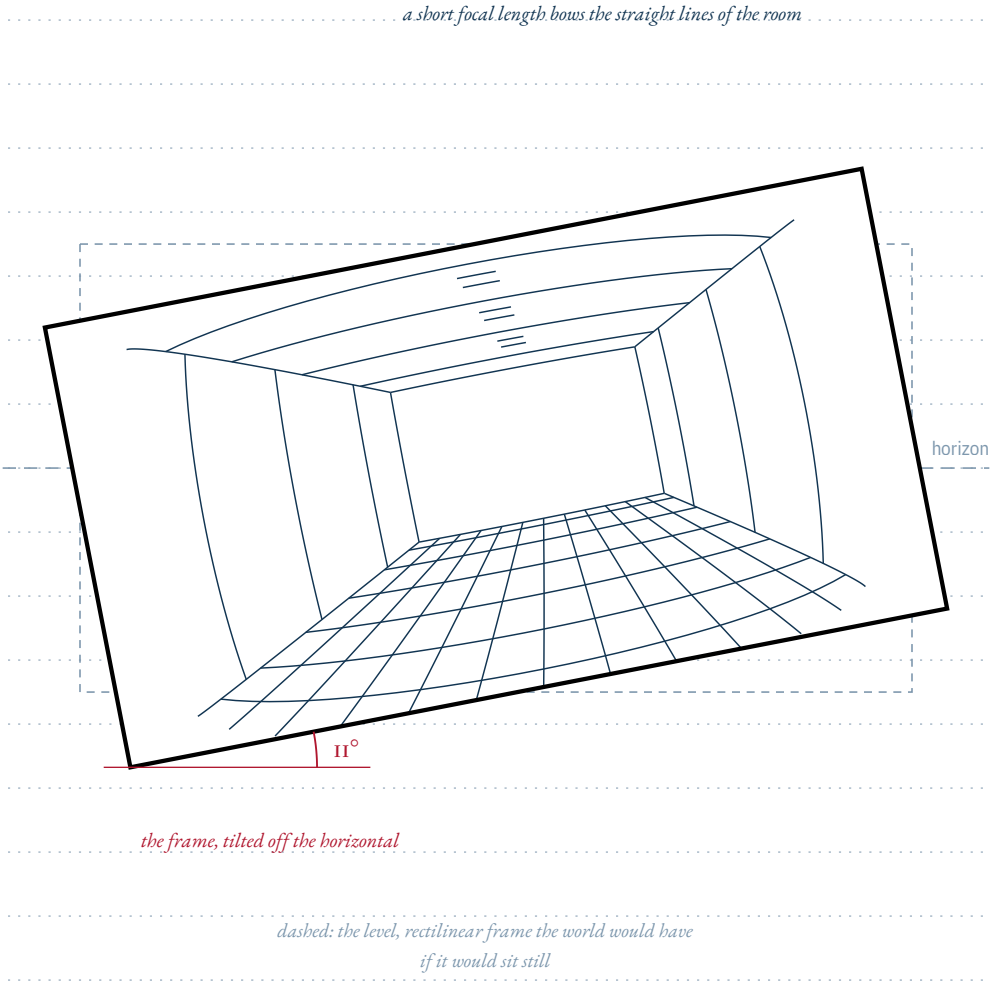
E.3
The Frame out of Joint

Expressionism, the Wide Angle, and the Tilted World

Gilliam's camera is the most immediately recognisable element of his style, and in *Twelve Monkeys*, photographed by Roger Pratt, it reaches a kind of perfection. The director favours short focal lengths, lenses wide enough to bow the edges of the frame and to exaggerate the depth of rooms, so that faces in the foreground swell while walls recede; he tilts the camera off the horizontal; he sets it low beneath his actors or high above them; he crowds the frame with objects. The effect is a world that will not sit still, whose verticals lean and whose spaces bulge, and which seems at every moment about to slide out of its own frame. In the asylum of 1990 and the bunker of 2035

Plate II

The frame out of joint



The wide angle and the tilted camera as a diagram: a corridor seen through a short focal length, whose straight lines bow outward, inside a frame turned off the horizontal. The dashed rectangle is the level frame that a world out of joint will not occupy.

alike, the tilted frame is the film's ordinary condition, not an exception reserved for crisis.

The tilted world has an ancestry in the cinema of madness. Robert Wiene's *The Cabinet of Dr. Caligari* (1920) painted its streets and rooms with oblique lines and impossible perspectives, and closed with the revelation that its narrator is an inmate of an asylum whose director he has cast as the murderous Caligari—a frame that leaves undecided whether the distorted world of the film is the madman's vision or the truth. Siegfried Kracauer's *From Caligari to Hitler* (1947) read the expressionist frame as the image of a collective soul in crisis, and Lotte Eisner's *The Haunted Screen* (1952) traced its sources in German Romanticism and the stagecraft of Max Reinhardt. *Twelve Monkeys* inherits Caligari's undecidability for most of its length and resolves it in the opposite direction. The world is distorted not because the man who sees it is mad but because the world is out of joint; and the asylum director whom the patient might accuse is no murderer, only a functionary whose categories cannot accommodate the truth.

A second ancestor deserves more attention than it has received. Carol Reed's *The Third Man* (1949), written by Graham Greene and photographed by Robert Krasker in canted frames among the ruins of occupied Vienna, is a film about a medical crime. Its villain, Harry Lime, has made a fortune selling diluted penicillin on the black market, and the children who received it have died or been left damaged in a hospital ward. Lime's apologia is delivered at the top of the giant Ferris wheel in the Prater, from which he invites his friend to look down at the people below—mere dots—and asks whether he would really feel pity if one of the dots stopped moving for ever, were he paid enough for each. In the speech Orson Welles is said to have written for himself, Lime adds that Italy under the Borgias, with its warfare, terror, and murder, produced Michelangelo and the Renaissance, while Switzerland, with its brotherly love and five hundred years of democracy and peace, produced the cuckoo clock. Every element of the actuarial argument of this dissertation is present in embryo: the medical vector of death; the view from the wheel, in which persons become dots on a statistical field; the theodicy that justifies slaughter by its products; the contempt for brotherly love, for *philadelphia*, and its emblem, the clock. Dr. Peters, looking down from his aircraft, is Harry Lime on the wheel, and the tilted frames through which Gilliam shows his world are Krasker's, inherited together with Lime's arithmetic.

Welles himself was the great precursor of Gilliam's lens. In *Touch of Evil* (1958) and *The Trial* (1962) the wide angle turned offices into hangars and corridors into abysses, and dwarfed individuals within architectures of power; the debt of *Brazil*'s ministries to Welles's Kafka is often remarked. In *Twelve Monkeys* the Wellesian lens serves a further purpose. It makes the institutions of the film—the panel room of 1990, the inter-

rogation chamber of 2035—into spaces in which the individual is seen from a distance that is also a judgment, a small body in a field of apparatus. The wide angle is the visual equivalent of the actuarial gaze. It sees the person as a figure in a population.

The tilted frame is, finally, the spatial form of Hamlet's complaint, which Chapter Three took, with Deleuze, as the formula of the film's time. A time out of joint produces a space out of true. The horizon that will not lie level, the wall that leans, the room that bulges toward the lens are the visible signs of a world whose temporal order has been dislocated, in which effects precede their causes and a child watches his own death. Gilliam's style is often described as a personal idiosyncrasy, the signature of a visionary temperament. In *Twelve Monkeys* it is something more precise: the translation of a metaphysics of closed time into the geometry of the frame.

E.4
The Grotesque Body and the Carnival

Bakhtin, Bruegel, and the Fools' Army

Mikhail Bakhtin's study of Rabelais, written in 1940 and published in 1965, gave modern criticism its theory of carnival: the festive suspension of hierarchy in which the world is turned upside down, the fool is crowned and the king uncrowned, the sacred is parodied, and the body is celebrated in its grotesque form—open, protruding, eating and drinking and excreting, dying and giving birth, never finished and never closed. Carnival laughter, for Bakhtin, was not satire, which stands outside its object, but a regenerative laughter in which the whole people participates and through which the old world dies and the new is born. Gilliam's work, from the grotesque bodies of his Python animations to the feasts and freaks of *Munchausen*, is among the purest cinematic heirs of this tradition; the giant foot that stamps down at the close of the *Flying Circus* titles was cut, famously, from Cupid's foot in Bronzino's *Allegory with Venus and Cupid* (c. 1545), a Renaissance allegory of desire and time turned into an instrument of slapstick.

Twelve Monkeys contains a carnival, and it is the domain of Jeffrey Goines. The day room of 1990, with its inmates wandering, shouting, and parading under the eyes of indifferent attendants, is a carnival square, and Jeffrey is its lord of misrule, the mock king who conducts Cole through the ward as though it were his kingdom. In 1996 his Army stages the carnival's two great rites. It uncrowns the king, locking the father, the eminent virologist, in a cage at the zoo; and it turns the world upside down, releasing the animals into the streets and confining the human in their place.

Late medieval Europe knew confraternities of fools organised as mock armies—the Infanterie Dijonnaise of the Mère Folle at Dijon is the best known—that paraded their banners and parodied the institutions of order at carnival time. The Army of the Twelve Monkeys is such a confraternity, and its banner is a ring of beasts.

Pieter Bruegel the Elder painted in 1559 *The Fight Between Carnival and Lent*, in which a fat Carnival astride a barrel jousts with a gaunt Lent seated on a cart and armed with a baker's paddle bearing two herrings, across a town square crowded with revellers and penitents. *Twelve Monkeys* restages the fight across its two centuries. Lent is 2035: austere, hungry, sealed, ruled by aged specialists in a world without pleasure. Carnival is 1996: Jeffrey's army, the songs on the car radio, the air Cole gulps, the zoo opened to the streets. Bruegel's painting leaves the outcome of the joust undecided, and the calendar decides it in any case, since Lent always follows Carnival. *Twelve Monkeys* knows the calendar. Its carnival is the last before an endless Lent; and a few years after his joust, Bruegel painted *The Triumph of Death*, in which an army of skeletons overruns the living.

The grotesque body is everywhere in the film, and it is never celebrated. Cole's body is shaved, scrubbed, hosed, drugged, bound, shot, and bled, and stripped of a tooth by his own hand; the scientists' bodies are aged and oddly costumed; Jeffrey's twitches with a mania it cannot contain. Bakhtin's grotesque was the body of the people in festive renewal, the body that dies and is reborn in its descendants. In *Twelve Monkeys* the grotesque body has been captured by institutions: it is the body of the patient and the prisoner, opened not by laughter but by examination. Only in the zoo and among the animals does something of the carnival's renewing energy survive, and the renewal it promises is not a human one.

The fool, carnival's central figure, has a long literary career as a teller of truths no one will heed, and Shakespeare gave him his bitterest line. "Truth's a dog must to kennel," says Lear's Fool; "he must be whipped out, when Lady the brach may stand by the fire and stink." Truth is an animal driven from the house while flattery, another animal, warms itself at the hearth. Gilliam's cinema is populated by such fools: Sam Lowry dreaming of flight, Munchausen lying his way to truth, Parry hunting the Red Knight, Cole insisting on a future no one wants. And Gilliam's own career supplies the fool's loop. For a quarter of a century he attempted to film *Don Quixote*, Cervantes's tale of a madman who sees giants in windmills. The first attempt collapsed in 2000, a catastrophe recorded by Keith Fulton and Louis Pepe—the makers of *The Hamster Factor*—in *Lost in La Mancha* (2002), and the film was finally completed as *The Man Who Killed Don Quixote* (2018). The director of *Twelve Monkeys* spent the decades after it reliving the failure of a mission he would not abandon, until, like the hero of a time-loop game, he completed it by persistence alone.

E.5
Bellows, Spirals, and a Song

The Sound of the Loop

The music of *Twelve Monkeys* is built on a borrowed loop. Paul Buckmaster's score takes as its main theme a figure from Astor Piazzolla's *Suite Punta del Este*, played on the bandoneon, the square button accordion of the Argentine tango. The figure turns and returns, climbing and falling back, never quite resolving, like the time of the film. The instrument has its own history of exile and return. It was developed in Germany in the middle of the nineteenth century—by tradition, as a portable substitute for the organ in churches too poor to own one—and it travelled with emigrants to the Río de la Plata, where it became the voice of the tango, the music of the port and the uprooted. Piazzolla, who renewed the tango in the second half of the twentieth century and was reproached by its traditionalists for betraying it, made the bandoneon an instrument of modern melancholy. Its sound is produced by bellows: it breathes. In a film whose catastrophe is airborne, and whose hero gulps the air of a world about to lose it, the theme is played on an instrument that breathes in and out, endlessly, around a figure that cannot resolve.

The film's second music is Bernard Herrmann's, heard in the cinema where Cole and Railly hide as the sequoia scene of *Vertigo* plays on the screen and Railly prepares the disguise that will make her the woman of Cole's dream. Herrmann's score for *Vertigo* is famous for its spiralling figures, which answer the spirals of Saul Bass's titles, and for a love theme that critics have long heard as an homage to the Liebestod of Wagner's *Tristan und Isolde*: the love-death, in which the consummation of desire is its dissolution into night. The thanatic primal scene of *La Jetée* and *Twelve Monkeys*, in which the beloved waits at the end of the jetty and the reunion is a death, is a Liebestod staged at an airport, and Herrmann's music carries Wagner's chromatic longing into the film like a memory from an older century of loss.

The third music is the popular song: the radio of 1996 that Cole hears with the ecstasy of a man who has never heard music on the air, and the Armstrong recording over the credits. Chapter Six read the closing song as the anthem of an actuarial theodicy. It is also an archive. For the man from 2035 the popular music of the twentieth century is what the arcades were for Benjamin: the dream-songs of a dead epoch, preserved in recordings, which a later age can hear only as the promise of a happiness it did not inherit. Cole's ecstasy at the car radio is the most Benjaminian moment in the film. The outmoded, heard by someone for whom it is already lost, flares up with the whole energy of the world that produced it.

Against these musics the film sets the soundscape of 2035: the hum of machines, the clang of cages, the voices of the scientists echoing through their monitors, the hiss of the hoses, the wind over the snow on the surface. It is a world without music, and therefore, in Nietzsche's terms, a world without the medium from which tragedy is born. The Dionysian music of the bandoneon belongs to the film, not to the future it depicts. That asymmetry is one more measure of what the actuarial future has lost: not only the surface, the air, and the animals, but the capacity to breathe in time.

The sound of *La Jetée* offers a final contrast. Marker's film is narrated by a single voice over still images, with music and the whispers of the experimenters beneath it; its moving image arrives with birdsong rather than with music, as though the return of life to the image were announced by animals. *Twelve Monkeys* reverses the proportion. It is full of voices, many of them talking at once—scientists completing one another's sentences, patients shouting over the television, Jeffrey outrunning his own thoughts—and its moments of silence are rare and precise. The last of them falls at the airport, between the shot and the song, in the interval in which a woman looks at a child. In a film so crowded with sound, the look is given the silence of a photograph.

E.6
Two Films in One Table, and a Note on Style as Thought

The comparisons this dissertation has drawn between *La Jetée* and *Twelve Monkeys* can be gathered in a single exhibit, which sets the two films side by side on the dimensions that have mattered most to the argument.

Read down its columns, the table shows two complete worlds; read across its rows, it shows a single history. Every row records a translation from Marker's idiom into Gilliam's, and every translation moves in the same direction: from the photograph to the monitor, from the bomb to the virus, from the camp to the clinic and the laboratory, from memory to machine, from the museum to the zoo, from the stuffed animal to the living one, and from an ending without commentary to an ending with an insurer. The direction is the passage, traced in Chapter One, from the sovereign instant to the actuarial curve.

Theodor Adorno wrote in his *Aesthetic Theory* (1970) that artworks are the unconscious historiography of their epoch. The aesthetics of *Twelve Monkeys* are that historiography in visible form. The used future records an age that could imagine the end of the world more easily than a new design for a chair; the tilted frame records a time out of joint; the carnival records the last festival before an endless Lent; the bandoneon records the breath of a world about to lose its air. Gilliam's style has been

Plate 12

Two films in one table

Dimension	La Jetée *(1962)*	Twelve Monkeys *(1995)*
Form	Photo-roman of black-and-white stills with one moving shot; about 28 minutes	Colour feature film; about two hours
Catastrophe	Third World War; Paris destroyed	Virus released from a laboratory in 1996; five billion dead
Survivors	Victors' camp in the galleries beneath the Palais de Chaillot	Underground society beneath Philadelphia, run by a council of scientists
Travellers	Prisoners used as experimental subjects	Convicts sent as "volunteers" for a reduced sentence
Criterion of selection	Fixation on an image from childhood	Strength of mind and powers of observation
Means of travel	Injection, blindfold, memory	Machine; arrival naked in the street
Destinations	Peacetime Paris; a future city	Baltimore, 1990; the trenches, 1917; Philadelphia, 1996
The woman	Unnamed; met in the past	Kathryn Railly, psychiatrist
Animals	A museum of stuffed animals	Laboratory monkeys, an opened zoo, beasts in a ruined city
Hitchcock	The sequoia of *Vertigo* restaged	*Vertigo* screened; the sequoia on a cinema screen
Death	Shot on the observation pier at Orly by an agent of the camp	Shot on an airport concourse by police
Coda	None	A scientist "in insurance" beside the carrier
Last image	The falling man	The boy's face following the aircraft

praised as fantasy and blamed as excess. In this film it is neither. It is the form in which the argument of the preceding chapters becomes visible: a world made of the debris of our own, seen through a lens that will not let it stand straight, scored for an instrument that breathes, and ending, as the outmoded always ends for Benjamin, in a flash of recognition before the dark.

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Keywords

Twelve Monkeys; *La Jetée*; Terry Gilliam; David Webb Peoples; Janet Peoples; Chris Marker; actuarial tragedy; Cassandra Policy; Cassandra complex; prophecy and disbelief; time travel; closed timelike curves; Novikov self-consistency principle; bootstrap paradox; fatalism and foreknowledge; eternal recurrence; *amor fati*; *Augenblick*; photogram; still and moving image; photo-roman; *Vertigo*; anterior future; dialectical image; optical unconscious; *Nachträglichkeit*; primal scene; repetition compulsion; history of madness; anti-psychiatry; looping effects of human kinds; biopolitics; risk society; insurance; actuarial reason; history of probability; enlightened catastrophism; apocalypse; eschatology; plague narrative; pandemic imaginary; zoonosis; animal studies; anthropological machine; zoetrope; zodiac; tragic chorus; Apollo Smintheus; hauntology; re-vision; film-philosophy; remake studies; media archaeology; Anthropocene

